

SPIE PRESS | Field Guide



**SPIE**

Field Guide to  
**Infrared  
Systems,  
Detectors,  
and FPAs**  
Second Edition

***Arnold Daniels***

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Field Guide to

# **Infrared Systems, Detectors, and FPAs**

Second Edition

Arnold Daniels

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## Introduction to the Series

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Welcome to the *SPIE Field Guides*—a series of publications written directly for the practicing engineer or scientist. Many textbooks and professional reference books cover optical principles and techniques in depth. The aim of the *SPIE Field Guides* is to distill this information, providing readers with a handy desk or briefcase reference that provides basic, essential information about optical principles, techniques, or phenomena, including definitions and descriptions, key equations, illustrations, application examples, design considerations, and additional resources. A significant effort will be made to provide a consistent notation and style between volumes in the series.

Each *SPIE Field Guide* addresses a major field of optical science and technology. The concept of these *Field Guides* is a format-intensive presentation based on figures and equations supplemented by concise explanations. In most cases, this modular approach places a single topic on a page, and provides full coverage of that topic on that page. Highlights, insights, and rules of thumb are displayed in sidebars to the main text. The appendices at the end of each *Field Guide* provide additional information such as related material outside the main scope of the volume, key mathematical relationships, and alternative methods. While complete in their coverage, the concise presentation may not be appropriate for those new to the field.

The *SPIE Field Guides* are intended to be living documents. The modular page-based presentation format allows them to be updated and expanded. We are interested in your suggestions for new *Field Guide* topics as well as what material should be added to an individual volume to make these *Field Guides* more useful to you. Please contact us at [fieldguides@SPIE.org](mailto:fieldguides@SPIE.org).

John E. Greivenkamp, *Series Editor*  
College of Optical Sciences  
The University of Arizona

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## **Field Guide to Infrared Systems, Detectors, and FPAs, 2<sup>nd</sup> Edition**

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*Field Guide to Infrared Systems, Detectors, and FPAs, Second Edition*, is written to clarify and summarize the theoretical and practical principles of modern infrared technology. It is intended as a reference for the practicing engineer and/or scientist who requires effective practical information to design, build, and/or test infrared equipment in a wide variety of applications.

This book combines numerous engineering disciplines necessary for the development of an infrared system. It describes the basic elements involving image formation and image quality, radiometry and flux transfer, and explains the figures of merit for detector performance. It considers the development of search infrared systems and specifies the main descriptors used to characterize thermal imaging systems. Furthermore, this guide clarifies, identifies, and evaluates the engineering tradeoffs in the design of an infrared system.

The 2<sup>nd</sup> edition provides the reader with an up-to-date view of the various third-generation infrared focal plane array (IRFPA) technologies currently in use or being actively researched. It also includes an overview of a new target acquisition model known as the “Targeting Task Performance (TTP) metric.” The applicability of this range performance model extends to sample imagers, digital image enhancement, and other features of modern imaging systems.

I would like to acknowledge and express my gratitude to my professor and mentor Dr. Glenn Boreman for his guidance, experience, and friendship. The knowledge that he passed on to me during my graduate studies at CREOL ultimately contributed to the creation of this book. I also would like to thank Dr. Boreman for reviewing a draft copy of the IRFPA 2<sup>nd</sup> edition manuscript.

I extend my sincere appreciation to Dr. Mel Friedman, NVESD, who took upon himself the onerous task of improving and clarifying the TTP metric concepts and its contents.

Above all, I voice a special note of gratitude to my kids Becky and Alex and my wife Rosa for their love and support.

## **Field Guide to Infrared Systems, Detectors, and FPAs, 2<sup>nd</sup> Ed. (cont.)**

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Lastly, I would particularly like to thank you, the reader, for selecting this book and taking the time to explore the topics related to this motivating and exciting field. I trust that the contents of this book will prove interesting and useful to engineers and scientists working in one of the various infrared fields.

This Field Guide is dedicated to the memory of my father and brothers.

*Arnold Daniels  
October 2010*

## Table of Contents

|                                               |           |
|-----------------------------------------------|-----------|
| <b>Glossary of Symbols</b>                    | <b>xi</b> |
| <b>Introduction</b>                           | <b>1</b>  |
| Electromagnetic Spectrum                      | 1         |
| Infrared Concepts                             | 2         |
| <b>Optics</b>                                 | <b>3</b>  |
| Imaging Concepts                              | 3         |
| Magnification Factors                         | 4         |
| Thick Lenses                                  | 5         |
| Stops and Pupils                              | 6         |
| F-number and Numerical Aperture               | 7         |
| Field of View                                 | 8         |
| Combination of Lenses                         | 9         |
| Afocal Systems and Refractive Telescopes      | 10        |
| Cold-Stop Efficiency and Field Stop           | 11        |
| Image Quality                                 | 12        |
| Image Anomalies in Infrared Systems           | 14        |
| Infrared Materials                            | 15        |
| Ceramic and Amorphous Materials               | 19        |
| GASIR Chalcogenide Glass Materials            | 20        |
| Material Dispersion                           | 21        |
| Atmospheric Transmittance                     | 23        |
| <b>Radiometry and Sources</b>                 | <b>24</b> |
| Solid Angle                                   | 24        |
| Radiometry                                    | 25        |
| Radiometric Terms                             | 26        |
| Flux Transfer                                 | 28        |
| Flux Transfer for Image-Forming Systems       | 29        |
| Source Configurations                         | 30        |
| Blackbody Radiators                           | 32        |
| Planck's Radiation Law                        | 33        |
| Stefan-Boltzmann and Wien's Displacement Laws | 35        |
| Rayleigh-Jeans and Wien's Radiation Laws      | 36        |
| Exitance Contrast                             | 37        |
| Emissivity                                    | 38        |
| Kirchhoff's Law                               | 39        |

## Table of Contents (cont.)

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| Emissivity of Various Common Materials                                     | 40        |
| Radiometric Measure of Temperature                                         | 41        |
| Collimators                                                                | 43        |
| <b>Performance Parameters for Optical Detectors</b>                        | <b>44</b> |
| Infrared Detectors                                                         | 44        |
| Primary Sources of Detector Noise                                          | 45        |
| Noise Power Spectral Density                                               | 46        |
| White Noise                                                                | 47        |
| Noise-Equivalent Bandwidth ( $NE\Delta f$ )                                | 48        |
| Shot Noise                                                                 | 50        |
| Signal-to-Noise Ratio: Detector and BLIP Limits                            | 51        |
| Generation-Recombination Noise                                             | 52        |
| Johnson Noise                                                              | 53        |
| $1/f$ Noise and Temperature Noise                                          | 54        |
| Detector Responsivity                                                      | 55        |
| Spectral Responsivity                                                      | 57        |
| Blackbody Responsivity                                                     | 58        |
| Noise-Equivalent Power                                                     | 59        |
| Specific or Normalized Detectivity                                         | 60        |
| Photovoltaic Detectors or Photodiodes                                      | 61        |
| Sources of Noise in PV Detectors                                           | 62        |
| Expressions for $D^*_{PV,BLIP}$ , $D^{**}_{PV,BLIP}$ , and $D^*_{PV,JOLI}$ | 63        |
| Photoconductive Detectors                                                  | 64        |
| Sources of Noise in PC Detectors                                           | 65        |
| Third-Generation Infrared Imagers                                          | 66        |
| Indium Antimonite (InSb) Photodiodes                                       | 67        |
| Mercury Cadmium Telluride (HgCdTe)                                         |           |
| Photodetectors                                                             | 68        |
| Control of the Alloy Composition                                           | 69        |
| HgCdTe Photodiodes and FPAs                                                | 70        |
| DLHJ Photodiodes                                                           | 71        |
| Dual-Band HgCdTe FPAs                                                      | 72        |
| HDVIP Photodiodes                                                          | 73        |
| Uncooled HgCdTe Photodiodes                                                | 75        |
| Quantum Well Infrared Photodetectors (QWIPs)                               | 77        |
| Types of QWIPs                                                             | 80        |

## Table of Contents (cont.)

|                                                |            |
|------------------------------------------------|------------|
| Superlattices (SLs)                            | 82         |
| Multispectral QWIPs                            | 83         |
| Light Couplers                                 | 85         |
| Pyroelectric Detectors                         | 88         |
| Pyroelectric Detectors—Mathematical Approach   | 90         |
| Microbolometers                                | 93         |
| Microbolometers—Mathematical Approach          | 96         |
| Infrared Dynamic Scene Simulators              | 99         |
| Thermoelectric Detectors                       | 100        |
| <b>Infrared Systems</b>                        | <b>101</b> |
| Raster Scan Format: Single-Detector            | 101        |
| Multiple-Detector Scan Formats:                |            |
| Serial Scene Dissection                        | 103        |
| Multiple-Detector Scan Formats: Parallel Scene |            |
| Dissection                                     | 104        |
| Staring Systems                                | 105        |
| Search Systems and Range Equation              | 106        |
| Noise-Equivalent Irradiance                    | 109        |
| Performance Specification: Thermal-Imaging     |            |
| Systems                                        | 110        |
| Modulation Transfer Function (MTF) Definitions | 111        |
| Optics MTF: Calculations                       | 114        |
| Electronics MTF: Calculations                  | 116        |
| MTF Measurement Setup and Sampling Effects     | 117        |
| MTF Measurement Techniques: PSF and LSF        | 118        |
| MTF Measurement Techniques: ESF and CTF        | 119        |
| MTF Measurement Techniques: Noiselike Targets  | 121        |
| MTF Measurement Techniques: Interferometry     | 123        |
| Noise-Equivalent Temperature Difference (NETD) | 124        |
| NETD Measurement Technique                     | 125        |
| Minimum Resolvable Temperature Difference      |            |
| (MRTD)                                         | 126        |
| MRTD: Calculation                              | 127        |
| MRTD Measurement Technique                     | 128        |
| MRTD Measurement: Automatic Test               | 129        |
| Johnson Metric Methodology                     | 130        |
| Johnson Criteria Flaws                         | 132        |

**Table of Contents (cont.)**

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|                                         |            |
|-----------------------------------------|------------|
| Targeting Task Performance (TTP) Metric |            |
| Methodology                             | 133        |
| Human Vision—Distribution of Retinal    |            |
| Photoreceptors                          | 134        |
| Contrast Threshold Function (CTF)       | 136        |
| Target Acquisition Performance          | 141        |
| Probability of Task Performance         | 144        |
| N50 to V50 Conversion (Example)         | 146        |
| Acquisition Level Definitions           | 147        |
| TTP Summary                             | 148        |
| <b>Equation Summary</b>                 | <b>149</b> |
| <b>Bibliography</b>                     | <b>163</b> |
| <b>Index</b>                            | <b>167</b> |



## Glossary of Symbols

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|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| $A$                       | Area                                                          |
| $A_d$                     | Detector area                                                 |
| $A_{\text{enp}}$          | Area of an entrance pupil                                     |
| $A_{\text{exp}}$          | Area of an exit pupil                                         |
| $A_{\text{footprint}}$    | Footprint area                                                |
| $A_{\text{img}}$          | Area of an image                                              |
| $A_{\text{lens}}$         | Lens area                                                     |
| $A_{\text{obj}}$          | Area of an object                                             |
| $A_{\text{opt}}$          | Area of an optical component                                  |
| $A_s$                     | Source area                                                   |
| AMTIR                     | Amorphous materials transmitting infrared radiation           |
| ATR                       | Automatic target recognition                                  |
| $B$                       | 3-db bandwidth                                                |
| B-B                       | Bound-to-bound                                                |
| B-C                       | Bound-to-continuum                                            |
| B-QB                      | Bound-to-quasi-bound                                          |
| $b.f.l$                   | Back focal length                                             |
| BLIP                      | Background-limited infrared photodetector                     |
| $c$                       | Speed of light in vacuum                                      |
| $C_d$                     | Detector capacitance                                          |
| $C_{\text{th}}$           | Thermal capacitance                                           |
| CQWIP                     | Corrugated quantum well infrared photodetector                |
| CTF                       | Contrast transfer function                                    |
| $\text{CTF}_{\text{eye}}$ | Contrast threshold function of the eye                        |
| $\text{CTF}_n$            | Contrast threshold function in the presence of external noise |
| $\text{CTF}_{\text{sys}}$ | Contrast threshold function of a system                       |
| CVD                       | Chemical vapor deposition                                     |
| $d_{\text{diff}}$         | Diameter of a diffraction-limited spot                        |
| <b>D</b>                  | Electrical displacement                                       |
| $D^*$                     | Normalized detectivity of a detector                          |
| $D_{\text{BF}}^*$         | Background fluctuation D-star                                 |
| $D_{\text{BLIP}}^*$       | D-star under BLIP conditions                                  |
| $D_{\text{TF}}^*$         | Temperature fluctuation D-star                                |
| $D^{**}$                  | Angle-normalized detectivity                                  |
| $D_{\text{enp}}$          | Diameter of an entrance pupil                                 |

## Glossary of Symbols (cont.)

---

|                           |                                              |
|---------------------------|----------------------------------------------|
| $D_{\text{exp}}$          | Diameter of an exit pupil                    |
| $D_{\text{img}}$          | Image diameter                               |
| $D_{\text{in}}$           | Input diameter                               |
| $D_{\text{lens}}$         | Lens diameter                                |
| $D_{\text{obj}}$          | Object diameter                              |
| $D_{\text{opt}}$          | Optics diameter                              |
| $D_{\text{out}}$          | Output diameter                              |
| DDCA                      | Detector-Dewar cooler assembly               |
| DEE                       | Digital emitter engine                       |
| DLHJ                      | Double-layer heterostructure junction        |
| DSS                       | Dynamic scene simulator                      |
| e                         | Energy-based unit subscript                  |
| $E_{\text{bkg}}$          | Background irradiance                        |
| $E_{\text{img}}$          | Image irradiance                             |
| $E_{\text{source}}$       | Source irradiance                            |
| EAPD                      | Electron-injected avalanche photodiode       |
| ESF                       | Edge spread function                         |
| $\epsilon$                | Energy of a photon                           |
| $\epsilon_{\text{gap}}$   | Energy gap of a semiconductor material       |
| $f$                       | Focal length                                 |
| $f_{\text{eff}}$          | Effective focal length                       |
| $f_0$                     | Center frequency of an electrical filter     |
| $f.f.l$                   | Front focal length                           |
| $f(x,y)$                  | Object function                              |
| $F_{\text{B}}$            | Back focal point                             |
| $F_{\text{F}}$            | Front focal point                            |
| $F(\xi,\eta)$             | Object spectrum                              |
| FOR                       | Field of regard                              |
| FOV                       | Full-angle field of view                     |
| FOV <sub>half-angle</sub> | Half-angle field of view                     |
| FPA                       | Focal plane array                            |
| $F/\#$                    | F-number                                     |
| $g(x,y)$                  | Image function                               |
| $G$                       | Gain of a photoconductive detector           |
| $G(\xi,\eta)$             | Image spectrum                               |
| GASIR                     | Germanium arsenic selenium infrared material |

## Glossary of Symbols (cont.)

---

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| $h$                  | Planck's constant                                          |
| $h_{\text{img}}$     | Image height                                               |
| $h_{\text{obj}}$     | Object height                                              |
| $h(x,y)$             | Impulse response                                           |
| $H$                  | Heat capacity                                              |
| $H(\xi,\eta)$        | Transfer function                                          |
| HDVIP                | High-density vertically integrated photodiode              |
| HIFOV                | Horizontal instantaneous field of view                     |
| HFOV                 | Horizontal field of view                                   |
| $i$                  | Electrical current                                         |
| $\bar{i}$            | Mean current                                               |
| $i_{1/f}$            | rms 1/f-noise current                                      |
| $i_{\text{avg}}$     | Average electrical current                                 |
| $i_{\text{bkg}}$     | Background rms current                                     |
| $i_{\text{dark}}$    | Dark current                                               |
| $i_j$                | rms Johnson noise current                                  |
| $i_{\text{G/R}}$     | Generation-recombination noise rms current                 |
| $i_{\text{noise}}$   | Noise current                                              |
| $i_o$                | Dark current                                               |
| $i_{\text{oc}}$      | Open circuit current                                       |
| $i_{\text{pa}}$      | Preamplifier noise rms current                             |
| $i_{\text{ph}}$      | Photogenerated current                                     |
| $i_{\text{rms}}$     | rms current                                                |
| $i_{\text{sc}}$      | Short-circuit current                                      |
| $i_{\text{shot}}$    | Shot noise rms current                                     |
| $i_{\text{sig}}$     | Signal current                                             |
| IC                   | Integrated circuit                                         |
| IRFPA                | Infrared focal plane array                                 |
| $J$                  | Current density                                            |
| $k$                  | Boltzmann's constant                                       |
| $K$                  | Thermal conductance                                        |
| $\mathcal{K}(\xi_t)$ | Spatial-frequency dependant MRTD<br>proportionality factor |
| $l_w$                | Width of a well                                            |
| $L$                  | Radiance                                                   |
| $L_{\text{bkg}}$     | Background radiance                                        |
| $L_v$                | Visual luminance                                           |
| $L_\lambda$          | Spectral radiance                                          |

## Glossary of Symbols (cont.)

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|                            |                                                                          |
|----------------------------|--------------------------------------------------------------------------|
| LPE                        | Liquid phase epitaxy                                                     |
| LSF                        | Line spread function                                                     |
| LWIR                       | Long-wave infrared                                                       |
| $M$                        | Exitance                                                                 |
| $M_{\text{meas}}$          | Measured exitance                                                        |
| $M_{\text{obj}}$           | Exitance of an object                                                    |
| $M_{\lambda}$              | Spectral exitance                                                        |
| MBE                        | Molecular beam epitaxy                                                   |
| MEMS                       | Micro-electro-mechanical systems                                         |
| MQW                        | Multiple quantum well                                                    |
| MRTD                       | Minimum resolvable temperature difference                                |
| MTF                        | Modulation transfer function                                             |
| $\text{MTF}_{\text{pre}}$  | Pre-sampled MTF (optics detector, and line-of-sight jitter)              |
| $\text{MTF}_{\text{d}}$    | Detector MTF                                                             |
| $\text{MTF}_{\text{post}}$ | Post-sampled MTF (display, digital processing, and the eye-brain filter) |
| $\text{MTF}_{\text{sys}}$  | System's MTF                                                             |
| MWIR                       | Midwave infrared                                                         |
| $m_e^*$                    | Effective mass of an electron                                            |
| $\mathcal{M}$              | Magnification                                                            |
| $\mathcal{M}_{\text{ang}}$ | Angular magnification                                                    |
| $n$                        | Refractive index                                                         |
| $n_{\text{cycles}}$        | Number of cycles to discriminate a target                                |
| $n_{\text{d}}$             | Number of detectors                                                      |
| $n_e$                      | Number of photogenerated electrons                                       |
| $n_{\text{lines}}$         | Number of lines                                                          |
| NEI                        | Noise-equivalent irradiance                                              |
| NEP                        | Noise-equivalent power                                                   |
| $\text{NE}\Delta f$        | Noise-equivalent bandwidth                                               |
| OTF                        | Optical transfer function                                                |
| $p$                        | Object distance                                                          |
| $\rho$                     | Pyroelectric coefficient                                                 |
| $p$                        | Momentum of an electron                                                  |
| $P$                        | Magnitude of internal polarization                                       |
| $P_{\text{avg}}$           | Average power                                                            |
| $P_{\text{chance}}$        | Probability of chance                                                    |
| $P_{\text{measured}}$      | Field-measured probability                                               |

## Glossary of Symbols (cont.)

---

|                           |                                     |
|---------------------------|-------------------------------------|
| $P_{\text{observer}}$     | Observer's probability              |
| PC                        | Photocurrent                        |
| PSD                       | Power spectral density              |
| PSF                       | Point spread function               |
| PV                        | Photovoltaic or photodiode          |
| $q$                       | Image distance                      |
| QW                        | Quantum well                        |
| QWIP                      | Quantum well infrared photodetector |
| $R$                       | Resistance                          |
| $R_d$                     | Detector resistance                 |
| $R_{\text{eq}}$           | Equivalent resistance               |
| $R_{\text{in}}$           | Input resistance                    |
| $R_L$                     | Load resistance                     |
| $R_{\text{out}}$          | Output resistance                   |
| $R_{\text{th}}$           | Thermal Resistance                  |
| RIIC                      | Read-in integrated circuit          |
| ROIC                      | Read-out integrated circuit         |
| $\mathcal{R}$             | Responsivity                        |
| $\mathcal{R}_i$           | Current responsivity                |
| $\mathcal{R}_v$           | Voltage responsivity                |
| $\mathcal{R}(\lambda)$    | Spectral responsivity               |
| $\mathcal{R}(T)$          | Blackbody responsivity              |
| $\text{SCN}_{\text{tmp}}$ | Scene contrast temperature          |
| SL                        | Superlattices                       |
| SNR                       | Signal-to-noise ratio               |
| SR                        | Strehl-intensity ratio              |
| $\text{SSR}_{\text{out}}$ | Out-of-band spurious response ratio |
| $t$                       | Time                                |
| $T$                       | Temperature                         |
| $T_B$                     | Brightness temperature              |
| $T_{\text{bkg}}$          | Background temperature              |
| $T_C$                     | Color temperature                   |
| $T_{\text{Curie}}$        | Curie temperature                   |
| $T_d$                     | Detector temperature                |
| $T_{\text{load}}$         | Load temperature                    |
| $T_{\text{rad}}$          | Radiation temperature               |
| $T_{\text{source}}$       | Source temperature                  |
| $T_{\text{target}}$       | Target temperature                  |

## Glossary of Symbols (cont.)

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|                             |                                             |
|-----------------------------|---------------------------------------------|
| TDMI                        | Time-division-multiplexed integration       |
| TIR                         | Total internal reflection                   |
| TLHJ                        | Triple-layer heterostructure junction       |
| TTP                         | Targeting task performance metric           |
| TTPF                        | Target transfer probability function        |
| $v_{\text{in}}$             | Input voltage                               |
| $v_j$                       | Johnson noise rms voltage                   |
| $v_n$                       | rms noise voltage                           |
| $v_{\text{oc}}$             | Open-circuit voltage                        |
| $v_{\text{out}}$            | Output voltage                              |
| $v_s$                       | Shot-noise rms voltage                      |
| $v_{\text{sc}}$             | Short-circuit voltage                       |
| $v_{\text{scan}}$           | Scan velocity                               |
| $v_{\text{sig}}$            | Signal voltage                              |
| $\bar{v}$                   | Mean voltage                                |
| $V$                         | Abbe number                                 |
| VIFOV                       | Vertical instantaneous field of view        |
| VFOV                        | Vertical field of view                      |
| $w$                         | proportionality factor                      |
| $x$                         | Alloy composition or molar fraction ratio   |
| $\alpha$                    | Thermal resistance coefficient              |
| $\alpha$                    | Coefficient of absorption                   |
| $\beta$                     | Blur angle caused by diffraction            |
| $\Delta f$                  | Electronic frequency bandwidth              |
| $\Delta t$                  | Time interval                               |
| $\Delta T$                  | Temperature difference                      |
| $\Delta \lambda$            | Wavelength interval                         |
| $\varepsilon$               | Emissivity                                  |
| $\varepsilon_o$             | Permeability of a material                  |
| $\eta$                      | Quantum efficiency                          |
| $\eta_{\text{scan}}$        | Scan efficiency                             |
| $\eta$                      | Spatial frequency in the vertical direction |
| $\theta$                    | Angle variable                              |
| $\theta_{\text{max}}$       | Maximum angle subtense                      |
| $\Theta$                    | Seebeck coefficient                         |
| $\lambda$                   | Wavelength                                  |
| $\lambda_{\text{cut}}$      | Cutoff wavelength                           |
| $\lambda_{\text{max}}$      | Maximum wavelength                          |
| $\lambda_{\text{max-cont}}$ | Maximum contrast wavelength                 |

## Glossary of Symbols (cont.)

---

|                         |                                               |
|-------------------------|-----------------------------------------------|
| $\lambda_{\text{peak}}$ | Peak wavelength                               |
| $\lambda_o$             | Fixed wavelength                              |
| $\Lambda$               | The de Broglie wavelength                     |
| $\mu$                   | Vertical sample frequency                     |
| $\nu$                   | Optical frequency                             |
| $\xi$                   | Spatial frequency in the horizontal direction |
| $\xi_{\text{cutoff}}$   | Spatial cutoff frequency                      |
| $\xi_J$                 | Johnson spatial frequency                     |
| $\rho$                  | Electric charge                               |
| $\rho$                  | Reflectance                                   |
| $\sigma$                | Standard deviation                            |
| $\sigma^2$              | Variance                                      |
| $\sigma_e$              | Stefan-Boltzmann constant in energy units     |
| $\sigma_{\text{eye}}$   | rms visual noise expressed at a display       |
| $\sigma_n$              | rms noise filtered on by a display            |
| $\sigma_p$              | Stefan-Boltzmann constant in photon units     |
| $\tau$                  | Transmittance                                 |
| $\tau_{\text{atm}}$     | Atmospheric transmittance                     |
| $\tau_{\text{dwell}}$   | Dwell time                                    |
| $\tau_{\text{ext}}$     | External transmittance                        |
| $\tau_{\text{frame}}$   | Frame time                                    |
| $\tau_{\text{int}}$     | Internal transmittance                        |
| $\tau_{\text{line}}$    | Line time                                     |
| $\tau_{\text{opt}}$     | Optical transmittance                         |
| $\tau_{\text{RC}}$      | Electrical time constant                      |
| $\tau_{\text{th}}$      | Thermal time constant                         |
| $\nu$                   | Horizontal sample frequency                   |
| $\phi$                  | Flux                                          |
| $\phi_{\text{abs}}$     | Absorbed flux                                 |
| $\phi_{\text{bkg}}$     | Background flux                               |
| $\phi_d$                | Detector flux                                 |
| $\phi_{\text{img}}$     | Flux incident on an image                     |
| $\phi_{\text{inc}}$     | Incident flux                                 |
| $\phi_{\text{obj}}$     | Flux radiated by an object                    |
| $\phi_{\text{sig}}$     | Signal flux                                   |
| $\phi_{\text{trans}}$   | Transmitted flux                              |
| $\phi_\lambda$          | Spectral flux                                 |
| $\psi$                  | Eigenfunction                                 |
| $\omega$                | Angular frequency                             |

**Glossary of Symbols (cont.)**

---

|                        |                            |
|------------------------|----------------------------|
| $\Omega$               | Solid angle                |
| $\Omega_d$             | Detector solid angle       |
| $\Omega_{\text{bkg}}$  | Background solid angle     |
| $\Omega_{\text{exp}}$  | Exit pupil solid angle     |
| $\Omega_{\text{enp}}$  | Entrance pupil solid angle |
| $\Omega_{\text{img}}$  | Image solid angle          |
| $\Omega_{\text{lens}}$ | Lens solid angle           |
| $\Omega_{\text{obj}}$  | Object solid angle         |
| $\Omega_s$             | Source solid angle         |



## Electromagnetic Spectrum

The **electromagnetic spectrum** is the distribution of **electromagnetic radiation** according to energy, frequency, or wavelength. Electromagnetic radiation can be described as a stream of **photons**, which are particles exhibiting wave-like properties that move at the speed of light.

| Type of Radiation     | Frequency Range (Hz)                      | Wavelength Range                      |
|-----------------------|-------------------------------------------|---------------------------------------|
| Gamma rays            | $<3 \times 10^{20}$                       | $<1$ fm                               |
| X rays                | $3 \times 10^{17}$ – $3 \times 10^{20}$   | 1 fm–1 nm                             |
| Ultraviolet           | $7.5 \times 10^{14}$ – $3 \times 10^{17}$ | 1 nm–400 nm                           |
| Visible               | $4 \times 10^{14}$ – $7.5 \times 10^{14}$ | 0.4 $\mu\text{m}$ –0.75 $\mu\text{m}$ |
| Near-infrared         | $10^{14}$ – $7.5 \times 10^{14}$          | 0.75 $\mu\text{m}$ –3.0 $\mu\text{m}$ |
| Midwave infrared      | $5 \times 10^{13}$ – $10^{14}$            | 3.0 $\mu\text{m}$ –6 $\mu\text{m}$    |
| Long-wave infrared    | $2 \times 10^{13}$ – $5 \times 10^{13}$   | 6.0 $\mu\text{m}$ –15 $\mu\text{m}$   |
| Extreme infrared      | $3 \times 10^{11}$ – $2 \times 10^{13}$   | 15 $\mu\text{m}$ –1 mm                |
| Micro and radio waves | $<3 \times 10^{11}$                       | $>1$ mm                               |

Spectral bands are commonly referred to in terms of their frequencies or wavelengths. Frequencies in the visible and infrared wavebands are measured in millions of megahertz. Wavelength can be measured interferometrically with great accuracy and is related to the optical frequency by

$$c = \lambda\nu,$$

where  $\lambda$  is the wavelength,  $\nu$  is the optical frequency, and  $c$  is the speed of light in free space ( $3 \times 10^8$  m/sec).

The difference between the wavebands of electromagnetic radiation is the amount of energy found in their photons. The energy of a photon is inversely proportional to the wavelength, and is given by

$$\mathcal{E} = h\nu = \frac{hc}{\lambda},$$

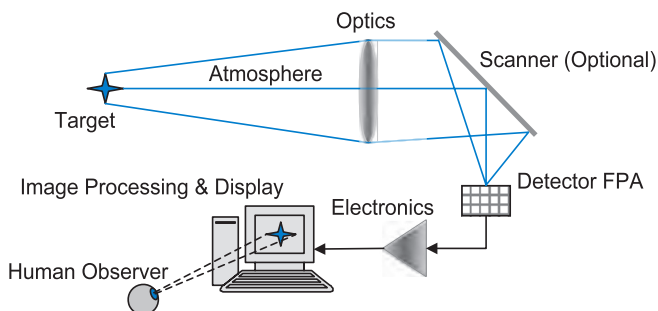
where  $h$  is Planck's constant ( $6.626 \times 10^{-34}$  J·sec).

Radio waves have photons with very low energies while gamma rays are the most energetic of all. The electromagnetic spectrum is classified based on the source, detector, and materials technologies employed in each of the spectral regions.

## Infrared Concepts

**Infrared-imaging systems** are often used to form images of targets under nighttime conditions. The target is seen because of **self-radiation** rather than the reflected radiation from the sun. Self-radiation is a physical property of all objects that are at temperatures above absolute zero (i.e.,  $0\text{ K} = -273.15^\circ\text{C}$ ).

In order to make this radiation visible, the infrared system depends on the interaction of several subsystems.



The self-radiation signature is determined by the **temperature** and the surface characteristics of the target. Gases in the atmosphere limit the frequencies at which this radiation is transmitted. The configuration of the optical system defines the **field of view (FOV)**, the **flux collection efficiency**, and the **image quality**. These parameters, along with the detector interface, impact the radiometric accuracy and **resolution** of the resulting image. The detector is a transducer that converts the optical energy into an electrical signal, and electronics amplify this signal to useful levels.

For typical terrestrial and airborne targets **Planck's equation** dictates that, within the range of temperatures of  $300\text{ K}$  to  $1000\text{ K}$ , emission of radiation occurs primarily in the infrared spectrum. However, the background is self-luminous as well, causing terrestrial targets to compete with background clusters of similar temperature. Infrared images have much lower **contrast** than corresponding visual images, which have orders of magnitude higher reflectance and emittance differences.

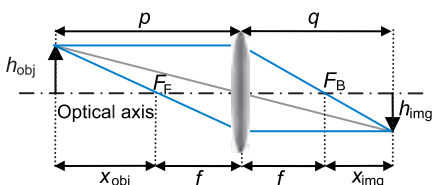
## Imaging Concepts

An object is a collection of independent source points, each emitting light rays into all forward directions. When refracted by a lens, (i.e., Snell's law) rays that diverge from each object point intersect at corresponding image-plane points. The image is built on a point-by-point basis. The power at any image location is proportional to the strength of the corresponding object point, causing a geometrical distribution of power.

The symmetry line that contains the centers of curvature of all optical surfaces is called the **optical axis**. Three ray-trace rules are used to find the image position with respect to the object:

1. rays entering parallel to the optical axis exit through the back focal point  $F_B$ ;
2. rays entering the lens through the front focal point  $F_F$  exit parallel to the optical axis; and
3. rays entering through the center (chief rays) of the lens do not change direction.

To determine the image-plane location and size, a thin lens and small-angle or **paraxial approximation** is used, which linearizes the ray-trace equations that determine the ray paths through the optical system (i.e.,  $\sin \theta \approx \tan \theta \approx \theta$ ).



**Gaussian lens equation:** 
$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q}$$

**Newtonian lens equation:** 
$$x_{obj} x_{img} = f^2$$

A **thin lens** has a thickness that is considered negligible in comparison with its focal length.

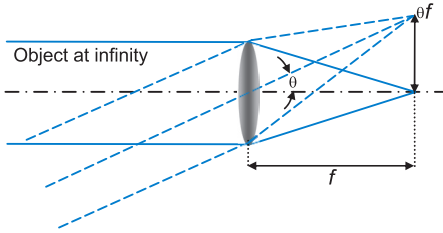
## Magnification Factors

As the object gets farther away, the image distance gets closer to  $f$  ( $f \rightarrow \infty, q \rightarrow f$ ); and as the object is placed closer to the front focus of the lens, the image gets farther away ( $p \rightarrow f, q \rightarrow \infty$ ).

The **lateral or transverse magnification** of an optical system is given by

$$\mathcal{M} = -\frac{q}{p} = \frac{h_{\text{img}}}{h_{\text{obj}}}.$$

By using the Gaussian lens equation, it can be verified that the minimum distance between a real object and its corresponding image is  $4f$  (i.e.,  $p = q = 2f$ , in which case  $\mathcal{M} = -1$ ).

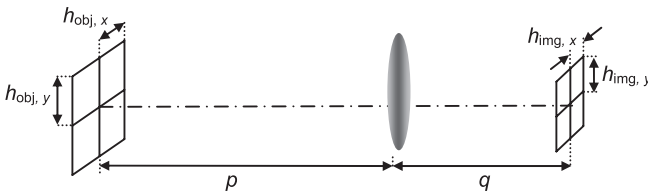


When an off-axis source is located at an infinite distance from the lens; an angle  $\theta$  [rad] exits between the direction of the collimated rays and the optical axis. The ray focuses at a distance  $\theta f$  away from the optical axis.

Squaring the lateral magnification, the **area or longitudinal magnification** is obtained by

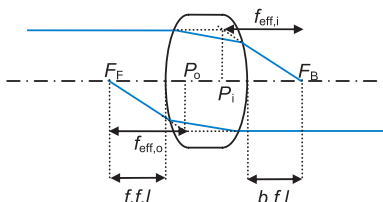
$$\mathcal{M}^2 = \frac{A_{\text{img}}}{A_{\text{obj}}} = \left( -\frac{q}{p} \right)^2,$$

which is used extensively in radiometric calculations.



## Thick Lenses

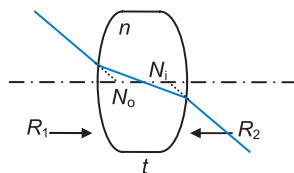
When the thickness of a lens cannot be considered negligible, the lens is treated as a thick lens.  $F_F$  and  $F_B$  are the **front** and **back focal points**, and when these focal points



are measured from the lens vertices they define the **front focal length (f.f.l)** and the **back focal length (b.f.l)** of the optical element. A diverging ray

from  $F_F$  emerges parallel to the optical axis, while a parallel incident ray is brought to  $F_B$ . In each case, the incident and emerged rays are extended to the point of intersection between the surfaces.

Transverse planes through these intersections are termed the **primary and secondary principal planes** and can either lie inside or outside the lens. Points where these planes intersect the optical axis are known as the **first and second principal points  $P_o$  and  $P_i$** . Incoming and outgoing chief rays that extend until they cross the optical axis locate the **nodal points  $N_o$  and  $N_i$** . These six points; two focal, two principal and two nodal, are named the **cardinal points** of the optical system.



$R_1$  and  $R_2$  are the lens radii, and  $n$  is the index of refraction of the lens material.

The **effective focal lengths**  $f_{eff,o}$  and  $f_{eff,i}$  are measured from the foci to their respective principal points, and are identical if the medium on each side has the same refractive index:

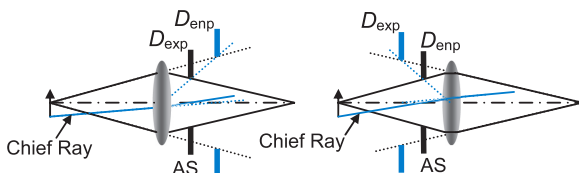
$$\frac{1}{f_{eff}} = (n-1) \left[ \frac{1}{R_1} - \frac{1}{R_2} + \frac{(n-1)t}{n} \frac{1}{R_1 R_2} \right].$$

A rule of thumb for ordinary glass lenses immersed in air is that the separation between the principal points roughly equals one third of the lens thickness ( $t$ ).

## Stops and Pupils

**Aperture stop (AS):** the physical opening that limits the angle over which the lens accepts rays from the axial object point.

**Entrance pupil ( $D_{\text{enp}}$ ):** the image of the AS as seen from the axial point on the object plane through the optical elements preceding the stop. If there are no elements between the object and the AS, the latter operates as the entrance pupil.



**Exit pupil ( $D_{\text{exp}}$ ):** the image of the AS as seen from the axial point on the image plane through those optical elements that follow the stop. If there are no elements between the AS and the image, the former serves as the exit pupil.

**Axial ray:** a ray that starts at the axial object point and ends on the axial image point.

**Marginal ray:** a special axial ray that starts at the axial object point, goes through the edge of the entrance pupil, and ends on the axial image point. The marginal ray is used to define the  $F/\#$  and the numerical aperture.

**Chief ray:** a ray that starts at the edge of the object, passes through the center of the entrance pupil, and defines the height of the image.

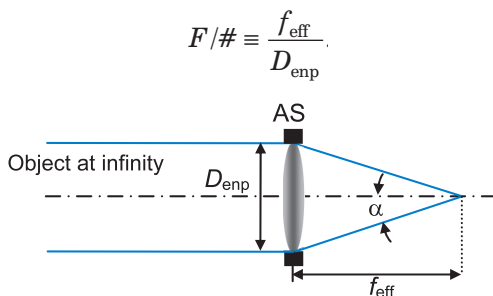
**Telecentric stop:** an aperture stop that is located at a focal point of the optical system. It is used to reduce the magnification error in the size of the projected image for a small departure from best focus.

**Telecentric system:** a system with the entrance or exit pupil located at infinity.

For any point in the object, the amount of radiation accepted by and emitted from the optical system is determined by the sizes and locations of the pupils. The location of the AS is determined by that stop or image of a stop that subtends the smallest angle as seen from the object axial point. An analogous procedure can be carried out from the image plane.

## F-number and Numerical Aperture

The **F-number** ( $F/\#$ ) is the parameter used to describe the ratio between the  $f_{\text{eff}}$  of an optical system and the **diameter** of the entrance pupil. It describes the image-spaced cone for an object at infinity:



Although the  $F/\#$  also exists in image space as  $q/D_{\text{enp}}$  for finite conjugate systems, the **numerical aperture (NA)** is usually the parameter used in these cases.

The refractive index  $n$  in air is approximately 1 and the numerical aperture is the axial cone of light in terms of the marginal ray angle  $\alpha$  and is defined as  $\text{NA} \equiv \sin \alpha$ .

The NA and the  $F/\#$  are related as follows:

$$\text{NA} \equiv \sin \left( \tan^{-1} \frac{1}{2F/\#} \right) \quad \text{or} \quad F/\# = \frac{1}{2 \tan(\sin^{-1} \text{NA})}$$

Assuming paraxial approximation,  $\sin \alpha \approx \alpha$  yielding

$$F/\# \approx \frac{1}{2\text{NA}}$$

**Example:** If an  $F/3$  system has its aperture made larger or smaller by 50% in diameter, what are the new  $F/\#$ s?

If  $D \uparrow \Rightarrow F/\# \downarrow$ ;  $F/\#_{\text{new}} = f_{\text{eff}}/1.5D = 2/3 F/\# = 2$ , resulting in faster optics.

If  $D \downarrow \Rightarrow F/\# \uparrow$ ;  $F/\#_{\text{new}} = f_{\text{eff}}/0.5D = 2 F/\# = 6$ , resulting in slower optics.

## Field of View

The **field of view (FOV)** is the angular coverage of an optical system. It can be defined either in full or half angles. Using the half-angle principle

$$\text{FOV}_{\text{half-angle}} = \theta_{1/2} = \left| \tan^{-1} \frac{h_{\text{obj}}}{p} \right| = \left| \tan^{-1} \frac{h_{\text{img}}}{q} \right|.$$

The element that limits the size of the object to be imaged is called the **field stop**, which determines the system's FOV. In an infrared camera, it is the edge of the detector array itself that bounds the image plane and serves as the field stop.

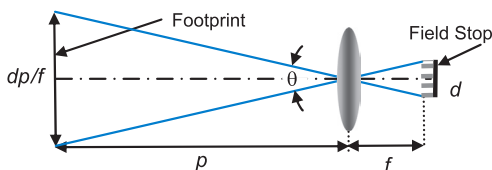
For an object at infinity, the full-angle FOV is determined by the ratio between the detector size and the system's focal length:

$$\text{FOV} = \theta = \frac{d}{f}.$$

The detector has a **footprint**; the image of the detector projected onto the object plane. The footprint defines the area of the object that contributes flux onto the detector. Given the focal length and the size of the detector, the resolution element at the object plane can be determined.

A smaller FOV is attained by increasing the focal length, causing an increment in the **magnification**; a shorter focal length widens the FOV but decreases the magnification.

The  $F/\#$  and FOV are inversely proportional, and are affected by both **flux transfer**

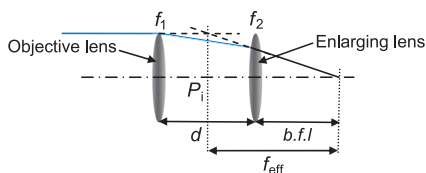


and **optical aberrations**. There is a tradeoff between the amount of light that reaches the detector, and the **image quality**. A system with a small  $F/\#$  and large FOV has high flux-transfer efficiency, but the image quality worsens. A large  $F/\#$  and small FOV restricts the system's flux, but the quality of the image is improved.



## Combination of Lenses

Consider two lenses separated by a distance  $d$ .



The  $f_{\text{eff}}$  of this optical system is given by the expression

$$\frac{1}{f_{\text{eff}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2},$$

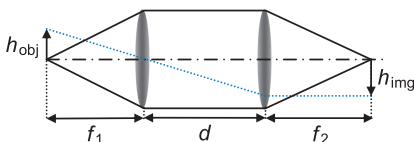
where  $f_1$  and  $f_2$  are the focal lengths of the **objective** and **enlarging lenses**, respectively. The back focal length is the distance from the last surface of the enlarging lens to the focal plane of the system, and is given by

$$b.f.l. = \frac{f_2(d - f_1)}{d - (f_1 + f_2)}.$$

When the two lenses are placed in contact (i.e.,  $d \rightarrow 0$ ), the combination acts as a single lens yielding

$$\frac{1}{f_{\text{eff}}} = \frac{1}{f_1} + \frac{1}{f_2}.$$

A special configuration of the two-lens combination system is the so-called “relay lens pair.” In this case, a source is placed at the front focal point of the optical system; the objective lens projects this source to infinity which is then imaged by the enlarging lens.



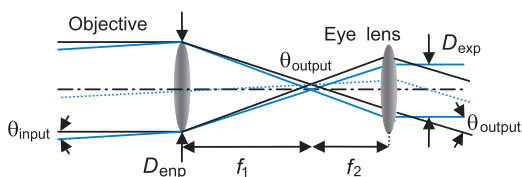
$$\mathcal{M} = \frac{h_{\text{img}}}{h_{\text{obj}}} = -\frac{f_2}{f_1}$$

The separation of the lenses affects the location of the principal planes and thereby the effective focal length of the system. Furthermore, as the lenses move apart, the size the detector lens must be increased to avoid vignetting.

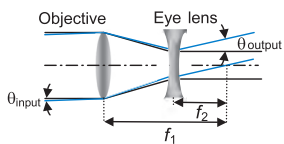
## Afocal Systems and Refractive Telescopes

**Afocal systems** do not have focal lengths. Telescopes are afocal systems with their object and image located at infinity. Their primary function is to enlarge the apparent size of a distant object. There are three main types of refractive telescopes as defined below.

**Astronomical** (or **Keplerian**) **telescopes** comprise two convergent lenses spaced by the sum of their focal lengths. The objective is usually an achromatic doublet forming a real inverted and reverted image at its focal point; the eye lens then reimages the object at infinity, where it may be erected by the use of an auxiliary lens. The aperture stop and the entrance pupil are located at the objective to minimize its size and cost.



**Galilean telescopes** comprise a positive objective and a negative eye lens, where spacing is the difference between the absolute values of the focal lengths since  $f_2$  is negative. There is no real internal image, and a reticle or crosshair



cannot be introduced into the optical system. The final image is erect. The aperture stop is usually the pupil of the viewer's eye, which is also the exit pupil.

**Terrestrial** (or **erecting**) **telescopes** are astronomical telescopes with an erecting system inserted between the eye lens and objective so that the final image is erected.

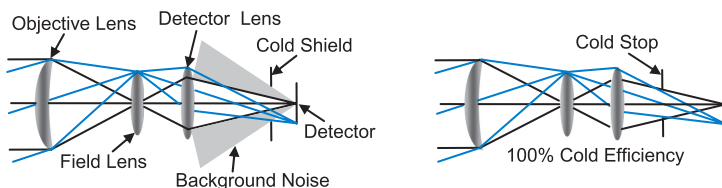
The **angular magnification** of these afocal systems is given by the ratio between the angle subtended by the image and the angle subtended by the object:

$$\mathcal{M}_{\text{angular}} = \frac{\theta_{\text{output}}}{\theta_{\text{input}}} = \left| \frac{f_2}{f_1} \right|.$$

## Cold-Stop Efficiency and Field Stop

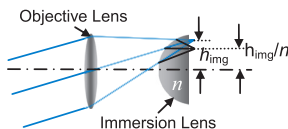
To reduce **thermal noise**, infrared photon detectors are cooled to **cryogenic temperatures**. The detector is housed in a vacuum bottle called a **Dewar**. An aperture stop is adjacent to the detector plane, which prevents **stray radiation** from reaching the detector. This **cold shield** is located inside the evacuated Dewar, and limits the angle over which the detector receives radiation.

The **cold-stop efficiency** is the percentage of the total scene source power reaching the detector. The perfect **cold stop** is defined as one that limits the reception of background radiation to the cone established by the  $F/\#$  (i.e., 100% cold-stop efficiency). This is achieved when the cold shield is located at the exit pupil of the infrared optical system.



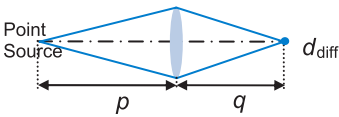
The FOV of an optical system may be increased without increasing the diameter of the detector lens by placing a **field lens** at the internal image of the system. This lens redirects ray bundles that would otherwise miss the detector back toward the optical axis. The insertion of this lens has no effect on the system's magnification. This arrangement is good for flux-collection systems (i.e., search systems), but not for imaging systems since the object is not imaged onto the detector, but rather into the field lens.

If the field lens is moved to the detector plane, it becomes an **immersion lens**, and the numerical aperture is increased by a factor of the **index of refraction** of the lens material, without modifying the characteristics of the system. This configuration allows the object to be imaged onto the detector array.



## Image Quality

The assumption thus far has been that all points in object space are mapped to points in image space. However, more detailed information such as the size of the image and its energy distribution is required to properly design an optical system. Due to effects of **diffraction** and **optical aberrations**, point sources in the object are seen in the image as **blur spots**, producing a blurred image.



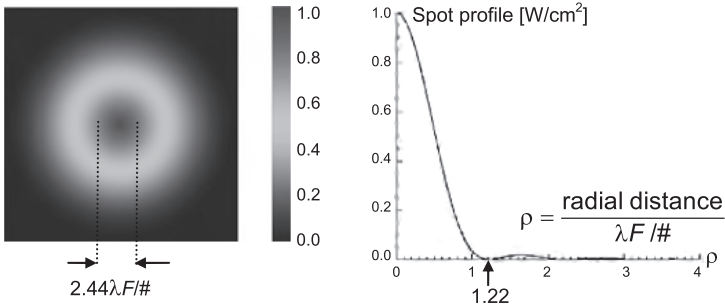
Better quality is achieved for smaller blur spots.

Diffraction is a consequence of the wave nature of radiant energy. It is a physical limitation of the optical system over which there is no control. On the other hand, optical aberrations are image defects that

arise due to the deviation from of the paraxial approximation; therefore, they can be controlled through proper design.

Even in the absence of optical aberrations, diffraction phenomena still cause a point to be imaged as a blur circle. Such an optical system is said to be **diffraction limited** and represents the best in optical performance.

The diffraction pattern of a point source appears as a bright central disk surrounded by several alternative bright and dark rings. The central disk is called the Airy disk and contains 84% of the total flux.



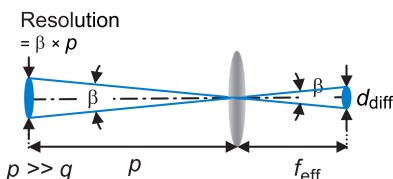
The linear diameter of the diffraction-limited blur spot is given by

$$d_{\text{diff}} = 2.44\lambda F/\#.$$

## Image Quality (cont.)

The effects of diffraction may also be expressed in angular terms. The full-angular blur is the diameter of the diffraction spot divided by  $f_{\text{eff}}$ , yielding

$$\beta = 2.44 \frac{\lambda}{D}.$$



It can also be defined as the angular subtense of the minimum resolution feature in the object space viewed from the entrance pupil.

The size of the blur spot depends on the  $F/\#$  and the spectral band in which the imaging system operates.

Low  $F/\#$  systems have the smallest diffraction-limited spot sizes, thus the best potential performance. However, the aberration effects generally become worse as the  $F/\#$  decreases. Therefore, low  $F/\#$  systems are harder to correct to diffraction-limited performance. Alternatively, longer wavelength systems have larger diffraction spots, and are easier to correct to a diffraction-limited level of performance.

For example, an  $F/1$  diffraction-limited system operating at  $10\text{ }\mu\text{m}$  forms a spot diameter of  $24.4\text{ }\mu\text{m}$ . The same system operating at  $F/7$  would form a spot diameter of  $170.8\text{ }\mu\text{m}$ . The same  $F/1$  system operating in the visible spectrum at  $0.5\text{ }\mu\text{m}$  forms a spot diameter of  $1.22\text{ }\mu\text{m}$ , while the  $F/7$  system produces a diffraction spot of  $8.5\text{ }\mu\text{m}$  in diameter.

Optical aberrations depend on the refractive and dispersive effects of the optical materials and on the geometrical arrangement of the optical surfaces. Inherent aberrations in the performance of the system with spherical surfaces include

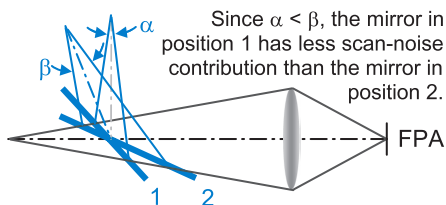
- spherical
- astigmatism
- field curvature
- distortion
- coma
- axial and lateral chromatic aberration

## Image Anomalies in Infrared Systems

Three common image anomalies are associated with IR systems.

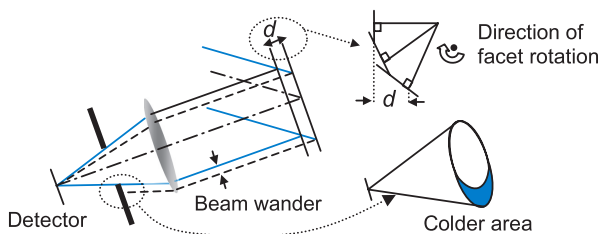
**Shading:** the gradual fall off in the scene radiance toward the edges of the detector, caused by  $\cos^4 \theta$  dependence of the effective exitance of a uniform source. Shading is controlled by optical design techniques that keep the angle of the chief ray small in image space.

**Scan noise:** the amount of self-radiation reaching the detector due to both room-temperature internal housing

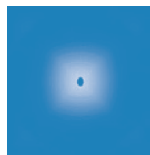


and optical elements as a function of scan position. Scan noise can also be caused by vignetting in an image-side scanner (e.g., rotating polygon). There is a final

displacement  $d$  from the center of the facet in the direction normal to the initial position. Then the interactive scanning moves in and out, and the exit beam wanders left and right, causing vignetting.



**The narcissus effect:** the result of a cold reflection of the detector array into itself, which appears as a dark spot at the center of the scan. It is controlled by using the appropriate antireflective coatings on the optical elements, and by optical design techniques that ensure the cold-reflected image is out of focus at the detector plane. Its magnitude is often expressed as a multiple of the system's noise level.



## Infrared Materials

The choice of infrared materials is dictated by the application. The most important material parameters to consider are its **transmission range** and its **material dispersion**. Other properties to be considered are the **absorption coefficient**, **reflection loss**, **rupture modulus**, **thermal expansion**, **thermal conductivity**, and **water-erosion resistance**.

The **refractive index**  $n$  is defined as

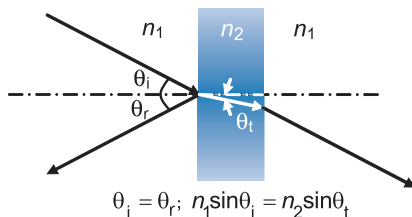
$$n = \frac{c}{v},$$

where  $c$  is the speed of light ( $3 \times 10^{10}$  cm/sec) in free space and  $v$  is the speed of light in the medium.

Whenever a ray crosses a boundary between two materials of different refractive indexes, there is some power transmitted, reflected, and absorbed (i.e., conservation of energy):

$$\phi_{\text{inc}} = \phi_{\text{trans}} + \phi_{\text{ref}} + \phi_{\text{abs}}; \quad \phi \equiv \text{power [watts]}.$$

The direction of the transmitted light is given by **Snell's law**, and the direction of the reflected beam is determined by the **law of reflection**.



The distribution of power from a plane-parallel plate at normal incidence is determined by the **Fresnel equations**:

$$\rho = \left( \frac{n_2 - n_1}{n_1 + n_2} \right)^2 \quad \text{and} \quad \tau = \frac{4n_1 n_2}{(n_1 + n_2)^2}, \quad \text{where}$$

$$\rho = \frac{\phi_{\text{ref}}}{\phi_{\text{inc}}} \quad \text{and} \quad \tau = \frac{\phi_{\text{trans}}}{\phi_{\text{inc}}}.$$

The **absorption** is described as the attenuation coefficient  $\alpha$  [1/cm] and takes power out of the beam radiation, raising the temperature of the material:

$$\phi(z) = \phi_{\text{inc}} e^{-\alpha z},$$

where  $z$  is the propagation distance within the material.

## Infrared Materials (cont.)

The **internal transmittance** is defined as the transmittance through a distance in the medium, excluding the Fresnel reflection loss at the boundaries:

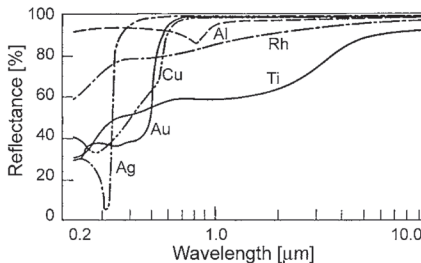
$$\tau_{\text{internal}} = \frac{\phi(z)}{\phi_{\text{inc}}} = e^{-\alpha z}.$$

Conversely, the **external transmittance** is the transmittance through a distance in the medium that includes the Fresnel loss at the boundaries:

$$\tau_{\text{external}} = \tau^2 e^{-\alpha z} = \tau^2 \tau_{\text{internal}}.$$

When examining material specifications from a vendor, it is necessary to take into account the distinction between internal and external transmittances.

**Mirrors** are characterized by their surface reflectivity and polish, as well as by the properties of the blanks on which these polishes are established. Plots of reflectance versus wavelength for commonly used metallic coating materials are shown below. Aluminum is widely used because it offers an average reflectance of 96% throughout the visible, near-infrared, and near-ultraviolet regions of the spectrum. Alternatively, silver exhibits higher reflectance (98%) through most of the visible and IR spectrum. However, it oxidizes faster, reducing its reflectance and causing light to scatter. Bare gold, on the other hand, combines good tarnish resistance with consistently high reflectance (99%) through the near, middle, and far-infrared regions. All of the metals exhibit higher reflectance at long wavelengths.



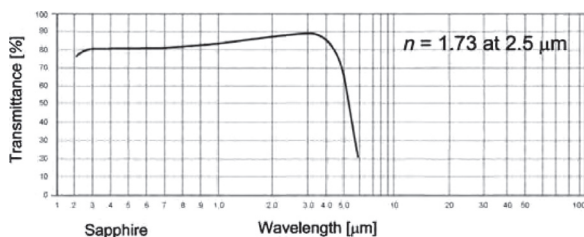
Data from Wolfe & Zissis, *The Infrared Handbook* (1989).



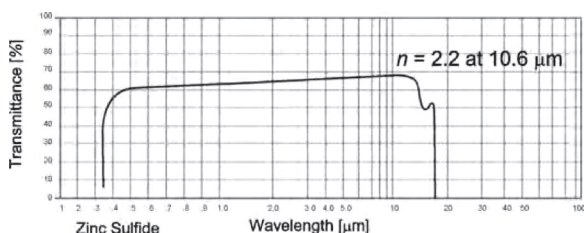
## Infrared Materials (cont.)

Metallic reflective coatings are delicate and require care during cleaning. Therefore, metallic coatings that are overcoated with a single, hard dielectric layer of half-wave optical thickness improve abrasion and tarnish resistance. Depending on the dielectric used, such coatings are referred to as **durable**, **protected**, or **hard coated**. The reflectance of metallic coatings can also be increased over the desired spectral range or for different angles of incidence by overcoating it with a quarter-wave stack of multilayer dielectric film, said to be **enhanced**.

The most versatile materials commonly used for systems operating in the MWIR and LWIR spectral regions are sapphire ( $\text{Al}_2\text{O}_3$ ), zinc sulfide (ZnS), zinc selenide (ZnSe), silicon (Si), and germanium (Ge).



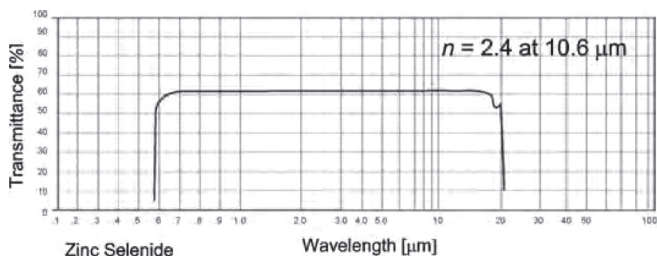
Sapphire is an extremely hard material, which is useful for visible, NIR, and IR applications through  $5 \mu\text{m}$ . It is practical for high-temperature and pressure applications.



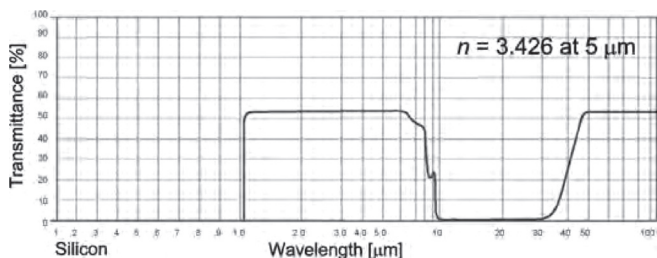
ZnS comes in two grades. The regular grade transmits in the  $1\text{--}12 \mu\text{m}$  spectral band with reasonable hardness and good strength. The other grade is water-clear called CLEARTRAN and transmits in the  $0.4\text{--}12 \mu\text{m}$  spectral band.

## Infrared Materials (cont.)

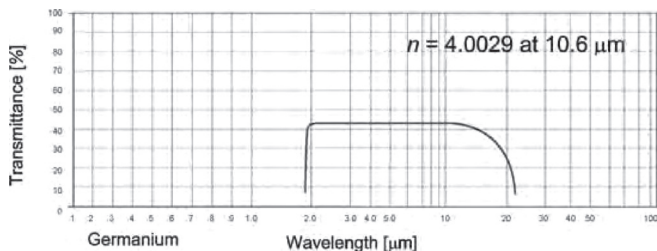
ZnSe has low absorbance and is an excellent material used in many laser and imaging systems. It transmits well from 0.6–18  $\mu\text{m}$ .



Si is a good substrate for high-power lasers due to its high thermal conductivity. It is useful in the 3–5  $\mu\text{m}$  and 48–100  $\mu\text{m}$  (astronomical applications) spectral bands.



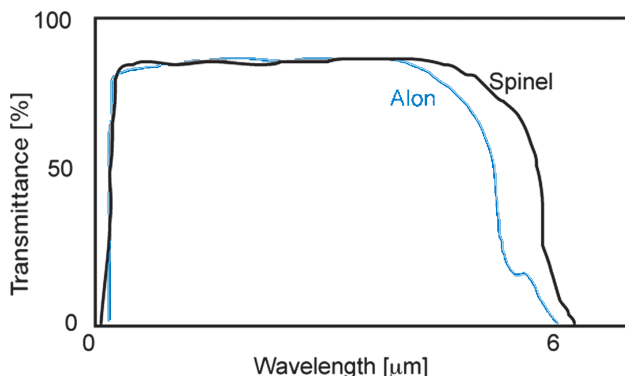
Ge has good thermal conductivity, excellent surface hardness, and good strength. It is used for IR instruments operating in the 2–14  $\mu\text{m}$  spectral band.



Other popular infrared materials are calcium fluoride ( $\text{CaF}_2$ ), magnesium fluoride ( $\text{MgF}_2$ ), barium fluoride ( $\text{BaF}_2$ ), gallium arsenide (GaAs), thallium bromoiodide (KRS-5), and cesium iodide (CsI).

## Ceramic and Amorphous Materials

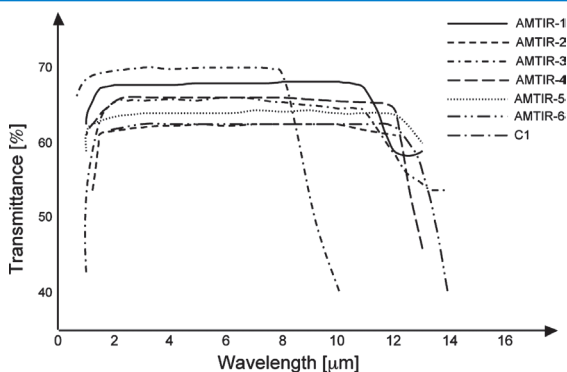
Alon ( $\text{Al}_{23}\text{O}_{27}\text{N}_5$ ) and spinel ( $\text{MgAl}_2\text{O}_4$ ) are transparent polycrystalline ceramic materials with optical and mechanical properties similar to sapphire. Their advantage is that these isotropic cubic crystal structures are non-birefringent and have a high degree of temperature stability. Their transmission window spans the near ultraviolet to  $5\text{ }\mu\text{m}$ . The refractive index of these materials is  $n = 1.7$  @  $4\text{ }\mu\text{m}$ .



Amorphous materials transmitting infrared radiation (AMTIR) began as a development for molding technology required to produce infrared lenses from chalcogenide glasses. The main optical properties of the different commercial AMTIR materials are listed below.

| Name    | Composition | $n$ @ $10\text{ }\mu\text{m}$ | Spectral Band [ $\mu\text{m}$ ] |
|---------|-------------|-------------------------------|---------------------------------|
| AMTIR 1 | GeAsSe      | 2.4981                        | 0.7 12                          |
| AMTIR 2 | AsSe        | 2.7613                        | 1.0 14                          |
| AMTIR 3 | GeSbSe      | 2.6027                        | 1.0 12                          |
| AMTIR 4 | AsSe        | 2.6431                        | 1.0 12                          |
| AMTIR 5 | AsSe        | 2.7398                        | 1.0 12                          |
| AMTIR 6 | AsS         | 2.3807                        | 0.6 8.0                         |
| C 1     | AsSeTe      | 2.8051                        | 1.2 14                          |

## GASIR Chalcogenide Glass Materials



GASIR materials are chalcogenide glasses that can be molded into finished optics with complex geometries, including asphero-diffractive surfaces. They can be produced in high volume at relatively low cost. Standard GASIR chalcogenide glass compositions are germanium arsenic selenium ( $\text{Ge}_{22}\text{As}_{20}\text{Se}_{58}$ , or GASIR-1), germanium antimony selenium ( $\text{Ge}_{20}\text{Sb}_{15}\text{Se}_{65}$ , or GASIR-2), and GASIR-3 (optimized for low cost with a reduced germanium content).

When heated up, GASIR glasses behave like plastics, (i.e., their viscosity progressively decreases). At certain temperatures, the viscosity is ideal for these glasses to precisely duplicate the surface of a mold. Molding is much less costly than diamond turning. The main disadvantage of GASIR glasses is their relatively high chromatic dispersion.

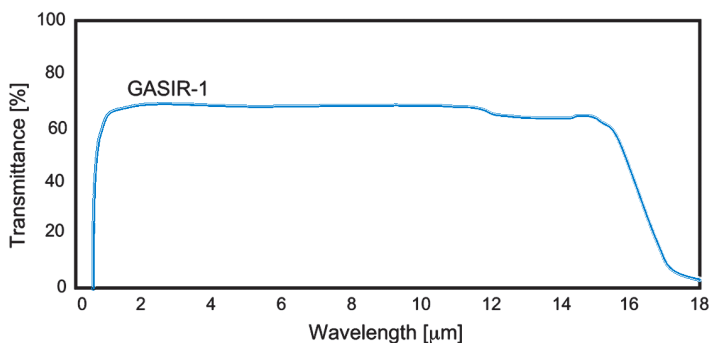


Figure adapted from Umicore Electro-Optic Materials [www.optics.umicore.com](http://www.optics.umicore.com).

## Material Dispersion

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**Dispersion** is the variation in the index of refraction with wavelength ( $dn/d\lambda$ ) and is an important material property when considering large-spectral-bandwidth optical systems. The dispersion is greater at short wavelengths and decreases at longer wavelengths; however, for most of the materials, it increases again when approaching the long infrared wavelength absorption band.

For historical reasons, dispersion is often quoted as a unitless ratio called the **reciprocal relative dispersion** or **Abbe number** defined by:

$$V = \frac{n_{\text{mean}} - 1}{\Delta n} = \frac{n_{\text{mean}} - 1}{n_{\text{final}} - n_{\text{initial}}},$$

where  $n_{\text{initial}}$  and  $n_{\text{final}}$  are the initial and final index of refraction values of the spectral band of interest, and  $n_{\text{mean}}$  is the mean center value.  $\Delta n$  is basically the measured value of the dispersion, and  $n_{\text{mean}} - 1$  specifies the refractive power of the material. The smaller the Abbe number, the larger the dispersion. For example, the  $V$  number for a germanium lens in the 8- to 12- $\mu\text{m}$  spectral band is

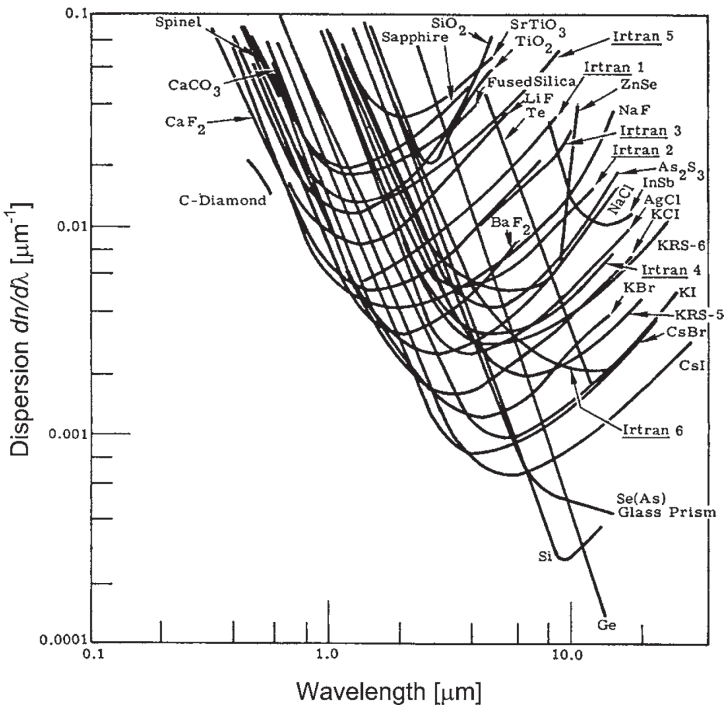
$$V = \frac{n(10\ \mu\text{m}) - 1}{n(12\ \mu\text{m}) - n(8\ \mu\text{m})} = \frac{4.0038 - 1}{4.0053 - 4.0023} = 1001.27.$$

Another useful definition is the **relative partial dispersion** given by

$$P = \frac{n_{\text{mean}} - n_{\text{initial}}}{n_{\text{final}} - n_{\text{initial}}},$$

which is a measure of the rate of change of the slope of the index as a function of wavelength (i.e.,  $d^2n/d\lambda^2$ ).

Material Dispersion (cont.)



The above chart shows the dispersion of the most common IR materials.

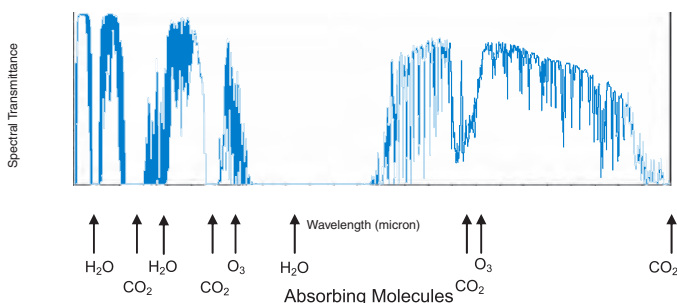
Figure adapted from Wolfe & Zissis, *The Infrared Handbook* (1989).

## Atmospheric Transmittance

The **absorption** and **emission** of radiation in the atmosphere are critical parameters that must be considered when developing infrared systems. Because of the sun's radiation, aerosols and molecular scattering are particularly important background sources in the visible spectrum. However, in the IR this effect is minimal since the wavelengths are much longer. According to the Rayleigh principle, the **scattered flux density** is inversely proportional to the fourth power of the driving wavelength.

The atmosphere is comprised primarily of  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , and  $\text{O}_3$ . High absorption occurs in different parts of the infrared spectrum due to molecular vibrations of these elements. For example, the NIR is greatly affected by water vapor as well as the short- and long-wave sides of the LWIR large window. The MWIR has two dips due to carbon dioxide and ozone.

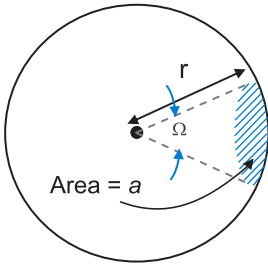
There are three main atmospheric windows in the infrared: NIR, 3–5  $\mu\text{m}$ , and 8–14  $\mu\text{m}$ . System technologies have evolved independently, optimizing the operation in each of these spectral bands.



The atmosphere is problematic for high-energy laser systems. Small temperature variations cause random changes in wind velocity, or turbulence motion. These changes in temperature give rise to small changes in the index of refraction of air, acting like little lenses that cause intensity variations. These fluctuations distort the laser beam wavefront, producing unwanted effects such as beam wander, beam spreading, and scintillation.

## Solid Angle

The **solid angle**  $\Omega$  in 3D space measures a range of pointing directions from a point to a surface. It is defined, assuming paraxial approximation, as the element of the area of a sphere divided by the square of the radius of the sphere. It is dimensionless and measured in square radians or **steradians** [ster].

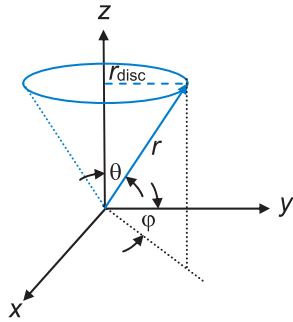


$$\Omega = \frac{a}{r^2} \quad a \ll r^2$$

For example, the area of a full sphere is given by  $4\pi r^2$ ; therefore, its solid angle is  $4\pi$ . A hemisphere contains half as many radians as a sphere (i.e.,  $\Omega = 2\pi$ ).

When large angles are involved, a more exact definition is required. Using spherical coordinates the solid-angular subtense can be expressed as a function of a flat disc or planar angle  $\theta$  as

$$d\Omega = \frac{da}{r^2} \sin\theta d\theta d\phi.$$

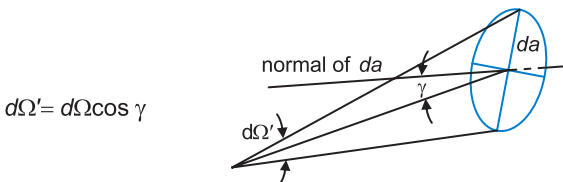


Integrating over the acceptance cone

$$\Omega = \int_0^{2\pi} d\phi \int_0^{\theta_{\max}} \sin\theta d\theta = 2\pi(1 - \cos\theta_{\max}) = 4\pi \sin^2 \frac{\theta_{\max}}{2}$$

is obtained.

If the disc is tilted at a selected angle  $\gamma$ , its differential solid angular subtense is decreased by a factor of  $\cos\gamma$ .



$$d\Omega' = d\Omega \cos\gamma$$



## Radiometry

**Radiometry** is a quantitative understanding of flux transfer through an optical system. Given the radiation from a thermal source transmitted through the optics of an infrared system, the fundamental question is how much of the source power is collected by the infrared sensor. Radiometric calculations predict the system's **signal-to-noise ratio (SNR)**.

Understanding the radiometric terms and their units are the key to performing radiometric calculations.

| Symbol   | Radiometric Term      | Units                            |
|----------|-----------------------|----------------------------------|
| $Q_e$    | Radiant energy        | Joule                            |
| $\phi_e$ | Radiant power or flux | Watt                             |
| $I_e$    | Radiant intensity     | Watt/ster                        |
| $M_e$    | Radiant exitance      | Watt/cm <sup>2</sup>             |
| $E_e$    | Irradiance            | Watt/cm <sup>2</sup>             |
| $L_e$    | Radiance              | Watt/cm <sup>2</sup> ·ster       |
| $Q_p$    | Photon energy         | Photon or quantum                |
| $\phi_p$ | Photon flux           | Photon/sec                       |
| $I_p$    | Photon intensity      | Photon/sec·ster                  |
| $M_p$    | Photon exitance       | Photon/cm <sup>2</sup> ·sec      |
| $E_p$    | Photon irradiance     | Photon/cm <sup>2</sup> ·sec      |
| $L_p$    | Photon radiance       | Photon/sec·cm <sup>2</sup> ·ster |

Subscript e = energy-derived units; subscript p = photon rate quantities.

Conversion between the two sets of units is done with the formula that determines the amount of energy carried per photon:

$$\mathcal{E} = \frac{hc}{\lambda}; \quad \Rightarrow \quad \phi_e[\text{watt}] = \phi_p[\text{photon/sec}] \cdot \mathcal{E},$$

where  $h$  is Planck's constant,  $c$  is the speed of light, and  $\lambda$  is the wavelength. Photon-derived units are useful when considering an infrared sensor (e.g., photovoltaic) that responds directly to photon events, rather than to thermal energy (e.g., microbolometer).

The energy carried per photon is inversely proportional to the wavelength; therefore, a short-wavelength photon carries more energy than a long-wavelength photon. The energy carried per photon can also be explained as the number of photons per second it takes to produce 1 W.

## Radiometric Terms

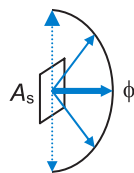
**Flux:** a quantity propagated or spatially distributed according to the laws of geometrical optics.

Both **irradiance** and **exitance** have units of spatial power density; however, the terms have different interpretations. The exitance is the power per unit area leaving a surface, thus describing a self-luminous source. It is defined as the ratio of the differential flux to the source area from which it is radiating as the area is reduced to a point:

$$M = \frac{\partial \phi}{\partial A_s},$$

where the total flux radiated into a hemisphere is given by:

$$\phi = \int M \cdot \partial A_s$$

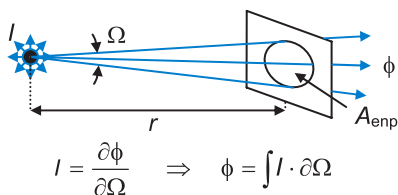


$\phi = MA_s$  for small but final quantities.

Equivalently, irradiance is a measure of the total incident flux per unit area on a passive receiver surface (e.g., sensor). It is defined as the ratio of power to the area upon which it is incident as the area is reduced to a specific position.

$$E = \frac{\partial \phi}{\partial A_d} \Rightarrow \phi = \int E \cdot \partial A_d.$$

The **intensity** is the radiant power per unit solid angle as the solid angle is reduced in value to a specific direction.



$$I = \frac{\partial \phi}{\partial \Omega} \Rightarrow \phi = \int I \cdot \partial \Omega$$

Intensity is used to characterize the amount of flux radiated from a **point source** that is collected by the entrance pupil.

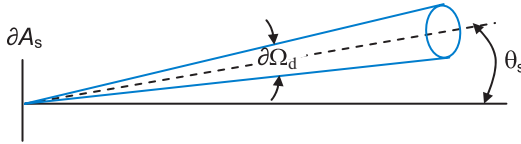
The intensity varies as a function of the view angle and can be written as  $I = I_0 \cos \theta$ , where  $I_0$  is the intensity in the direction normal to the surface.

Both the flux and the irradiance decrease as “one-over-squared fall off”:

$$\phi = I \cdot \Omega = I \frac{A_{\text{enp}}}{r^2}; \quad E = \frac{\phi}{A_{\text{enp}}} = \frac{I}{r^2}.$$

### Radiometric Terms (cont.)

**Radiance** is the most general term to describe source flux, because it includes both positional and directional characterization. It is used to characterize extended sources; that is, sources that have appreciable area compared to the square of the viewing distance. The visual equivalent of the radiance is the term “brightness.”



The radiance is defined for a particular ray direction as the radiant power per unit projected source area (perpendicular to the ray) per unit solid angle:

$$L \equiv \frac{\partial^2 \phi}{\partial A_s \cos \theta_s \partial \Omega_d}; \Rightarrow \quad \partial^2 \phi = L \partial A_s \cos \theta_s \partial \Omega_d,$$

which is the fundamental equation of **radiation transfer**. The term  $\partial^2 \phi$  is the power radiated into the cone and is incremental with respect to both the area of the source and the solid angle of the receiver.

For small but final source area and detector solid angle quantities,

$$\phi \cong L A_s \cos \theta_s \Omega_d.$$

A **Lambertian radiator** emits radiance that is independent of angle (i.e., the radiance is isotropic and is equally uniform in each direction within the hemisphere). The transfer equation can be applied to a Lambertian emitter to obtain the relationship between radiance and exitance:

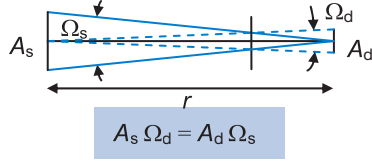
$$M = \frac{\partial \phi}{\partial A_s} = \int_{\Omega_d} L \cos \theta_s \partial \Omega_d = \int_0^{2\pi} d\varphi \int_0^{\pi/2} L \cos \theta_s \sin \theta_s d\theta_s = \pi L.$$

Similarly, the intensity can be obtained by integrating the fundamental equation with respect to the source area:

$$I = \frac{\partial \phi}{\partial \Omega_d} = \int_{A_s} L \cos \theta_s dA_s = L A_s \cos \theta_s.$$

## Flux Transfer

The fundamental equation of radiation transfer states that an element of power is given by the radiance times the product of two projected areas, divided by the square of the distance between them. Assuming normal angles of incidence (i.e.,  $\theta_s = \theta_d = 0$ ),



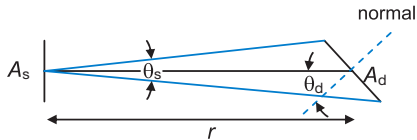
$$\phi = L A_s \Omega_d = L \frac{A_s A_d}{r^2} = L \Omega_s A_d.$$

Two equivalent expressions in terms of area solid angle product are obtained by grouping either the source or the detector area with  $r^2$ .

This relationship, defined as the so-called **AΩ product**, is completely symmetrical, and can be used to calculate the power in either direction of the net power flow. It is also known as the **optical invariant**, **throughput**, or **étendue**.

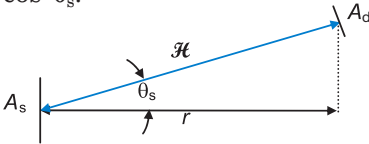
In the case where the detector is tilted ( $\theta_d \neq 0$ ), the flux decreases by the cosine of the angle:

$$\phi = L A_s \frac{A_d \cos \theta_d}{r^2}.$$



Consider the case for

$\theta_s \neq 0$  but  $\theta_d = 0$ , where the flux decreases in proportion to  $\cos^3 \theta_s$ :

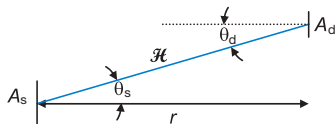


$$\phi = L A_s \cos^3 \theta_s \frac{A_d}{r^2}$$

The last case is when  $\theta_d = \theta_s \equiv \theta \neq 0$ , which leads to the most realistic situation, the so-called

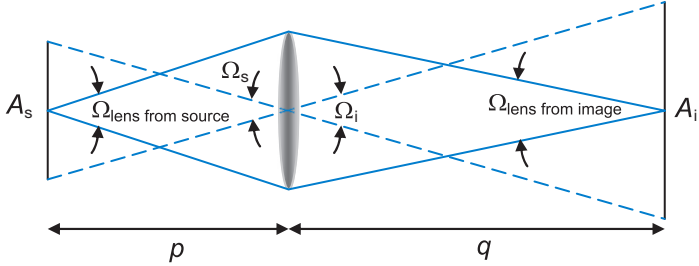
**cosine fourth law**:

$$\phi = L A_s \cos^4 \theta \frac{A_d}{r^2}.$$



## Flux Transfer for Image-Forming Systems

For purposes of simplification, the cosine projections are dropped (i.e., paraxial approximation), in which case the flux transfer is simply described in terms of the  $A\Omega$  product.



Recalling the area or longitudinal magnification equation, the total flux collected by the optical system may be calculated by one of the following flux-transfer equations:

$$\phi = L_s A_s \Omega_{\text{lens from source}} = L_s A_s \frac{A_{\text{lens}}}{p^2},$$

$$\phi = L_s A_i \Omega_{\text{lens from image}} = L_s A_i \frac{A_{\text{lens}}}{q^2},$$

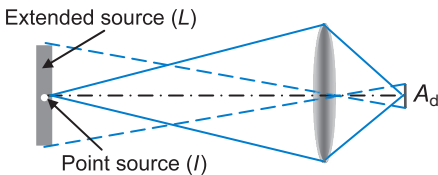
$$\phi = L_s A_{\text{lens}} \Omega_s = L_s A_{\text{lens}} \frac{A_s}{p^2},$$

$$\phi = L_s A_{\text{lens}} \Omega_i = L_s A_{\text{lens}} \frac{A_i}{q^2},$$

In paraxial approximation, the lens aperture acts as the intermediate receiver. In more complex optical systems, the entrance pupil is the intermediate receiver.

## Source Configurations

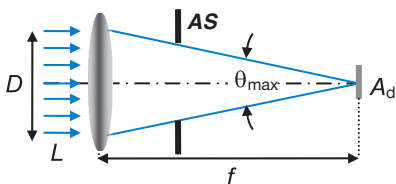
The source may be idealized as either a **point source** or a uniform **extended-area source**. A point source cannot be resolved by the optical system. It is smaller than the projection of the resolution spot at the object plane. An extended source, on the other hand, has an appreciable area compared to the detector's footprint.



It is appropriate to use **radiance** in characterizing radiant flux from an ex-

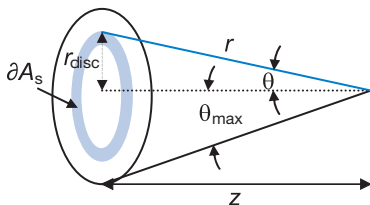
tended-area source, while it is appropriate to use **intensity** to characterize radiation from a point source.

The source power collected by the lens is imaged onto the detector focal plane. The **image irradiance** for a distant extended-area source can be calculated directly using a large **Lambertian disc**, which can represent the actual extended source or an intermediate source such as a lens.



An extended source fills the FOV such that the solid angle is bounded by the marginal rays and limited by the aperture stop.

Using a Lambertian disc, the following relationships are derived:



$$r_{\text{disc}} = z \tan \theta \Rightarrow dr_{\text{disc}} = d(z \tan \theta) = z \sec^2 \theta d\theta = \frac{z}{\cos^2 \theta} d\theta,$$

$$A_s = \pi r_{\text{disc}}^2 \Rightarrow dA_s = 2\pi r_{\text{disc}} dr_{\text{disc}} = 2\pi z^2 \tan \theta \frac{d\theta}{\cos^2 \theta}.$$

### Source Configurations (cont.)

The transfer flux is obtained by integrating the fundamental equation of radiation transfer over the source and detector areas:

$$\phi_d = L \iint_{A_s A_d} \partial A_s \cos \theta_s \partial \Omega_d = L \iint_{A_s A_d} \partial A_s \cos \theta_s \frac{\partial A_d \cos \theta_d}{r^2},$$

$$\phi_d = L \iint_{\theta A_d} 2\pi z^2 \tan \theta \frac{d\theta}{\cos^2 \theta} \cos^2 \theta \left( \frac{\cos^2 \theta}{z^2} \right) \partial A_d,$$

$$\phi_d = 2\pi L A_d \int_0^{\theta_{\max}} \sin \theta \cos \theta d\theta = \pi L A_d \sin^2 \theta_{\max}.$$

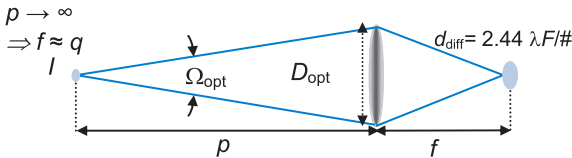
The irradiance on a detector from an extended-area source is then obtained by dividing the flux transferred by the area of the detector:

$$E_{\text{extended source}} = \frac{\phi}{A_d} = \pi L \sin^2 \theta_{\max} = \frac{\pi L}{4(F/\#)^2 + 1}.$$

For an extended-area source the image irradiance depends only on the source radiance and the  $F/\#$  of the optical system.

In the case of a point source, the collected power is defined by the intensity times the solid angle of the optics:

$$\phi = I \cdot \Omega_{\text{opt}} = I \cdot \frac{A_{\text{opt}}}{p^2}.$$



If the optics is diffraction limited, 84% of the flux transfer is concentrated into the image spot; therefore, the average irradiance of a point source at the detector plane is given by

$$E_{\text{point source}} = \frac{\phi}{A_d} \times 0.84 = \frac{0.84 I \Omega_{\text{opt}}}{\frac{\pi}{4} d_{\text{diff}}^2} = \frac{0.84 I \Omega_{\text{opt}}}{\frac{\pi}{4} [2.44 \lambda (F/\#)]^2}.$$

## Blackbody Radiators

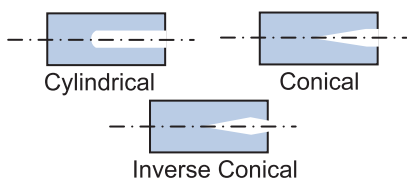
Empirically, it is found that solid bodies heated to incandescent temperatures emit radiation primarily in the infrared portion of the spectrum, and these incandescent sources emit their radiation in a continuous wavelength rather than at discrete spectral lines.

To describe radiation that emits a finite total power distributed continuously over wavelength, **spectral radiometric quantities** with units per micron of wavelength interval are used. The weighted spectral quantities are denoted with a subscript  $\lambda$  (e.g.,  $M_{e,\lambda}$  is the spectral exitance in  $\text{W}/\text{cm}^2 \mu\text{m}$ ). In-band nonspectral quantities are obtained by integrating the spectral terms over any spectral interval, for example,

$$L = \int_{\lambda_1}^{\lambda_2} L_{\lambda} d\lambda, \quad M = \int_{\lambda_1}^{\lambda_2} M_{\lambda} d\lambda, \quad \phi = \int_{\lambda_1}^{\lambda_2} \phi_{\lambda} d\lambda.$$

The term **blackbody (BB)** is used to describe a perfect radiator (i.e., an idealized thermal source). A blackbody absorbs all the radiant energy; and as a consequence, it is the perfect emitter. BBs have the maximum spectral exitance possible for a body at a specified temperature, either over a particular spectral region, or integrated over all wavelengths. They are a convenient baseline for radiometric calculations, since any thermal source at a specified temperature is constrained to emit less radiation than a blackbody source at the same temperature.

BB radiation is also called **cavity radiation**. Virtually any heated cavity with a small aperture produces high-quality radiation. These blackbody simulators are used primarily as laboratory calibration standards. The most popular



blackbody cavities are cylinders and cones, the latter being the most common. The aperture of the cavity defines the area of the source.

Some commercial blackbodies have an aperture wheel that allows a choice of source area.



## Planck's Radiation Law

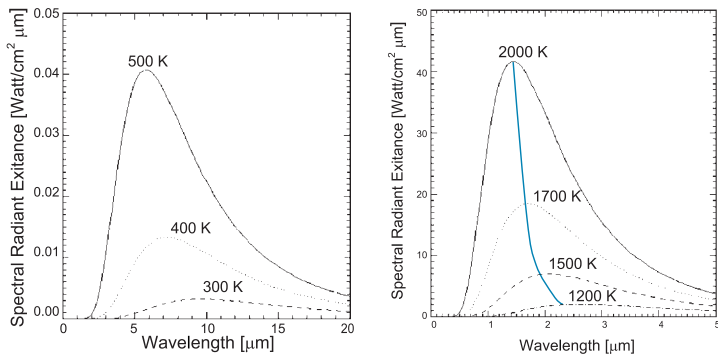
The radiation characteristics of ideal blackbody surfaces are completely specified if the temperature is known. Blackbody radiation is specified by Planck's equation and defines the spectral exitance as a function of absolute temperature and wavelength:

$$M_{e,\lambda} = \frac{2\pi hc^2}{\lambda^5} \frac{1}{\exp(hc/\lambda kT) - 1}$$

$$= \frac{c_1}{\lambda^5} \frac{1}{\exp(c_2/\lambda T) - 1} [\text{W/cm}^2 \cdot \mu\text{m}],$$

where  $h$  is Planck's constant:  $6.626 \times 10^{-34}$  Joule·sec;  $k$  is Boltzmann's constant:  $1.3806 \times 10^{-23}$  Joule/K;  $T$  is the absolute temperature in degrees Kelvin [K];  $\lambda$  is the wavelength in centimeters [cm];  $c$  is the speed of light in vacuum:  $2.998 \times 10^{10}$  cm/sec;  $c_1$  is the first radiation constant:  $2\pi hc^2 = 3.7415 \times 10^4$  W/cm<sup>2</sup> μm<sup>4</sup>; and  $c_2$  is the second radiation constant:  $hc/k = 1.4382$  cm · K.

Planck's equation generates spectral exitance curves that are quite useful for engineering calculations.



Planck's curves illustrate the following characteristics:

- The shape of the blackbody curves does not change for any given temperature.

### Planck's Radiation Law (cont.)

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- The temperature is inversely proportional to wavelength; the peak exitance shifts toward shorter wavelengths as the temperature increases. Therefore, at low temperatures, most of the exitance is at higher wavelengths, while at high temperatures, most of the exitance is at lower wavelengths.
- The slope of the Planck curve is sharper to the left of the peak exitance than to the right where the curve becomes more asymptotic. Consequently, the sensitivity (i.e.,  $\partial M_\lambda / \partial T$ ) of an infrared system increases at lower wavelengths.
- The individual curves never cross one another; the exitance increases rapidly for increased temperatures at all wavelengths.
- Other factors that must be taken into consideration are atmospheric conditions, aerosols in the atmosphere, and scattering.

The Planck radiation formula models system design and analysis problems. For example, the radiant emittance of a blackbody at temperatures from 400 to 900 K includes the temperature of the hot metal tailpipes of jet aircrafts.

## Stefan-Boltzmann and Wien's Displacement Laws

The **Stefan-Boltzmann law** relates the total blackbody exitance at all wavelengths to source temperature and is obtained by integrating out the wavelength dependence of Planck's radiation law:

$$M_e(T) = \int_0^\infty M_{e,\lambda}(\lambda, T) d\lambda = \int_0^\infty \frac{2\pi hc^2}{\lambda^5} \frac{d\lambda}{\exp(hc/\lambda kT) - 1} = \frac{2\pi^5 k^4 T^4}{15c^2 h^3}$$

$$M_e(T) = \sigma_e T^4,$$

where  $\sigma_e$  is the **Stefan-Boltzmann constant** and has a value of  $5.7 \times 10^{-12} \text{ W/cm}^2 \text{ K}^4$ . The Stefan-Boltzmann law only holds for exitance integrated from zero to infinity at the interval of wavelength.

The total exitance at all wavelengths multiplied by the source area results in the total power radiated, which increases as the fourth power of the absolute source temperature in Kelvin. For example, at room temperature (300 K), a perfect blackbody of an area equal to  $1 \text{ cm}^2$  emits a total power of  $4.6 \times 10^{-2} \text{ W}$ . Doubling the temperature to 600 K, the total power increases 16 fold to 0.74 W.

The derivative of Planck's equation with respect to wavelength yields **Wien's displacement law**, which gives the wavelength for which the peak of the spectral-exitance function occurs as a function of temperature:

$$\frac{\partial M_{e,\lambda}(\lambda, T)}{\partial \lambda} = 0 \Rightarrow \lambda_{\max} T = 2898 [\mu\text{m} \cdot \text{K}].$$

Thus, the wavelength at which the maximum spectral radiant exitance varies is inversely proportional to the absolute temperature. The plot of  $\lambda_{\max}$  as a function of temperature is a hyperbola (see plots on page 33).

For example, a blackbody source at 300 K has its maximum exitance at  $9.7 \mu\text{m}$ ; however, if the temperature of this source is changed to 1000 K, the maximum value at which the peak exitance occurs is at  $2.9 \mu\text{m}$ . The sun is a blackbody source at approximately 6000 K. Applying Wien's law it is found that its maximum wavelength occurs at  $0.5 \mu\text{m}$ , which corresponds to the peak response of the human eye.

## Rayleigh-Jeans and Wien's Radiation Laws

Two well-known approximations to Planck's radiation law are the Rayleigh-Jeans and Wien's radiation laws. The former applies at long wavelengths:

$$hc / \lambda kT \ll 1 \Rightarrow M_{e,\lambda} \cong \frac{2\pi c kT}{\lambda^4},$$

while the latter is valid only at short wavelengths:

$$hc / \lambda kT \gg 1 \Rightarrow M_{e,\lambda} \cong \frac{2\pi hc^2}{\lambda^5} \exp\left(-\frac{hc}{\lambda kT}\right).$$

As the temperature increases, the peak wavelength decreases (Wien's law), and the area under the Planck curve increases much faster (Stefan-Boltzmann law).

### Thermal Equations in Photon-Derived Units

In photon-derived units, Planck's radiation, Boltzmann and Wien's displacement, Wien's radiation, and Rayleigh-Jeans formulae are given by:

#### Planck's radiation equation:

$$M_{p,\lambda} = \frac{2\pi c}{\lambda^4} \frac{1}{\exp(hc / \lambda kT) - 1} [\text{photon/sec} \cdot \text{cm}^2 \cdot \mu\text{m}]$$

#### Stefan-Boltzmann law:

$$M_p(T) = \sigma_p T^3$$

where  $\sigma_p$  has a value of  $1.52 \times 10^{-11}$  photons/sec·cm<sup>2</sup>·K<sup>3</sup>

#### Wien's displacement law:

$$\lambda_{\max} T = 3662 \quad [\mu\text{m} \cdot \text{K}]$$

#### Rayleigh-Jeans radiation law:

$$hc / \lambda kT \ll 1 \Rightarrow M_{p,\lambda} \cong \frac{2\pi kT}{\lambda^3 h}$$

#### Wien's radiation law:

$$hc / \lambda kT \gg 1 \Rightarrow M_{p,\lambda} \cong \frac{2\pi c}{\lambda^4} \exp\left(-\frac{hc}{\lambda kT}\right)$$

## Exitance Contrast

In terrestrial infrared systems, the target and background are often of similar temperatures, in which case the target has very low contrast. The proper choice of spectral pass band wavelength  $\Delta\lambda$  becomes essential to maximize the visibility of the target. This pass band should straddle the wavelength for which the exitance changes the most as a function of temperature.

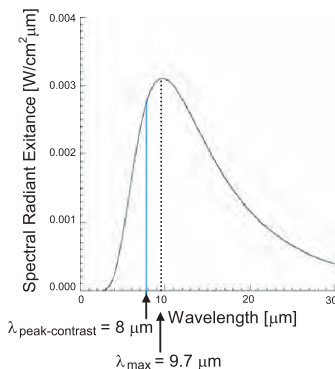
This consideration of exitance contrast involves the following second-order partial derivative, where the system operating within a finite pass band is most sensitive to small changes in temperature:

$$\text{Finds } \lambda_{\text{peak-contrast}} \quad \frac{\partial}{\partial \lambda} \left[ \frac{\partial M_{\lambda}(\lambda, T)}{\partial T} \right] = 0 \quad \text{Finds best sensitivity (steepest slope)}$$

Carrying out these derivatives yields a constraint on wavelength and temperature in a fashion similar to Wien's displacement law, yielding

$$\lambda_{\text{peak-contrast}} T = 2410 \text{ } [\mu\text{m} \cdot \text{K}].$$

For a given blackbody temperature, the maximum exitance contrast occurs at shorter wavelengths than the wavelength of the peak exitance. For example, at 300 K the peak exitance occurs at 9.7  $\mu\text{m}$  while the peak exitance contrast occurs at 8  $\mu\text{m}$ .



## Emissivity

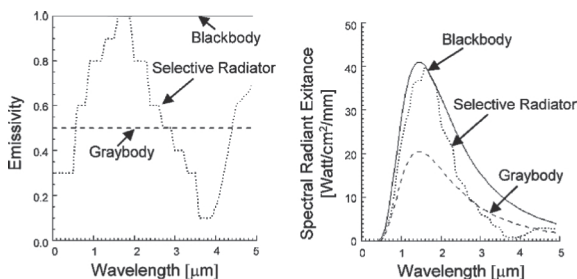
**Emissivity** is the ratio of the spectral exitance of a real source at a given temperature to that of a blackbody at the same temperature:

$$\varepsilon(\lambda, T) = \frac{M_{\lambda, \text{source}}(\lambda, T)}{M_{\lambda, \text{BB}}(\lambda, T)}.$$

The spectral exitance of any real source at a given temperature is bound by the spectral exitance of a perfect radiator at the same kinetic temperature; hence  $\varepsilon$  is constrained between zero and unity.

The emissivity characterizes how closely the radiation spectrum of a real heated body corresponds to that of a blackbody. It is a spectrally varying quantity and can be referred to either as a spectral (measured over a finite pass band) or a total (measured over all wavelengths) quantity. This emission efficiency parameter depends also on the surface temperature. However, emissivity data for most materials are typically a constant and are seldom given as a function of  $\lambda$  and  $T$  unless it is an especially well-characterized material.

There are three types of sources that can be differentiated depending on how their spectral emissivity varies: (1) a blackbody or perfect radiator, for which its emissivity is equal to unity for all wavelengths; (2) a graybody when its emissivity is a constant fraction of what the corresponding blackbody would radiate at the same temperature ( $\varepsilon < 1$ ). Its emissivity is independent of wavelength and has the same spectral shape as a blackbody; (3) a selective radiator where  $\varepsilon$  is an explicit function of  $\lambda$ .



## Kirchhoff's Law

If a solid body of a certain mass is located within a colder isothermal cavity, according to the second law of thermodynamics, there will be a net flow of heat from the object to the walls of the hollow space. Once the body reaches thermal equilibrium with its surroundings, the first law of thermodynamics or conservation of energy requires that

$$\phi_{\text{inc}} = \phi_{\text{absorbed}} + \phi_{\text{transmitted}} + \phi_{\text{reflected}},$$

where  $\phi_{\text{inc}}$  is the incident flux on the solid body. Dividing both sides of the equation by  $\phi_{\text{inc}}$  yields

$$1 = \alpha + \tau + \rho,$$

where  $\alpha$  is the absorbance,  $\tau$  the transmittance, and  $\rho$  is the reflectance. For an opaque body (i.e.,  $\tau = 0$ ) the incident radiation is either absorbed or reflected, yielding

$$\alpha = 1 - \rho,$$

which indicates that surfaces with low reflectance are high emitters.

If the body absorbs only a portion of the radiation that is incident on it, then it emits less radiation in order to remain in thermal equilibrium; that is

$$E = \epsilon M,$$

which leads to **Kirchhoff's law**, which states that the absorbance of a surface is identical to the emittance of that surface. Kirchhoff's law also holds for spectral quantities; it is a function of temperature and can vary with the direction of measurement. This law is sometimes verbalized as “**good absorbers are good emitters.**”

$$\text{Integrated absorbance} = \alpha(\lambda, T) \equiv \epsilon(\lambda, T) = \text{Integrated emittance}$$

For polished metals, the emissivity is low; however, it increases with temperature, and may increase substantially with the formation of an oxide layer on the object surface. A thin film of oil and surface roughness can increase the emissivity by an order of magnitude, compared to a polished metal surface. The emissivity of nonmetallic surfaces is typically greater than 0.8 at room temperature, and it decreases as the temperature increases.

## Emissivity of Various Common Materials

| Metals and Other Oxides                      | Emissivity |
|----------------------------------------------|------------|
| <b>Aluminum:</b> Polished sheet              | 0.05       |
| Sheet as received                            | 0.09       |
| Anodized sheet, chromatic-acid process       | 0.55       |
| Vacuum-deposited                             | 0.04       |
| <b>Brass:</b> highly polished                | 0.03       |
| Rubbed with 80-grit emery                    | 0.20       |
| Oxidized                                     | 0.61       |
| <b>Copper:</b> Highly polished               | 0.02       |
| Heavily oxidized                             | 0.78       |
| <b>Gold:</b> Highly polished                 | 0.21       |
| <b>Iron:</b> Cast-polished                   | 0.21       |
| Cast, oxidized                               | 0.64       |
| Sheet, heavily rusted                        | 0.69       |
| <b>Nickel:</b> Electroplated, polished       | 0.05       |
| Electroplated, not polished                  | 0.11       |
| Oxidized                                     | 0.37       |
| <b>Silver:</b> polished                      | 0.03       |
| <b>Stainless Steel:</b> type 18-8 buffed     | 0.16       |
| Type 18-8, oxidized                          | 0.85       |
| <b>Steel:</b> Polished                       | 0.07       |
| Oxidized                                     | 0.79       |
| <b>Tin:</b> Commercial tin-plated sheet iron | 0.07       |

| Nonmetallic Materials                             | Emissivity       |
|---------------------------------------------------|------------------|
| <b>Brick:</b> Red common                          | 0.93             |
| <b>Carbon:</b> Candle soot                        | 0.95             |
| Graphite, filed surface                           | 0.98             |
| <b>Concrete</b>                                   | 0.92             |
| <b>Glass:</b> Polished plate                      | 0.94             |
| <b>Lacquer:</b> White                             | 0.92             |
| Matte black                                       | 0.97             |
| <b>Oil, lubricant</b> (Thin film of nickel base): |                  |
| Nickel base alone                                 | 0.05             |
| Oil film: 1, 2, $5 \times 10^{-3}$ in             | 0.27, 0.46, 0.72 |
| Thick coating                                     | 0.82             |
| <b>Paint, oil:</b> Average of 16 colors           | 0.94             |
| <b>Paper:</b> White bond                          | 0.93             |
| <b>Plaster:</b> Rough coat                        | 0.91             |
| <b>Sand</b>                                       | 0.90             |
| <b>Human skin</b>                                 | 0.98             |
| <b>Soil:</b> Dry                                  | 0.92             |
| Saturated with water                              | 0.95             |
| <b>Water:</b> Distilled                           | 0.96             |
| Ice, smooth                                       | 0.96             |
| Frost crystals                                    | 0.98             |
| Snow                                              | 0.90             |
| <b>Wood:</b> Planed oak                           | 0.90             |

Data from Wolfe & Zissis, *The Infrared Handbook* (1989).



## Radiometric Measure of Temperature

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There are many applications in the infrared where the actual kinetic temperature of a distant object must be known. However, infrared systems can only measure the apparent spectral exitance emitted by targets and/or backgrounds, which are a function of both temperature and emissivity. The temperature of the infrared source can be measured if emissivity of the viewing source within the appropriate spectral region is known. Discrepancies in emissivity values produce built-in errors in the calculation of this kinetic temperature.

Three main types of temperature measurements are discussed below.

**Radiation temperature ( $T_{\text{rad}}$ ):** a calculation based on the Stefan-Boltzmann law because it is estimated over the whole spectrum:

$$M_{\text{meas}} = \sigma T_{\text{rad}}^4,$$

where  $M_{\text{meas}}$  is the measured exitance. If the source is a graybody with a known emissivity, then  $T_{\text{true}}$  can be calculated from  $T_{\text{rad}}$ :

$$M_{\text{meas}} = \sigma T_{\text{rad}}^4 = \epsilon \sigma T_{\text{true}}^4 \Rightarrow T_{\text{true}} = \frac{T_{\text{rad}}}{\sqrt[4]{\epsilon}}.$$

Due to its strong dependence on emissivity,  $T_{\text{rad}}$  cannot be corrected to find  $T_{\text{true}}$  if  $\epsilon$  is an unknown. Similarly,  $T_{\text{rad}}$  is affected by the attenuation in the optical system, especially in harsh environments where the optical elements might be dirty.

**Brightness temperature ( $T_b$ ):** a measurement performed using Planck's radiation law because it is estimated over a single wavelength  $\lambda_0$ , or in a narrow spectral region ( $\Delta\lambda$ ) around a fixed wavelength  $\lambda_0$ . For a blackbody source,  $T_b = T_{\text{true}}$ ; therefore, the Planck equation can be solved for  $T_b$ :

$$T_b = \frac{c_2}{\lambda_0 \ln \left\{ 1 + \frac{c_1}{\lambda_0^5 M_\lambda(\lambda_0, T_b)} \right\}},$$

where  $c_1 = 3.7415 \times 10^4 \text{ W/cm}^2 \mu\text{m}^4$  and  $c_2 = 1.4382 \text{ cm}\cdot\text{K}$ .

## Radiometric Measure of Temperature (cont.)

If the source is a graybody with a known emissivity, then  $T_{\text{true}}$  can be calculated from  $T_b$  as follows:

$$\frac{c_1}{\lambda_0^5} \frac{1}{\exp(c_2 / \lambda_0 T_b) - 1} = \frac{c_1}{\lambda_0^5} \frac{\varepsilon}{\exp(c_2 / \lambda_0 T_{\text{true}}) - 1},$$

yielding

$$T_{\text{true}} = \frac{c_2}{\lambda_0 \ln\{1 + \varepsilon[\exp(c_2 / \lambda_0 T_b) - 1]\}}.$$

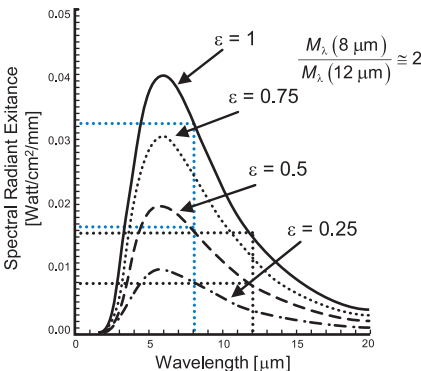
The exitance levels measured for  $T_b$  are lower than the levels for  $T_{\text{rad}}$  because of the narrowband filtering involved. The best sensitivity for this measurement is obtained by choosing  $\lambda_0$  near the wavelength of the peak exitance contrast, where  $M_\lambda$  changes most with temperature.

The brightness temperature measures the correct temperature as long as the emissivity is known and there is no spurious attenuation in the optical system. Lack of knowledge of either of these two parameters will produce a large error in the radiometric measurement.

**Color temperature ( $T_c$ ):** the temperature of a blackbody that best matches the spectral composition of the target source. This spectral composition is defined as the ratio of the measured spectral exitance at two different wavelengths given by

$$\frac{M_\lambda(\lambda_1)}{M_\lambda(\lambda_2)} = \frac{\lambda_2^5 \exp(c_2 / \lambda_2 T_c) - 1}{\lambda_1^5 \exp(c_2 / \lambda_1 T_c) - 1}.$$

Under these circumstances, the emissivity cancels out be-

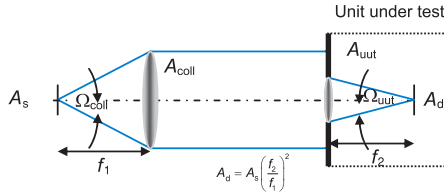


cause of the ratio, and  $T_c = T_{\text{true}}$  for both black and graybodies.

This method is strongly affected when the target source is a selective radiator. In this case, the measurement of the spectral exitance must be performed at many wavelengths.

## Collimators

A collimator is an optical assembly that places a target at infinity and produces a controllable irradiance that is independent of distance. It is widely used for testing the sensitivity and the resolution of infrared systems.



Assuming a Lambertian source that is radiating at all wavelengths, the total flux emitted can be written using the Stefan-Boltzmann law:

$$\phi = L_s A_s \Omega_{\text{coll}} = \frac{\sigma T^4}{\pi} \frac{A_s A_{\text{coll}}}{f_1^2}.$$

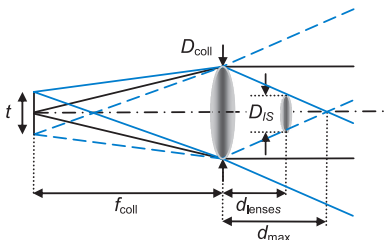
The source exitance and/or the irradiance falling on the detector surface can be obtained by dividing the radiant flux over the area of the collimator, yielding

$$M = \frac{\sigma T^4}{\pi} \frac{A_s}{f_1^2} = \frac{\sigma T^4}{\pi} \frac{A_d}{f_2^2} = E.$$

An extended source placed at the focal plane of a collimator can be seen only in a well-defined region. The maximum distance at which the infrared imaging system can be placed from the collimator is

$$d_{\text{max}} = \frac{f_{\text{coll}}}{t} D_{\text{coll}}.$$

If the imaging system is placed at a distance greater than  $d_{\text{max}}$ , the target's outer edges are clipped, and only the central portion of the target is seen.



The distance between the collimating and imaging system optical components is

$$d_{\text{lenses}} = \frac{f_{\text{coll}}}{t} (D_{\text{coll}} - D_{\text{IS}}).$$

## Infrared Detectors

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Infrared detectors are transducers that sample the incident radiation and produce an electrical signal proportional to the total flux incident on the detector surface. There are two main classes of infrared detectors: **thermal** and **photon detectors**. Both types respond to absorbed photons, but use different response mechanisms, which lead to variations in speed, spectral responsivity, and sensitivity. Thermal detectors depend on the changes in the electrical or mechanical properties of the sensing materials (e.g., resistance, capacitance, voltage, mechanical displacement) that result from temperature changes caused by the heating effect of the incident radiation. The change in these electrical properties with input flux level is measured by an external electrical circuit. The thermal effects do not depend on the photonic nature of the incident infrared radiation; they have no inherent long-wavelength cutoff. Their sensitivity limitation is due to thermal flux and/or the spectral properties of the protective window in front of them. The response rate of a thermal detector is slow because of the time required for the device to heat up after the incident flux has been absorbed. Examples of different classes of thermal detectors are **bolometers**, **pyroelectric detectors**, **thermopiles**, **Golay cells**, and **superconductors**.

The two basic types of semiconductor photon detectors are photoconductors and photovoltaics (or photodiodes). The photonic effects in these devices result from direct conversion of incident photons into conducting electrons within the material. An absorbed photon excites an electron from the nonconducting state into a conducting state instantaneously, causing a change in the electrical properties of the semiconductor material that can be measured by an external circuit. Photon detectors are very fast; however, their response speed is generally limited by the resistance-capacitance (RC) product of the readout circuit.

Detector performance is described in terms of **responsivity**, **noise-equivalent power**, or **detectivity**. These figures of merit enable a quantitative prediction and evaluation of the system and allow for comparison of relative performance between different detector types.

## Primary Sources of Detector Noise

Noise is a random fluctuation in electrical output from a detector, and must be minimized to increase the performance sensitivity of an infrared system. Sources of optical-detector noise can be classified as either external or internal. However, the focus here is on the internal generated detector noises that include shot, generation-recombination, one over frequency ( $1/f$ ), and temperature fluctuation, which are a function of the detector area, bandwidth, and temperature. It is possible to determine the limits of detector performance set by the statistical nature of the radiation to which it responds. Such limits set the lower level of sensitivity, and can be ascertained by the fluctuations in the signal or background radiation falling on the detector.

Random noise is expressed in terms of an electrical variable such as a voltage, current, or power. If the voltage is designated as a random-noise waveform  $v_n(t)$  and a certain probability-density function is assigned to it, its statistics are found as a function of the following statistical descriptors:

**Mean:**  $\overline{v_n} = \frac{1}{T} \int_0^T v_n(t) dt$  [volts],

**Variance or mean-square:**

$$\overline{v_n^2} = \overline{(v_n(t) - \overline{v_n})^2} = \frac{1}{T} \int_0^T [v_n(t) - \overline{v_n}]^2 dt \quad [\text{volts}^2],$$

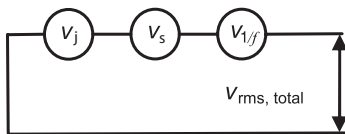
**Standard deviation:**

$$v_{\text{rms}} = \Delta v_n = \sqrt{\frac{1}{T} \int_0^T [v_n(t) - \overline{v_n}]^2 dt} \quad [\text{volts}],$$

where  $T$  is the time interval. The standard deviation represents the rms noise of the random variable.

Linear addition of independent intrinsic noise sources is carried out in power (variance) not in voltage noise (standard deviation); the total rms noise of random quantities are added in quadrature:

$$v_{\text{rms, total}} = \sqrt{v_{\text{rms, 1}}^2 + v_{\text{rms, 2}}^2 + \cdots + v_{\text{rms, n}}^2}$$



Assuming three sources of noise are present; Johnson, shot, and  $1/f$  noise:

$$v_{\text{rms, total}}^2 = v_j^2 + v_s^2 + v_{1/f}^2$$

## Noise Power Spectral Density

Noise can also be described in the frequency domain. The **power spectral density (PSD)**, or the mean-square fluctuation per unity frequency range, provides a measurement of frequency distribution of the mean-square value of the data (i.e., distribution of power).

For random processes, frequency can be introduced through the autocorrelation function. The time-average **autocorrelation** function of a voltage waveform may be defined as

$$c_n(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} v_n(t) v_n(t + \tau) dt,$$

where the autocorrelation is the measure of how fast the waveform changes in time. The PSD of a wide-sense stationary random process is defined as the Fourier transform of the autocorrelation function (Wiener-Khinchine theorem):

$$\text{PSD} = N(f) = \mathcal{F}\{c_n(\tau)\} = \int_{-\infty}^{\infty} c_n(\tau) e^{-j2\pi f\tau} d\tau.$$

The inverse relation is

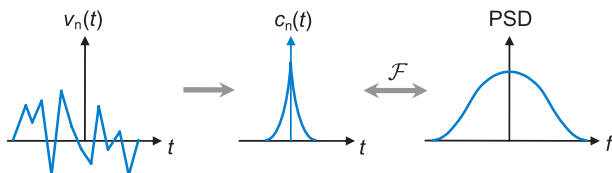
$$c_n(\tau) = \mathcal{F}^{-1}\{N(f)\} = \int_{-\infty}^{\infty} N(f) e^{j2\pi f\tau} df.$$

Using the central-ordinate theorem yields

$$c_n(0) = \int_{-\infty}^{\infty} N(f) df = \int_{-\infty}^{\infty} v_n^2(t) dt = \overline{v_n^2(t)}.$$

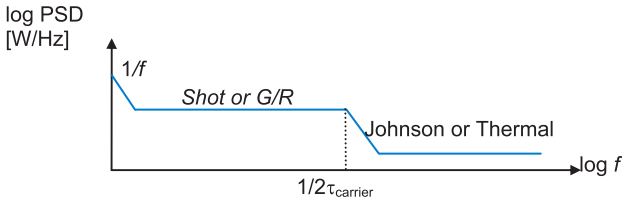
The average power of the random voltage waveform is obtained by integrating the PSD over its entire range of definition.

Uncorrelated noise such as white noise implies that its autocorrelation function is a delta function. The PSD of such random processes is a constant over the entire frequency range, but in practice, the PSD is constant over a wide but final range (i.e., it is bandpass limited).



## White Noise

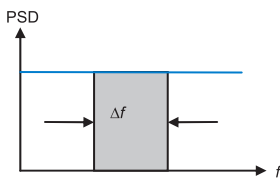
Detector noise curves have different frequency contents. A typical PSD plot for a sensor system is shown.



- $1/f$  noise can be partly excluded by ac coupling (i.e., cut off the system dc response by using a high pass filter with a cutoff between 1 Hz and 1 KHz).
- Shot noise and generation-recombination (G/R) noise have a roll-off frequency in the midband range ( $\approx 20 \text{ KHz} - 1 \text{ MHz}$ ), proportional to the inverse of carrier lifetime.
- Johnson noise and amplifier noise are usually flat to high frequencies past  $1/2\tau_{\text{carrier}}$ .
- In a photon sensor, the charge carriers transmit both signal and noise; therefore, the upper cutoff frequency of the electronics bandwidth should not be higher than  $1/2\tau_{\text{carrier}}$ ; i.e., wider bandwidth includes more noise, but not more signal.

Most system-noise calculations are done in the **white noise** region, where the PSD is flat over sufficient broadband relative to the signal band of interest. Shot and G/R noises are white up to  $f \approx 1/2\tau_{\text{carrier}}$ ; beyond that, they roll off. For white noise, the noise power is directly proportional to the detector bandwidth; consequently, the rms noise voltage is directly proportional to the square root of the detector bandwidth.

Detectors have temporal impulse response with a width of  $\tau$ .



Noise power:  $\text{PSD} \propto \Delta f = 1/2\tau$

rms noise voltage:  $v_{\text{rms}} \propto \sqrt{\Delta f} = \sqrt{1/2\tau}$

This response time or integration time is related to its frequency bandwidth as  $\Delta f = 1/2\tau$ .

If the input noise is white, then the noise-equivalent bandwidth of the filter determines how much noise power is passed through the system.

As the integration time shortens, the noise-equivalent bandwidth widens and the system becomes noisier, while using a measurement circuit with longer response time yields a system with less noise.

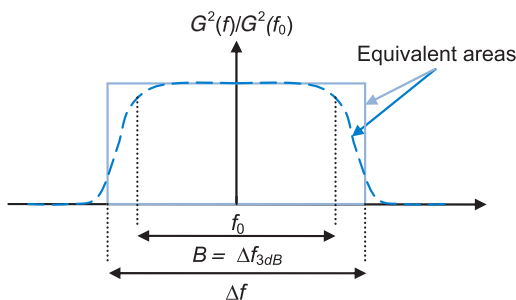
## Noise-Equivalent Bandwidth ( $NE\Delta f$ )

The **noise-equivalent bandwidth** ( $NE\Delta f$  or simply  $\Delta f$ ) of an ideal electronic amplifier has a constant power-gain distribution between its lower and upper frequencies, and zero elsewhere. It can be represented as a rectangular function in the frequency domain. However, real electronic frequency responses do not have such ideal rectangular characteristics; so it is necessary to find an equivalent bandwidth that would provide the same amount of noise power.

The noise-equivalent bandwidth is defined as

$$NE\Delta f \equiv \frac{1}{G^2(f_0)} \int_{-\infty}^{\infty} |G(f)|^2 df,$$

where  $G(f)$  is the power gain as a function of frequency, and  $G(f_0)$  is the maximum value of the power gain. The most



common definition of bandwidth is the frequency interval within which the power gain exceeds one half of its maximum value (i.e., 3-dB

bandwidth, which is usually denoted by the symbol  $B$ ).

The above definition of noise-equivalent bandwidth assumes white noise; that is, the power spectrum of noise is flat. However, if the noise power spectrum exhibits strong frequency dependence, the noise-equivalent bandwidth should be calculated from

$$NE\Delta f \equiv \frac{1}{G^2(f_0)\overline{v_n^2}} \int_{-\infty}^{\infty} \overline{v_n^2(f)} G^2(f) df,$$

where  $\overline{v_n^2(f)}$  is the mean-square noise voltage per unit bandwidth, and  $\overline{v_n^2}$  is the mean-square noise voltage per unit bandwidth measured at a frequency high enough that the PSD of the noise is flat.



**NE $\Delta f$  (cont.)**

For noise-equivalent bandwidth calculations, two impulse response forms are commonly considered: square and exponential. The square impulse response is most commonly used to relate response time and noise-equivalent bandwidth. The exponential impulse response arises from charge-carrier lifetime or by measuring the RC time constants in electrical circuits.

A square impulse response with a pulse width  $\tau$  can be expressed as a rectangular function as

$$v(t) = v_0 \text{ rect}[(t - t_0) / \tau].$$

Applying the Fourier transform, the normalized voltage transfer function is obtained:

$$\frac{V(f)}{V_0} = e^{-j2\pi f t_0 / \tau} \text{sinc}(\pi f \tau).$$

Substituting into the NE $\Delta f$  equation and solving the integral,

$$\text{NE}\Delta f = \int_{-\infty}^{\infty} |e^{-j2\pi f t_0 / \tau} \text{sinc}(\pi f \tau)|^2 df = \frac{1}{2\tau}.$$

The calculation of the noise-equivalent bandwidth from an exponential impulse response is obtained as follows:

The exponential impulse response can be specified as

$$v(t) = v_0 \exp(t/\tau) \text{ step}(t).$$

Fourier transforming gives:

$$\frac{V(f)}{V_0} = \frac{1}{1 + j2\pi f \tau},$$

and taking the absolute value squared yields:

$$\left| \frac{V(f)}{V_0} \right|^2 = \frac{1}{1 + (2\pi f \tau)^2}.$$

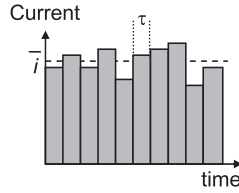
Integrating yields the following noise-equivalent bandwidth:

$$\text{NE}\Delta f = \int_{-\infty}^{\infty} \frac{df}{1 + (2\pi f \tau)^2} = \frac{1}{4\tau}.$$

## Shot Noise

The **shot noise** mechanism is associated with the nonequilibrium conditions in a potential energy barrier of a photovoltaic detector through which a dc current flows. Shot noise is a result of the discrete nature of the current carriers and therefore of the current-carrying process. The dc current is viewed as the sum of many short and small current pulses, each of which is contributed by the passage of a single electron or hole through the junction depletion layer. This type of noise is practically white, considering the spectral density of a single narrow pulse.

The generation of the carriers is random according to photon arrival times. However, once a carrier is generated, its recombination is no longer random; it is actually determined by transit-time considerations that obey Poisson statistics, which state that the variance equals the mean.



$\tau$  = interval of integration.

To determine the expression for the mean-square current fluctuation at the output of the measuring circuit, the current is measured during a time interval  $\tau$ :  
Faster circuit  $\rightarrow$  bigger  $\Delta f \rightarrow$  more noise

$$\bar{i} = \frac{n_e q}{\tau},$$

where  $n_e$  is the number of photogenerated electrons within  $\tau$ . The average current can be related to the average number of electrons created by

$$\bar{i} = \frac{\bar{n}_e q}{\tau} \Rightarrow \bar{n}_e = \frac{\bar{i} \tau}{q}.$$

The mean-square fluctuation averaged over many independent measuring times  $\tau$  is

$$\overline{i_{n,shot}^2} = \overline{(i - \bar{i})^2} = \frac{q^2}{\tau^2} \overline{(n_e - \bar{n}_e)^2} = \frac{q^2}{\tau^2} \bar{n}_e,$$

which yields the expression of the shot noise:

$$\overline{i_{n,shot}^2} = \frac{\bar{i} q}{\tau} = 2q\bar{i}\Delta f \Rightarrow i_{n,shot} = \sqrt{2q\bar{i}\Delta f}.$$

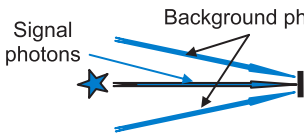
## Signal-to-Noise Ratio: Detector and BLIP Limits

Is the **signal-to-noise ratio (SNR)** predominantly generated by the signal or the background?

Considering the current generated primarily by signal photons, without any extraneous radiation present, the signal current from the detector is

$$i_{\text{sig}} = \phi_{\text{p,sig}} \eta q,$$

where  $\eta$  is the quantum efficiency (electrons per photon). Similarly, the current generated by the background flux is



$$\text{Detector current: } i_{\text{total}} = i_{\text{sig}} + i_{\text{bkg}} \quad i_{\text{bkg}} = \phi_{\text{p,bkg}} \eta q.$$

The SNR is that of the signal current to the shot noise, namely

$$\text{SNR} = \frac{i_{\text{sig}}}{i_{\text{n,shot}}} = \frac{\phi_{\text{p,sig}} \eta q}{\sqrt{2q i_{\text{total}} \Delta f}} = \frac{\phi_{\text{p,sig}} \eta q}{\sqrt{2q(\phi_{\text{p,sig}} \eta q + \phi_{\text{p,bkg}} \eta q) \Delta f}}.$$

Assuming that the dominant noise contribution is the shot noise generated by the signal-power envelope (i.e.,  $\phi_{\text{sig}} \gg \phi_{\text{bkg}}$ ),

$$\text{SNR} \cong \frac{\phi_{\text{p,sig}} \eta}{\sqrt{2\phi_{\text{p,sig}} \eta \Delta f}} = \sqrt{\phi_{\text{p,sig}} \eta \tau},$$

which states that the SNR increases as the square root of the signal flux, thus improving the sensitivity of the system.

With a weak signal source detected against a large background, which is the most common situation in infrared applications, the dominant noise contribution is associated with the shot noise of the background (i.e.,  $\phi_{\text{bkg}} \gg \phi_{\text{sig}}$ ). The photodetector is said to be background limited, yielding

$$\text{SNR}_{\text{BLIP}} \cong \frac{\phi_{\text{p,sig}} \eta}{\sqrt{2\phi_{\text{p,bkg}} \eta \Delta f}} = \phi_{\text{p,sig}} \sqrt{\frac{\eta \tau}{\phi_{\text{p,bkg}}}}.$$

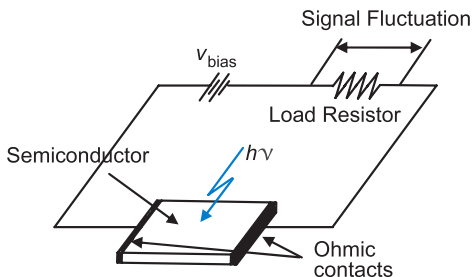
$\text{SNR}_{\text{BLIP}}$  is inversely proportional to the square root of the background flux; therefore, reducing the background photon flux increases the SNR **background-limited infrared photodetector (BLIP)**.

## Generation-Recombination Noise

**Generation-recombination (G/R) noise** is caused by fluctuations in the rates of thermal generation and recombination of free carriers in semiconductor devices without potential barriers (e.g., photoconductors), thereby giving rise to a fluctuation in the average carrier concentration. Thus, the electrical resistance of the semiconductor material changes, which can be observed as fluctuating voltage across the sample when the bias current flows through it.

The transverse photoconductivity geometry and circuit are shown.

Once a carrier is generated, it travels under the influence of an applied electric field. The carrier lasts until recombination occurs at a random time consistent with the mean carrier lifetime  $\tau_{\text{carrier}}$ . The statistical fluctuation in the concentration of carriers produces white noise.



The current noise expression for generation-recombination noise is given by

$$i_{n,G/R} = 2qG\sqrt{\eta E_p A_d \Delta f + g_{th} \Delta f},$$

where  $G$  is the photoconductive gain and  $g_{th}$  is the thermal generation of carriers. Since photoconductors are cooled cryogenically, the second term in the above equation can be neglected. Assuming that  $G$  equals unity,

$$i_{n,G/R} = 2q\sqrt{\eta E_p A_d \Delta f},$$

$$i_{n,G/R} = \sqrt{2}i_{n,shot}.$$

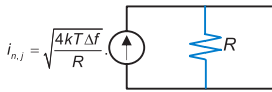
Note that the rms G/R noise is  $\sqrt{2}$  larger than for shot noise, since both generation and recombination mechanisms are random processes.

## Johnson Noise

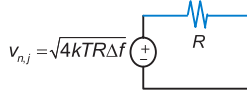
If the background flux is reduced enough, the noise floor is determined by the **Johnson noise**, also known as Nyquist or thermal noise. The fluctuation is caused by the thermal motion of charge carriers in resistive materials, including semiconductors, and occurs in the absence of electrical bias as a fluctuating rms voltage or current.

Johnson noise is an ideal noise-free resistor of the same resistance value, combined either in series with an rms noise voltage or in parallel with an rms noise current.

Norton model:  $R$  in parallel with rms noise current



Thevenin model:  $R$  in series with rms noise voltage



When multiplying the Johnson noise in terms of noise voltage and current spectra:

$$\sqrt{4kTR} \left[ \text{volt}/\sqrt{\text{Hz}} \right] \text{ and } \sqrt{\frac{4kT}{R}} \left[ \text{amp}/\sqrt{\text{Hz}} \right],$$

the power spectral density of the Johnson noise depends only on temperature, and not on resistance:

$$\text{PSD} = 4kT \text{ [watt/Hz]}.$$

Often the detector is cooled to cryogenic temperatures, while the load resistance is at room temperature. In combination, both parallel resistors are added as usual, but the rms noises are added in quadrature, yielding

$$i_j = \sqrt{i_d^2 + i_L^2} = \sqrt{4k\Delta f \left[ \frac{T_d}{R_d} + \frac{T_L}{R_L} \right]}.$$

The SNR in the Johnson noise limit is given by

$$\text{SNR}_j \cong \frac{\phi_{p,\text{sig}} \eta q}{\sqrt{\frac{4kT}{R} \Delta f}}.$$

Johnson noise is independent of the photogeneration process, thus the SNR is directly proportional to the quantum efficiency.

## 1/f Noise and Temperature Noise

**1/f noise** is found in semiconductors and increases at low frequencies. It is characterized by a spectrum in which the noise power depends approximately inversely on the frequency. The general expression for 1/f noise current is given by

$$i_{n,f} = \sqrt{\mathcal{K} \frac{\bar{i}^\alpha \Delta f}{f^\beta}},$$

where  $\mathcal{K}$  is a proportionality factor,  $\bar{i}$  is the average dc current,  $f$  is the frequency,  $\Delta f$  is the measuring bandwidth,  $\alpha$  is a constant of  $\sim 2$ , and  $\beta$  ranges from  $\sim 0.5$  to  $1.5$ .

Although the cause of 1/f noise is not fully understood, it appears to be associated with the presence of potential barriers at the nonohmic contacts and the surface of the semiconductor.

1/f noise is typically the dominant noise to a few hundred hertz, and is often significant to several kilohertz. It is always present in microbolometers and photoconductors, because there is always a dc-bias current flowing within the detector material. However, 1/f noise can be eliminated in photovoltaic detectors operating in open-circuit voltage mode, when no dc-bias current is allowed to flow through the diode.

**Temperature noise** is caused by fluctuations in the temperature of the detector due to fluctuations in the rate at which heat is transferred from the sensitive element to its surroundings (i.e., radiative exchange and/or conductance with a heat sink). The spectrum of the mean-square fluctuation in temperature is given by

$$\overline{\Delta T^2} = \frac{4kKT^2}{K^2 + (2\pi f)^2 C^2},$$

where  $k$  is Boltzmann's constant,  $K$  is the thermal conductance,  $C$  is the heat capacity, and  $T$  is the temperature.

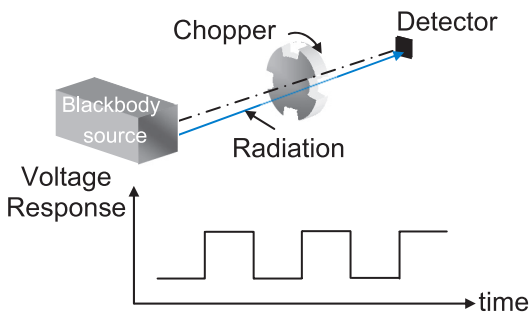
Temperature noise is primarily observed in thermal detectors, and being temperature-noise limited, provides superior performance. The power spectrum of the temperature noise is flat.

## Detector Responsivity

**Responsivity** gives the response magnitude of the detector. It provides information on gain, linearity, dynamic range, and saturation level. Responsivity is a measure of the transfer function between the input signal photon power or flux and the detector electrical signal output:

$$\mathcal{R} = \frac{\text{output signal}}{\text{input flux}},$$

where the output can be in volts or amperes, and the input in watts or photons/sec.  $\mathcal{R}_i$  is **current responsivity** and  $\mathcal{R}_v$  is **voltage responsivity**. A common technique in detection is to modulate the radiation to be detected and to measure the modulated component of the electrical output of the detector. This technique provides some discrimination against electrical noise since the signal is contained only within the Fourier component of the electrical signal at the modulation frequency, whereas electrical noise is often broadband. Furthermore, it avoids baseline drifts that affect electronic amplifiers due to ac coupling. The output voltage would vary from peak-to-valley.



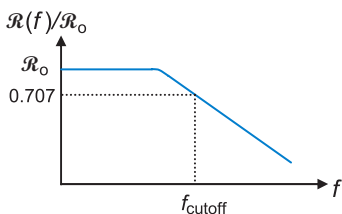
An important characteristic of a detector is how fast it responds to a pulse of optical radiation. The voltage response to radiation modulated at frequency  $f$  is defined as

$$\mathcal{R}_v(f) = \frac{v_{\text{sig}}(f)}{\phi_{\text{sig}}(f)},$$

where  $\phi_{\text{sig}}(f)$  is the rms value of the signal flux contained within the harmonic component at frequency  $f$ , and  $v_{\text{sig}}(f)$  is the rms output voltage within this same harmonic component.

## Detector Responsivity (cont.)

In general, the responsivity  $\mathcal{R}(f)$  of a detector decreases as the modulation frequency  $f$  increases. Changing the angular speed of the chopper, the responsivity can be obtained as a function of frequency. A typical responsivity versus frequency curve is plotted.



The cutoff frequency  $f_{\text{cutoff}}$  is defined as the modulation frequency at which  $|\mathcal{R}_v(f_{\text{cutoff}})|^2$  falls to one-half its maximum value, and is related to response time as

$$f_{\text{cutoff}} = \frac{1}{2\pi\tau}.$$

The response time of a detector is characterized by its **responsive time constant**, the time required for the detector output to reach 63% ( $1 - 1/e$ ) of its final value after a sudden change in the irradiance. For most sensitive devices, the response to a change in irradiance follows a simple exponential law. As an example, if a delta-function pulse of radiation  $\delta(t)$  is incident on the detector, an output voltage signal (i.e., the impulse response) of the form

$$v(t) = v_0 e^{-t/\tau} \quad t \geq 0$$

is produced, where  $\tau$  is the time constant of the detector. Transforming this time-dependent equation into the frequency domain by using the corresponding Fourier transform yields

$$V(f) = \frac{v_0 \tau}{1 + j2\pi f \tau},$$

which can be extended to responsivity as

$$\mathcal{R}_v(f) = \frac{\mathcal{R}_0}{1 + j2\pi f \tau},$$

where  $\mathcal{R}_0 = v_0 \tau$  is the dc value of the responsivity. The modulus is written as

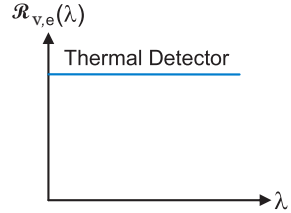
$$|\mathcal{R}_v(f)| = \frac{\mathcal{R}_0}{\sqrt{1 + (2\pi f \tau)^2}}.$$



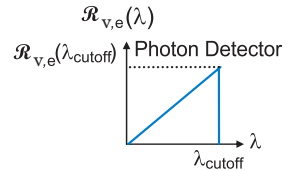
## Spectral Responsivity

Responsivity depends on the wavelength of the incident radiation beam, thus the spectral response of a detector can be specified in terms of its responsivity as a function of wavelength. **Spectral responsivity**  $\mathcal{R}(\lambda, f)$  is the output signal response to monochromatic radiation incident on the detector, modulated at frequency  $f$ . It determines the amplifier gain required to bring the detector output up to acceptable levels. To measure the spectral responsivity of a detector, a tunable narrowband source is required.

In energy-derived units, the spectral responsivity for a thermal detector is independent of wavelength (i.e., 1 W of radiation produces the same temperature rise at any spectral line). Therefore, its spectral response is limited by the spectral properties of the material of the window placed in front of the detector.



In a photon detector, the ideal spectral responsivity is linearly proportional to the wavelength:



$$\mathcal{R}_{v,e}(\lambda) = \mathcal{R}_{v,e}(\lambda_{\text{cutoff}}) \frac{\lambda}{\lambda_{\text{cutoff}}}.$$

Photons with  $\lambda > \lambda_{\text{cutoff}}$  are not absorbed by the material of the detector and are therefore not detected.

Long-wavelength cutoff is the longest wavelength detected by a sensor made of a material with a certain energy gap, and it is given by

$$\lambda_{\text{cutoff}} = hc / \mathcal{E}_{\text{gap}}$$

An expression to determine the energy gap in electron-volt units is

$$\mathcal{E}_{\text{gap}} [\text{eV}] = \frac{hc}{\lambda_{\text{cutoff}}} \cdot \frac{1 \text{ eV}}{1.6 \times 10^{-19} \text{ J}} = \frac{1.24}{\lambda_{\text{cutoff}}}.$$

The energy gap of silicon is 1.12 eV; therefore, photons with  $\lambda > 1.1 \mu\text{m}$  are not detected.

## Blackbody Responsivity

**Blackbody responsivity**  $\mathcal{R}(T, f)$  is interpreted as the output produced in response to a watt of input optical radiation from a blackbody at temperature  $T$  modulated at electrical frequency  $f$ . Since a BB source is widely available and rather inexpensive compared to a tunable narrowband source, it is more convenient to measure  $\mathcal{R}(T)$  and calculate its corresponding  $\mathcal{R}(\lambda)$ .

For a blackbody that produces a spectral flux  $\phi_\lambda(\lambda)$ , the **detector output voltage** is calculated from the following integral:

$$v_{\text{out,det}} = \int_0^{\lambda_{\text{cutoff}}} \phi_\lambda(\lambda) \mathcal{R}_v(\lambda) d\lambda \text{ [volt]},$$

which determines the contribution to the detector output in those regions where the spectral flux and the voltage spectral responsivity overlap.

When measuring blackbody responsivity, the radiant power on the detector contains all wavelengths of radiation, independent of the spectral response curve of the detector. Using the Stefan-Boltzmann law and some of the basic radiometric principles presented previously, the blackbody responsivity is

$$\mathcal{R}(T) = \frac{v_{\text{out,det}}}{\phi_e} = \frac{\pi \int_0^{\lambda_{\text{cutoff}}} \phi_{e,\lambda}(\lambda) \mathcal{R}_v(\lambda) d\lambda}{\sigma T^4 A_{\text{source}} \Omega_{\text{det}}} = \frac{\int_0^{\lambda_{\text{cutoff}}} M_{e,\lambda}(\lambda) \mathcal{R}_v(\lambda) d\lambda}{\sigma T^4}.$$

Substituting the ideal response of a photon detector yields

$$\mathcal{R}(T) = \frac{\frac{\mathcal{R}_v(\lambda_{\text{cutoff}})}{\lambda_{\text{cutoff}}} \int_0^{\lambda_{\text{cutoff}}} M_{e,\lambda}(\lambda) \lambda d\lambda}{\sigma T^4}.$$

The ratio of  $\mathcal{R}(T)$  to  $\mathcal{R}(\lambda_{\text{cutoff}})$  is defined by the  $\mathcal{W}$ -factor:

$$\mathcal{W}(\lambda_{\text{cutoff}}, T) = \frac{\mathcal{R}_v(\lambda_{\text{cutoff}})}{\mathcal{R}(T)} = \frac{\lambda_{\text{cutoff}} \sigma T^4}{\int_0^{\lambda_{\text{cutoff}}} M_{e,\lambda}(\lambda) \lambda d\lambda} = \frac{\lambda_{\text{cutoff}} \sigma T^4}{hc \int_0^{\lambda_{\text{cutoff}}} M_{p,\lambda}(\lambda) d\lambda}.$$

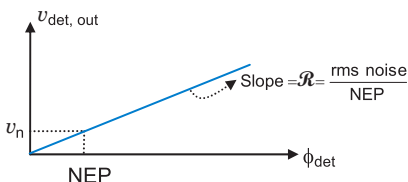
Two standard blackbody temperatures are used to evaluate detectors: (1) 500 K for midwave and long-wave infrared measurements; and (2) 2850 K for visible and NIR measurements.

## Noise-Equivalent Power

Although the responsivity is a useful measurement to predict a signal level for a given irradiance, it gives no indication of the minimum radiant flux that can be detected. In other words, it does not consider the amount of noise at the output of the detector that ultimately determines the SNR.

The ability to detect small amounts of radiant energy is inhibited by the presence of noise in the detection process. Since noise produces a random fluctuation in the output of a radiation detector, it can mask the output that is produced

by a weak optical signal. Noise thus sets limits on the minimum input spectral flux that can be detected under given conditions.



One convenient description of this minimum detectable signal is the **noise-equivalent power (NEP)**, which is defined as the radiant flux necessary to give an output signal equal to the detector noise. In other words, it is the radiant power  $\phi_e$  incident on the detector that yields an  $\text{SNR} = 1$ , and is expressed as the rms noise level divided by the responsivity of the detector:

$$\text{NEP} = \frac{v_n}{\mathcal{R}_v} = \frac{v_n}{v_{\text{sig}} / \phi_{\text{sig}}} = \frac{\phi_{\text{sig}}}{\text{SNR}} \quad [\text{watt}],$$

where  $v_n$  denotes the rms voltage produced by a radiation detection system. A smaller NEP implies better sensitivity.

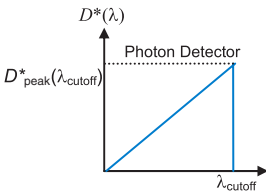
The disadvantage of using the NEP to describe detector performance is that the NEP does not allow a direct comparison of the sensitivity of different detector mechanisms or materials. This is because of its dependence on both the square root of the detector area and the square root of the electronic bandwidth. A descriptor that circumvents this problem is called  $D^*$  (pronounced “dee star”), which normalizes the inverse of the NEP to a  $1\text{-cm}^2$  detector area and 1 Hz of noise bandwidth.

## Specific or Normalized Detectivity

NEP is a situation-specific descriptor useful for design purposes, but it does not allow direct comparison of the sensitivity of different detector materials. The **specific or normalized detectivity ( $D^*$ )** is often used to specify detector performance. It normalizes the noise-equivalent bandwidth and the area of the detector; however, in order to predict the SNR, the sensor area and bandwidth must be chosen for a particular application.

$D^*$  is independent of detector area and electronic bandwidth, because the NEP is also directly proportional to the square root of these parameters.  $D^*$  is directly proportional to the SNR as well as to the responsivity:

$$D^* = \frac{\sqrt{A_d \Delta f}}{\text{NEP}} = \frac{\sqrt{A_d \Delta f}}{\phi_d} \text{SNR} = \frac{\sqrt{A_d \Delta f}}{v_n} \mathcal{R}_v \left[ \text{cm} \sqrt{\text{Hz}} / \text{watt} \right].$$



Plots of spectral  $D^*$  for photon detectors have the same linear dependency with  $\lambda$ :

$$D^*(\lambda) = D^*_{\text{peak}}(\lambda_{\text{cutoff}}) \frac{\lambda}{\lambda_{\text{cutoff}}}.$$

The same  $\mathcal{W}$ -factor applies between  $D^*_{\text{peak}}(\lambda_{\text{cutoff}})$  and  $D^*(T)$ .

The cutoff value of  $D^*(\lambda)$  is defined as the peak spectral value  $D^*_{\text{peak}}(\lambda_{\text{cutoff}})$ , and corresponds to the largest potential SNR. In addition, any optical radiation incident on the detector at a wavelength shorter than the cutoff wavelength  $\lambda_{\text{cutoff}}$  has  $D^*(\lambda)$  reduced from the peak  $D^*_{\text{peak}}(\lambda_{\text{cutoff}})$  in proportion to the ratio  $\lambda/\lambda_{\text{cutoff}}$ .

Unlike the NEP, this descriptor increases with the sensitivity of the detector. Depending on whether the NEP is spectral or blackbody,  $D^*$  can also be either spectral or blackbody.  $D^*(\lambda, f)$  is the detector's SNR when 1 W of monochromatic radiant flux (modulated at  $f$ ) is incident on a 1-cm<sup>2</sup> detector area, within a noise-equivalent bandwidth of 1 Hz. The blackbody  $D^*(T, f)$  is the signal-to-noise output when 1 W of blackbody radiant power (modulated at  $f$ ) is incident on a 1-cm<sup>2</sup> detector, within a noise-equivalent bandwidth of 1 Hz.

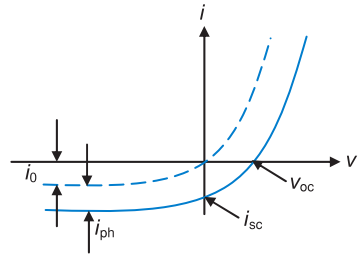
## Photovoltaic Detectors or Photodiodes

In **photovoltaic (PV) detectors**, more commonly called **photodiodes**, the optical radiation is absorbed at the p-n junction, which produces an output current or voltage.

The photodiode equation is given by:

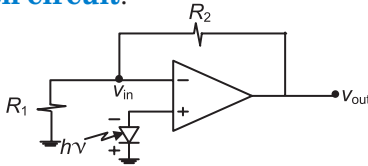
$$i = i_{\text{diode}} - i_{\text{ph}} = i_0 \left[ \exp\left(\frac{qv}{kT}\right) - 1 \right] - \eta q \phi_p,$$

where  $i_{\text{ph}}$  is the photo-generated current,  $i_0$  is the dark current,  $q$  is the charge of an electron,  $v$  is the voltage,  $k$  is Boltzmann's constant, and  $T$  is the temperature.



The photodiode operating in each quadrant has different electro-optical characteristics. The most common operating points are open circuit, reverse bias, and short circuit.

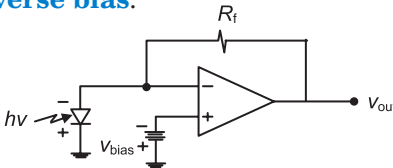
### Open circuit:



$$V_{\text{oc}} = \frac{kT}{q} \ln \left( \frac{i_{\text{ph}}}{i_0} + 1 \right)$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$

### Reverse bias:



$$V_{\text{out}} = V_{\text{bias}} + R_f i_{\text{ph}}$$

If capacitance (C) decreases then RC decreases, the junction widens, and the detector response increases.

**Short circuit:** the voltage is zero across the photodiode, and  $i_{\text{ph}}$  is forced to flow into an electrical short circuit:

$$i = i_{\text{ph}} = \eta q \phi_p = \eta q E_p A_d.$$

## Sources of Noise in PV Detectors

The sources of noise in a PV detector are

1. shot noise due to the dark current;
2. shot noise due to the feedback noise of the signal and background;
3. Johnson noise due to detector resistance;
4. Johnson noise due to the load resistors;
5.  $1/f$  noise associated with the current flow;
6. preamplifier circuit noise; and
7. preamplifier voltage noise.

The noise expression for the photodiode is then

$$i_n^2 = 4q^2 i_0^2 \Delta f + 2q^2 \eta \phi_{p, \text{sig}} \Delta f + 2q^2 \eta \phi_{p, \text{bkg}} \Delta f + 4k\Delta f \left( \frac{T_d}{R_d} + \frac{T_L}{R_L} \right) + \frac{\beta_0 \bar{i} \Delta f}{f} + i_{pa}^2 + v_{pa}^2 (R_L \parallel R_d),$$

where  $R_L$  and  $T_L$  are the load resistance and load temperature, respectively. The preamplifier noise can be made negligible by using low-noise transistors or by cooling the preamplifier to cryogenic temperatures.  $1/f$  noise is almost zero if the device is operating in either open or reverse circuit modes. It is assumed that the dark current is also negligible compared to both the background and signal currents (i.e.,  $i_0 \ll i_{\text{sig}} + i_{\text{bkg}}$ ), and that BLIP conditions are in effect. Under these conditions, the rms shot noise current is approximately

$$i_n \cong \sqrt{2q^2 \eta \phi_{p, \text{bkg}} \Delta f} = \sqrt{2q^2 \eta \phi_{e, \text{bkg}} \frac{\lambda}{hc} \Delta f}.$$

Recalling that the peak signal current generated by a photovoltaic detector is

$$i_{\text{signal}} = \phi_{p, \text{sig}} \eta q = \phi_{e, \text{sig}} \eta q \frac{\lambda}{hc},$$

the SNR for a photovoltaic detector is then

$$\text{SNR}_{\text{PV}} = \frac{\phi_{e, \text{sig}} \eta q \frac{\lambda}{hc}}{\sqrt{2q^2 \phi_{e, \text{bkg}} \eta \frac{\lambda}{hc} \Delta f}}.$$

Setting the  $\text{SNR}_{\text{PV}} = 1$ , the spectral  $\text{NEP}_{\text{PV, BLIP}}$  is obtained:

$$\text{NEP}_{\text{PV, BLIP}}(\lambda) = \sqrt{\frac{hc}{\lambda} \frac{2\phi_{e, \text{bkg}} \Delta f}{\eta}}.$$

### Expressions for $D_{\text{PV,BLIP}}^*$ , $D_{\text{PV,BLIP}}^{**}$ , and $D_{\text{PV,JOLI}}^*$

Spectral  $D_{\text{PV,BLIP}}^*$  for a PV detector is obtained from the definition of  $D^*$  in terms of NEP:

$$D_{\text{PV,BLIP}}^*(\lambda) = \frac{\sqrt{A_d \Delta f}}{\text{NEP}_{\text{PV,BLIP}}} = \frac{\sqrt{\frac{\lambda}{hc} \frac{\eta}{2E_{\text{e,bkg}}}}} = \frac{\lambda}{hc} \sqrt{\frac{\eta}{2E_{\text{p,bkg}}}}.$$

If the background is not monochromatic,  $E_{\text{bkg}}$  needs to be integrated over the sensor response from 0 to  $\lambda_{\text{cutoff}}$ :

$$\begin{aligned} D_{\text{PV,BLIP}}^*(\lambda_{\text{cutoff}}) &= \frac{\lambda_{\text{cutoff}}}{hc} \sqrt{\frac{\eta}{2 \int_0^{\lambda_{\text{cutoff}}} E_{\text{p,bkg}}(\lambda) d\lambda}} \\ &= (F/\#) \frac{\lambda_{\text{cutoff}}}{hc} \sqrt{\frac{2\eta}{\pi \int_0^{\lambda_{\text{cutoff}}} L_{\text{p,bkg}}(\lambda) d\lambda}}, \end{aligned}$$

where  $E_{\text{p,bkg}} = \pi L_{\text{p,bkg}} \sin^2 \theta \cong \pi L_{\text{p,bkg}} [1/2(F/\#)]^2$ .

$D_{\text{BLIP}}^*$  increases when increasing the  $F/\#$ , which illuminates the detector with a smaller cone of background radiation.

$D^{**}$  normalizes the dependence on  $\sin \theta$ , allowing a comparison of detectors normalized to hemispherical background:

$$D_{\text{PV,BLIP}}^{**}(\lambda) = \sin \theta D_{\text{PV,BLIP}}^*(\lambda).$$

**Johnson-limited noise performance (JOLI)** occurs in deep space applications when the shot noise has been decreased. It becomes negligible compared to the Johnson noise:

$$2q^2 \eta (\phi_{\text{p,sig}} + \phi_{\text{p,sig}}) \Delta f \ll 4k\Delta f \left( \frac{T_d}{R_d} + \frac{T_L}{R_L} \right),$$

where  $T_d$  and  $R_d$  are the temperature and resistance of the detector, respectively. The SNR of the PV detector under this condition becomes:

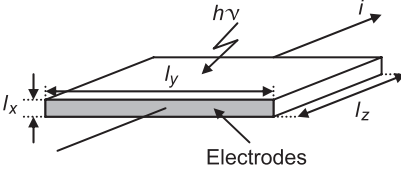
$$\text{SNR}_{\text{PV,JOLI}} = \frac{q\eta\phi_{\text{e,sig}} \frac{\lambda}{hc}}{4k\Delta f \left( \frac{T_d}{R_d} + \frac{T_L}{R_L} \right)} \cong \frac{q\eta\phi_{\text{e,sig}} \frac{\lambda}{hc}}{4k\Delta f \frac{T_d}{R_d}} R_L \gg R_d.$$

Setting  $\text{SNR}_{\text{PV,JOLI}} = 1$ , converting the noise-equivalent photon flux to NEP<sub>PV,JOLI</sub> and substituting the NEP expression in terms of  $D^*$ ,

$$D_{\text{PV,JOLI}}^* = \frac{\lambda q \eta}{2hc} \sqrt{\frac{R_d A_d}{k T_d}}.$$

## Photoconductive Detectors

Shot noise occurs in diodes and other devices with a potential-energy barrier for which the generation is a random process, while recombination is a deterministic process. In devices



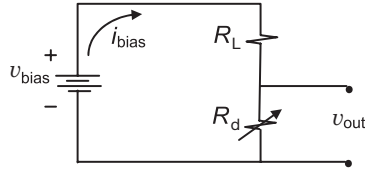
such as **photoconductive (PC) detectors** without junctions or other potential barriers, both generation and recombination are random processes.

**Photoconductors** respond to light by changing the resistance or conductance of the detector's material:

$$R_d \propto \frac{1}{\phi_p} \Rightarrow dR_d \propto \frac{d\phi_p}{\phi_p^2},$$

where  $\phi_p$  is the photon flux.

Photoconductive detectors have no junction; therefore, there is no intrinsic field. They cannot operate under open circuit conditions and do not generate voltage independently (i.e., they must use bias current). In order to detect the change in resistance, a biasing circuit with an applied field must be utilized.



The output voltage is given by

$$v_{out} = \frac{R_d}{R_d + R_L} v_{bias},$$

where a small change in the detector resistance produces a change in signal voltage, which is directly proportional to the change in the photon flux incident on the photoconductive detector:

$$dv_{out} = \frac{dR_d}{(R_d + R_L)^2} v_{bias} \propto \frac{d\phi_p}{(R_d + R_L)^2 \phi_p^2} v_{bias}.$$



## Sources of Noise in PC Detectors

The sources of noise in **photoconductors** are listed as

1.  $1/f$  noise associated with the current flow;
2. generation-recombination (G/R) noise;
3. Johnson noise due to detector resistance;
4. Johnson noise due to the load resistors;
5. preamplifier circuit noise; and
6. preamplifier voltage noise.

The noise expression for the photoconductive detector is then

$$i_n^2 = 4q^2 \eta E_p A_d G^2 \Delta f + 4q^2 g_{th} G^2 \Delta f + 4k\Delta f \left( \frac{T_d}{R_d} + \frac{T_L}{R_L} \right) + \frac{\beta_0 \bar{i} \Delta f}{f} + i_{pa}^2 + v_{pa}^2 (R_L \parallel R_d),$$

where  $g_{th}$  is the thermal generation rate of carriers and  $G$  is the photoconductive gain.  $G$  is proportional to the number of times an electron can transit the detector electrodes in its lifetime (i.e., excess signal electrons through the PC). The photoconductive gain depends on the detector size, material, and doping and can vary between  $1 < G < 10^5$ . If  $G < 1$ , the electron does not reach the electrode before recombining. Usually PC detectors are cooled cryogenically, in which case the thermal term in the generation recombination noise is negligible.

A photoconductor for BLIP- and Johnson-limited conditions are defined in the table.

| BLIP                                                                              | JOLI                                                                                                                  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $NEP_{PC, BLIP} = \frac{2hc}{\lambda G} \sqrt{\frac{E_{bkg} A_d \Delta f}{\eta}}$ | $NEP_{PC, JOLI} = \frac{i_j}{\mathcal{R}_{i, PC}} = \frac{\sqrt{4k\Delta f T / R_{eq}}}{\frac{\lambda q \eta}{hc} G}$ |
| $D_{PC, BLIP}^* = \frac{\lambda}{2hc} G \sqrt{\frac{\eta}{E_{bkg}}}$              | $D_{PC, JOLI}^* = \frac{\lambda q \eta}{2hc} G \sqrt{\frac{R_{eq} A_d}{kT}}$                                          |

where:

$$T \equiv T_d \approx T_L \text{ and } R_{eq} = R_d \parallel R_L$$

For a given photon flux and a photoconductive gain of unity, the generation-recombination noise is larger than the shot noise by a factor of  $\sqrt{2}$ .

## Third-Generation Infrared Imagers

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Third-generation **infrared focal plane arrays (IRFPAs)** are being actively developed by the U.S. Department of Defense. The proliferation of second-generation cooled and uncooled IRFPAs threatens to nullify the advantage that U.S. soldiers have in night combat. Third-generation FPA technology is being defined and developed to maintain this current advantage enjoyed by U.S. and allied armed forces.

This new generation of IRFPA technology is expected to provide advanced features such as higher resolution (e.g.,  $1024 \times 1024$  or  $2048 \times 1024$  and  $2048 \times 2048$  FPA formats) and new functions such as multispectral band capability, higher frame rates, better thermal resolution, and smart **read-out integrated circuits (ROICs)**.

Third-generation tactical IRFPAs ought to have some, or all, of the following characteristics:

- Single-color and large FPA formats with high spatial resolution (mega-pixel) cooled imagers in order to improve global system performance.
- High-performance multispectral and hyperspectral cooled detectors with the ability of each pixel to detect two or more infrared spectral bands.
- Thermal performance improvements of cryogenic technologies in order to improve thermal resolution.
- Fast frame rate operation to collect a higher percentage of available signals, and to provide more data throughput for advanced signal processing functions such as automatic target recognition (ATR) and missile tracking.
- FPA operation at higher temperatures to reduce cryogenics constraints and to improve reliability (driven by the system's portability and cost).
- Reduction of power dissipation and the power consumption of cryogenic coolers.
- 3D active imagery with high-speed detectors and low noise to improve identification range and efficiency.
- Medium-high performance uncooled imagers with small ( $\sim 25 \mu\text{m}$ ) pixels and power reduction.
- Very low cost, expendable uncooled imagers.

## Indium Antimonite (InSb) Photodiodes

**Indium antimonite (InSb)** is a crystalline compound made from the elements indium and antimony. It is a narrow-gap intrinsic III-V semiconductor detector material with a direct bandgap.

The relationship between the energy gap and operating temperature from 4 K to 300 K is given by

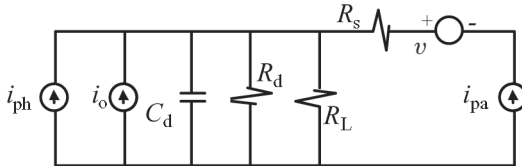
$$\mathcal{E}_{\text{gap}}(x) = 0.233 - (6.498 \times 10^{-5})T - (5.0826 \times 10^{-7})T^2,$$

and its current vs. voltage characteristics are described by

$$i = i_o [\exp(qv/kT) - 1] - i_{\text{ph}} + \frac{v}{R_s},$$

where  $R_s$  represents a shunt resistance of a leakage current path in the PN junction, and  $i_{\text{ph}}$  is the photogenerated current.

The diode equivalent circuit with simplified pre-amplifier and noise sources is depicted below.



InSb FPAs are highly uniform multichannel imaging devices that provide sensitive detection between 1–5.3  $\mu\text{m}$  (80K). These imaging devices have very high external quantum efficiencies ( $> 90\%$ ), while operabilities exceeding 99.5% are routine. Large-format resolutions ( $1280 \times 1024$  with a 12- $\mu\text{m}$  pitch and  $1024 \times 1024$  with a 25- $\mu\text{m}$  pitch) are commercially available.

The spectral response in these types of devices shifts to longer wavelengths with increasing temperature. Their thermionic noise also increases rapidly with temperature.

Distinct advantages to advanced digital FPA technology include low-noise data output and immunity to external noise sources for **non-uniformity correction (NUC)** stability, high-speed digital data output, and virtually no crosstalk.

## Mercury Cadmium Telluride (HgCdTe) Photodetectors

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**Mercury cadmium telluride (HgCdTe)** photodetectors are intrinsic PC and PV devices. The PC HgCdTe detector is a mature product that has been in production for many years. They are typically fabricated from n-type material with a high intrinsic concentration of carriers, which can achieve background-limited sensitivities at operating temperatures substantially higher than extrinsic detectors. Even though HgCdTe PC detectors are excellent single-element sensor devices, they are not conducive to producing 2D FPAs. Producing two identical HgCdTe PC detectors is rather difficult because of the material's non-uniformity. In addition, their large current densities (i.e., low resistance) can lead to overheating problems, and their potential wells fill up too quickly because of background currents. HgCdTe photodiodes also have difficulties associated with producing homogeneous material. However, modern fabrication methods can be applied to produce device-quality PV superlattice structures consisting of thin CdTe/HgTe stacked layers. This approach controls and improves the non-uniformity of the photodiodes and reduces the  $1/f$  noise contributions of the detector FPAs.

$\text{Hg}_{1-x}\text{Cd}_x\text{Te}$  is a ternary semiconductor compound that exhibits a wavelength cutoff proportional to the alloy composition (or molar fraction ratio  $x$  of Cd to Hg). Pure CdTe ( $x = 1$ ) will detect radiation wavelengths out to  $\sim 0.83 \mu\text{m}$ ; the greater the percentage of Hg, the longer the wavelength response. HgCdTe photodiodes can operate up to  $13 \mu\text{m}$ , while HgCdTe PC detectors can be tailored to operate further in the infrared to about  $30 \mu\text{m}$ .

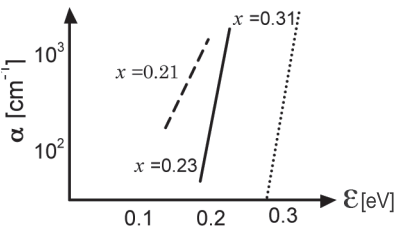
An empirical expression that relates the bandgap energy  $\mathcal{E}_{\text{gap}}$ , the temperature  $T$ , and the alloy composition  $x$  is given by

$$\mathcal{E}_{\text{gap}}(x) = -0.302 + 1.93x + (5.35 \times 10^{-4})T(1 - 2x) - 0.801x^2 + 0.832x^3$$

This equation is almost linear for molar values between  $0.2 \leq x \leq 0.6$ , and fits  $\mathcal{E}_{\text{gap}}$  data for the temperature range  $4.2 \text{ K} \leq T \leq 300 \text{ K}$ .

## Control of the Alloy Composition

The optical absorption coefficient  $\alpha$  of HgCdTe ranges from  $10^2$ – $10^5$   $\text{cm}^{-1}$  so virtually all photons are absorbed within 10  $\mu\text{m}$  of the incident surface, which means that the detector thickness can be much less than the thickness required of other detector materials for similar wavelengths. For example, a plot of  $\alpha$  vs. photon energy for  $x = 0.21$ , 0.23, and 0.31 is shown. The steepness in this curve leads to a rapid decline in detectivity for wavelengths larger than the cutoff wavelength.



The downside of this steep variability is that the molar fraction  $x$  is not well controlled in fabrication, so the alloy composition and cutoff wavelength must be determined by infrared transmission maps at a temperature of 300 K, usually at several points on the wafer. The fact that  $x$  is not able to be controlled leads to a non-uniform material, thus the responsivity varies as a function of position.

The variation of  $x$  is related to the cutoff wavelength by

$$\lambda_c = \frac{1.24}{\mathcal{E}_{\text{gap}}(x)},$$

$$\lambda_c = \frac{1}{-0.2435 + 1.5565x + (4.3145 \times 10^{-4})T(1 - 2x) - 0.646x^2 + 0.671x^3}.$$

To determine the impact of the uncertainty of  $x$  in manufacturing on variations in  $\lambda_c$ , we take the differential, yielding

$$\frac{\partial \lambda_c}{\lambda_c^2} = (1.5565 - 8.629 \times 10^{-4}T - 1.292x + 2.013x^2) \partial x.$$

This variation in  $x$  is typical of good-quality HgCdTe material. The variation in cutoff wavelength is negligible for SW/MWIR spectral bands; however, in the LWIR the uncertainty is quite large and cannot be neglected. This variation causes radiometric calibration issues, in which the radiation is sensed over a different spectral region.

## HgCdTe Photodiodes and FPAs

An uncertainty in cutoff wavelength for  $\partial x = 0.2\%$  is shown in the table below.

| $x$   | $\lambda_c$ [ $\mu\text{m}$ ] @ 77 K | $\partial\lambda_c$ [ $\mu\text{m}$ ] |
|-------|--------------------------------------|---------------------------------------|
| 0.395 | 3                                    | 0.023                                 |
| 0.295 | 5                                    | 0.064                                 |
| 0.21  | 10                                   | 0.26                                  |
| 0.196 | 14                                   | 0.51                                  |

Integrating the exitance of a blackbody at temperature  $T$  for  $x = 0.196$  from 0–13.5  $\mu\text{m}$  will yield the same response as integrating from 0–14  $\mu\text{m}$  of a source at the temperature  $T - \Delta T$ . Therefore, if a FPA is manufactured, a different output voltage for a uniform background would appear due to the non-uniformities of the semiconductor.

HgCdTe FPAs are made up of two components: an HgCdTe detector array and a silicon ROIC multiplexer. The ROIC circuit reads the photocurrent from each pixel of the detector FPA and outputs the signal in a desired sequence that is used to form the 2D image.

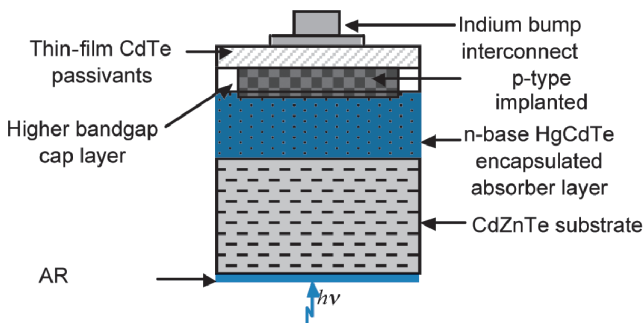
The highest detector performance is achieved by growing HgCdTe on lattice-matched substrate CdZnTe. The coefficient of thermal expansion (CTE) mismatch between CdZnTe and Si affects the size of the FPA and its reliability during repeated thermal cycling from room temperature to cryogenic operating temperatures. This temperature-cycling fatigue is addressed by using a shim between the Si and the Dewar cold plate, thus constraining the detector array and the ROIC to expand and contract at the same rate during thermal cycling. HgCdTe growth on alternate substrates such as silicon and sapphire are also being developed to reduce this thermal expansion mismatch. These alternate substrate materials suffer from a large lattice mismatch with HgCdTe, leading to a higher defect density and consequently reducing the detector performance. As a result, the trade-off between FPA performance, size, and cost dictates the selection of the substrate material.

The fabrication of HgCdTe photodiodes is usually based on DLHJ, TLHJ, and HDVIP technologies, defined next.

## DLHJ Photodiodes

### The double-layer heterostructure junction (DLHJ)

photodiode's fabrication process starts with in-situ molecular beam epitaxy (MBE) growth of a double-layer heterostructure consisting of an absorbing base layer and a higher bandgap cap layer. The formation of p-on-n photodiodes is achieved by selective area p-type arsenic (As) ion implantation through the cap layer into the narrow gap base layer. Each implanted area defines an individual pixel of the detector array. The implantation step is followed by a two-step thermal anneal under Hg over-pressure. A high temperature anneal in Hg vapor activates the dopant by substituting arsenic atoms on the Te sub-lattice. A lower temperature anneal is performed immediately after the high temperature anneal, annihilating the Hg vacancies formed in the HgCdTe lattice during growth and the high-temperature annealing step. Individual photodiodes are delineated by a mesa etch through the thin cap layer. The exposed surface is then passivated, and metal contacts and metal interconnect bumps are formed. The thin polycrystalline MBE-deposited CdTe layer provides the device passivation, which reduces surface recombination as well as the  $1/f$  noise and renders the surface impermeable to environmental conditions.

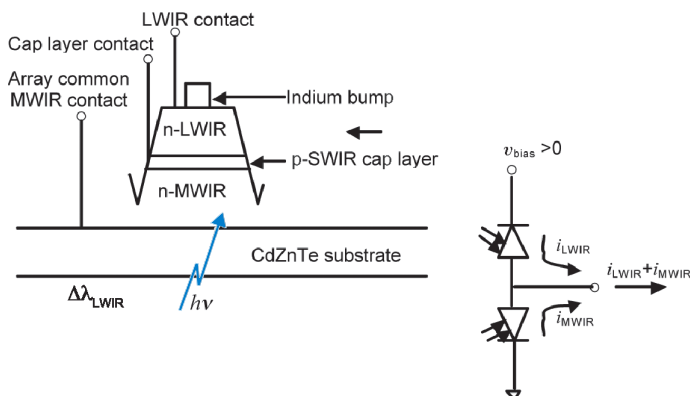


The goal is to have the p-type cap layer thin enough and of sufficiently wide bandgap so that it contributes a negligible amount of thermally generated diffusion current compared with that from the n-type encapsulated absorber layer.

## Dual-Band HgCdTe FPAs

Multicolor third-generation IRFPAs are desirable for numerous applications. For example, by using two infrared wavebands, spurious information such as background clutter and sun glint may be subtracted from an infrared image, leaving only the targets of interest. Thus, the effective SNR of two-color IRFPAs exceeds that of single-color IRFPAs for specific targets.

Two-color detector arrays are usually based on n-p-n HgCdTe **triple layer heterostructure junction (TLHJ)** design. The TLHJ detectors consist of back-to-back photodiode p-n junctions. This device architecture is realized by placing a longer-wavelength HgCdTe photodiode behind a shorter-wavelength photodiode. An additional electrical contact to the shared-type centered layer is employed so that each junction can be accessed independently with both signal channels integrated simultaneously.

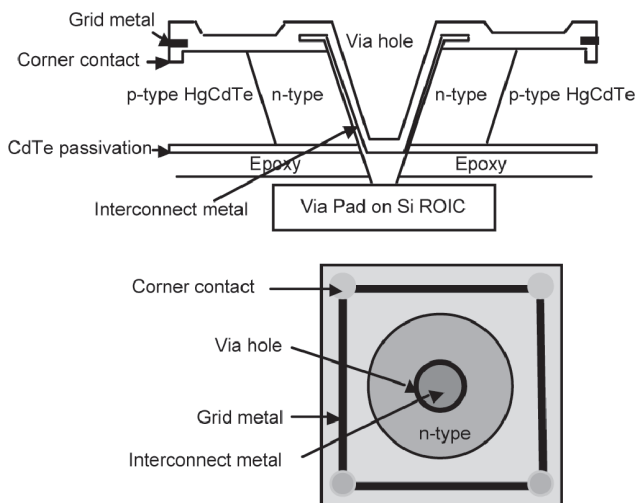


High-quality  $1280 \times 720$  MW/LWIR IRFPAs for third-generation systems have been fabricated from MBE-grown HgCdTe TLHJ wafers with 100% optical fill factor and 60-Hz frame rate. These devices contain cutoff wavelengths greater than  $10.5 \mu\text{m}$  at 78 K. They provide simultaneous temporal detection in the MWIR and LWIR bands using time-division-multiplexed integration (TDMI) incorporated into the ROIC. They have operability of  $>99.9\%$  and  $>98\%$  in the midwave and long-wave spectral bands, respectively.



## HDVIP Photodiodes

**The high-density vertically integrated photodiode (HDVIP)** represents a fairly new approach in the development of infrared focal plane array architecture. The monocolor HDVIP consists of a thin layer of extrinsically doped p-HgCdTe, passivated on both surfaces with an interdiffused layer of CdTe, which is glued to an underlying silicon ROIC with a uniformly thin bond-line of epoxy. The stress in the HgCdTe is uniform throughout the whole layer except for a small region around the periphery about the size of a layer thickness. Contact to the silicon ROIC is achieved by etching a via hole through the HgCdTe down to landing pads on the silicon. The  $n^+/n^-$  region of the diode is formed around the via hole, both by the etching process itself and a subsequent ion-implant step. The other notable feature of this architecture technology is the  $p^+/p$  non-injecting contact form in each unit cell of the FPA. These contacts are joined electrically by a top surface metal grid, which results in negligible de-biasing across a large area array, even at relatively high background flux currents. The fill factor of the unit cell is greater than 80%, with resulting unit cell collection efficiencies in excess of 60%.

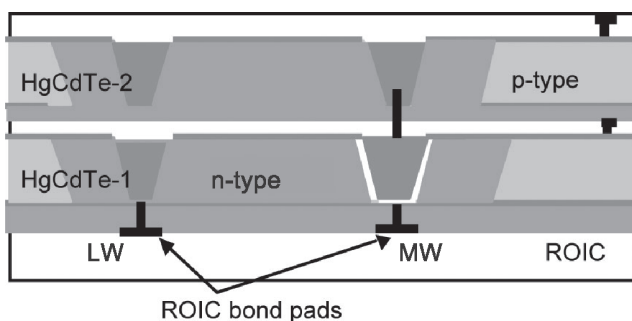


## HDVIP Photodiodes (cont.)

Liquid Phase Epitaxy (LPE) is used to grow the material, resulting in inherently purer HgCdTe epitaxial layers, and therefore the depletion region presents a smaller amount of threading dislocations. Applying the hybridization technique to silicon renders the device immune to rapid thermal cycling. Furthermore, the photodiode is front-side illuminated, leading to high quantum efficiencies and near theoretical MTF performance.

HDVIP technology is currently employed for production manufacturing of a wide variety of FPAs and detector-dewar cooler assemblies (DDCAs).

This technology has been extended to two-color by gluing two mono-color layers together into a composite and forming an insulated via hole through the lower layer in order to read out the upper color. This approach has been used to fabricate MW/MW and MW/LW 320 × 240 FPAs on a 50- $\mu\text{m}$  pitch, with co-located and simultaneous readout of the two colors using existing 640 × 480 ROIC. The two color signals are read out individually. This approach is readily adaptable to three or four colors.



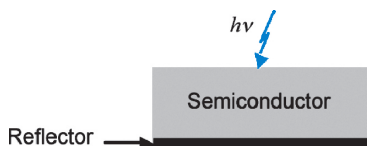
**Electron-injected avalanche photodiode (EAPD)** detectors are based on HDVIP architecture. This fairly new technology has been demonstrated to be one of the most promising paths to FPAs because of their exponential gains of 100–1000 for low values of reverse bias (5–10V), and high bandwidth associated with excess noise factors ( $f = \text{SNR}_{\text{in}}/\text{SNR}_{\text{out}}$ ) of 1.0–1.3.

## Uncooled HgCdTe Photodiodes

When operating at midwave and long wave, infrared photodetectors require cryogenic cooling to achieve useful performance. The need for cooling is a major limitation of these devices that prevents more widespread use of infrared technology. Thermal detectors operate at room temperature, but they cannot replace photon detectors in most applications due to their slow response and ineffectiveness in applications requiring multispectral detection.

There are some ways to improve the performance of photodetectors without cooling. One of them is the suppression of the Auger generation-recombination process. The Auger recombination and its counterpart impact ionization. It involves three particles, an electron, and a hole, which recombine in a band-to-band transition and give off the resulting excess kinetic energy to another electron-hole pair. A moderate p-type doping of the absorber region is widely used for some suppression of the Auger mechanism. Another way to improve the performance of uncooled infrared photon detectors is to reduce the physical volume of the semiconductor, thus reducing the amount of thermal generation. This must be done without decreasing the quantum efficiency, optical area, or FOV of the detector.

Thickness of the active region of the detector can be reduced significantly by enhancing the absorption of the radiation. This can be achieved by making use of multiple-pass radiation with a backside reflector. Even more efficient is the use of interference phenomena to set up a resonant cavity within the photodetector.



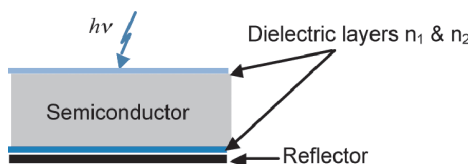
Interference occurs between the waves reflected at the rear and at the front surface of the semiconductor.

### Uncooled HgCdTe Photodiodes (cont.)

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The thickness of the semiconductor is selected to set up the thermal standing waves in the structure with peaks at the front and nodes at the back surface. The quantum efficiency oscillates with the thickness of the structure, with the peaks at a thickness corresponding to an odd multiple of  $\lambda/4n$ , where  $n$  is the refractive index of the semiconductor.

Higher gain can be obtained in structures immersed between two dielectric layers and supplied with a backside reflector. The gain in quantum efficiency increases with  $n$ . Increased absorption allows for improved collection efficiency in photodiodes since thickness can be smaller than the diffusion length.



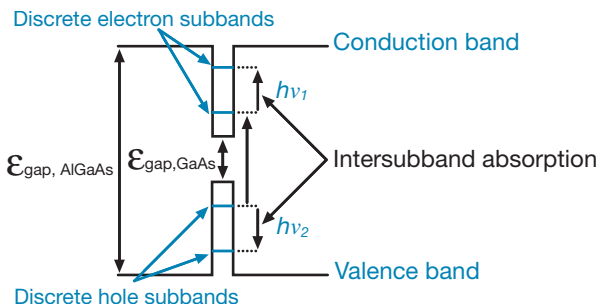
Interference effects modify the spectral response of the device, thus the gain within the optical cavity is achieved only in narrow spectral regions. Another limitation is that optical resonance occurs mostly for perpendicular incidence and is less effective for oblique incidence, thus limiting the use of these devices with fast optics.

Fundamental considerations dictate that HgCdTe is the optimal material for high-performance FPAs. The semiconductor's direct bandgap translates to high-absorptive efficiency, and the material's composition can be tailored to absorb infrared radiation at all infrared spectral regions. These attributes translate to flexibility and capability for BLIP detection. Unfortunately, insufficient demand to transition to high-rate production has led to difficulties in meeting the ultimate limit of HgCdTe detectors. Nevertheless, the fundamental performance limits of HgCdTe photodiodes have not been reached yet. Continued development of the in-situ epitaxial techniques will allow bandgap engineering devices of increasing quality and complexity.

## Quantum Well Infrared Photodetectors (QWIPs)

**Quantum wells (QWs)** are potential wells that confine particles that were originally free to move in 3D into 2D, forcing them to occupy a planar region. **Quantum well infrared photodetectors (QWIPs)** are infrared photodetectors made from semiconductor materials that contain one or more QWs. QWIP development is showing that the bulk of semiconductors addressing the 8–14  $\mu\text{m}$  spectral region (mainly HgCdTe) have a number of disadvantages in terms of fabrication and non-uniformity. Detection at these long wavelengths requires materials with small bandgaps, which are often difficult to grow and process. QWIP technology is based on large bandgap III-V semiconductor materials such as **GaAs**, which is simpler than HgCdTe, leading to high uniformity and reproducibility, as well as low-cost photodetectors.

QWs are formed in semiconductors by having a material, like GaAs, sandwiched between two layers of a material with a wider bandgap, like **AlGaAs**. These structures can be grown by molecular beam epitaxy (MBE) or chemical vapor deposition (CVD) with control of the layer thickness down to a monolayer ( $<10 \text{ \AA}$ ). QWIPs operate by the photoexcitation of carriers between the ground and first excited state subbands of a GaAs/AlGaAs QW.



**Multiple quantum wells (MQWs)** are engineered by placing as many as 50 alternating thin layers of two different, high-bandgap semiconductor materials. The bandgap discontinuity of these materials creates quantized intersubbands in the potential wells associated with conduction bands or valence bands.

## QWIPs (cont.)

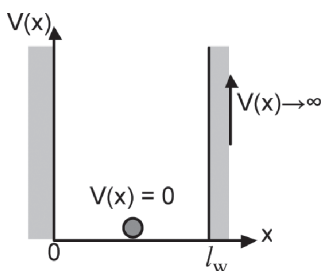
The position of these energy levels is established by adjusting the thickness and height of the QW. The design operating wavelength of intersubband photodetectors can be tailored from 3–20  $\mu\text{m}$  offering great flexibility. The lattice-matched GaAs/ $\text{Al}_x\text{Ga}_{1-x}\text{As}$  detector material is an excellent choice for QW structures because the bandgap of  $\text{Al}_x\text{Ga}_{1-x}\text{As}$  can be changed continuously by varying the composition index or molar ratio  $x$  (and hence the height of the quantum well). QW structures are designed so that the photo-excited carriers can escape from the QW and be collected as photocurrent.

Unlike intrinsic photodetectors, which use interband transition, QWIPs must be extrinsic in order for the photon to have sufficient energy to create photo-carriers (i.e.,  $h\nu > \mathcal{E}_{\text{gap}}$ ).

The effects of quantum confinement take place when the QW thickness approaches the **de Broglie wavelength  $\Lambda$**  of the carriers, leading to energy levels called “energy inter-subbands” (i.e., the carriers can only have discrete energy values).  $\Lambda$  is defined as

$$\Lambda = \frac{h}{2\pi p},$$

where  $h$  is Planck’s constant and  $p$  is the momentum of the electron. The condition for a bound energy state is that  $\Lambda/2$  must be an integral multiple of the well width, and that the energy must be smaller than the energy-band offset.



A heterostructure such as a QW is equivalent to a single point particle in an immovable box, which experiences zero potential energy. At the barriers of the well, the potential rises to infinity, thus constraining the particle to remain in the well.

### QWIPs (cont.)

The quantum behavior (boundary conditions) within the well includes energy quantization and zero-point energy (i.e., lowest possible ground state energy).

The idea of using multiple quantum well structures to detect IR radiation can be explained by using the basic principles of quantum mechanics and the Schrödinger equation. The time-independent Schrödinger equation for the particle inside the well (i.e.,  $V(x) = 0$ ) is given by

$$-\frac{h^2}{8\pi^2 m_e^*} \frac{d^2 \psi}{dx^2} = \mathcal{E} \psi,$$

where  $m_e^*$  is the effective mass of an electron and  $\psi$  is the eigenfunction of the system. The possible eigenfunctions and discrete energy levels for a QW with infinitely high barriers are obtained by solving the Schrödinger equation:

$$\psi_i = \sqrt{\frac{2}{l_w}} \sin\left(\frac{\pi x}{l_w} i\right) \quad \mathcal{E}_i = \frac{h^2}{8m_e^* l_w^2} i^2,$$

where  $i$  is an integer and  $l_w$  is the width of the well. The intersubband transition between the ground energy level and the first excited state is

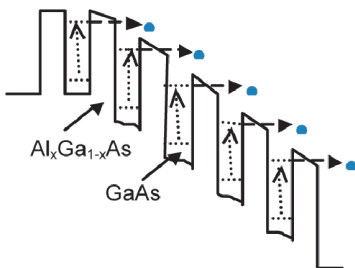
$$\Delta \mathcal{E} = \mathcal{E}_2 - \mathcal{E}_1 = \frac{3h^2}{8m_e^* l_w^2}.$$

The transition energy is inversely proportional to the square of the thickness of the well. The infinite potential-well solution gives us slightly higher energies than the energy levels of a finite potential well, but it is a good approximation to a more complex quantum problem.

When QWIPs are fabricated with thin barriers, the wavefunctions in neighboring wells overlap, causing energy-level perturbation and level shifting. The thin barriers allow tunneling of carriers between wells, providing a new degree of freedom in the material for optical properties. QWIPs are photoconductors, and their performance depends for the most part on minimizing the dark current (i.e., tunneling leakage and thermionic emissions), as well as improving the quantum efficiency.

## Types of QWIPs

**The bound-to-bound (B-B)** QWIPs contain two bound states. The interaction of a photon of energy  $\mathcal{E}_2 - \mathcal{E}_1$  and an electron in the ground state causes a transition to the first excited state. The two processes critical to the operation of this device as an IR detector are the intersubband absorption and tunneling. A bias voltage is applied across the device, changing the square wave potential to a staircase-type potential. This external electric field lowers the barrier height enough so that tunneling processes allow electrons to escape the well.



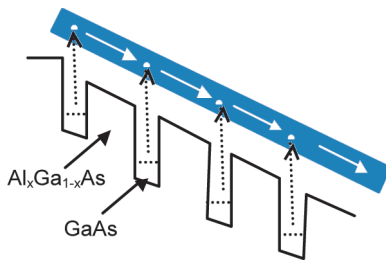
Once the electron has escaped the confines of the well, it accelerates toward the side with the applied voltage until it is captured by collision with a ground-state electron. The distance that the electron travels before it collides with the other electron is called the transport length  $L_t$ . If the kinetic energy of the first electron is large enough, it can cause the transition of the other electron to its first excited state. This process repeated many times down the chain of wells will cause an avalanche multiplication.

By reducing the width of the QW, it is possible to push the photo-excited carriers directly into the continuum, resulting in a strong **bound-to-continuum (B-C)** intersubband absorption. The major advantage of the B-C QWIP is that the photo-excited electrons can escape to the continuum transport without tunneling through the barrier. As a result, the bias required to efficiently collect photo-electrons can be decreased substantially, thus reducing the tunneling leakage dark current. In addition, the width of the  $\text{AlGaAs}$  barrier can be incremented without reducing the photo-electron collection, and hence lower the ground state sequential tunneling by an order of magnitude.

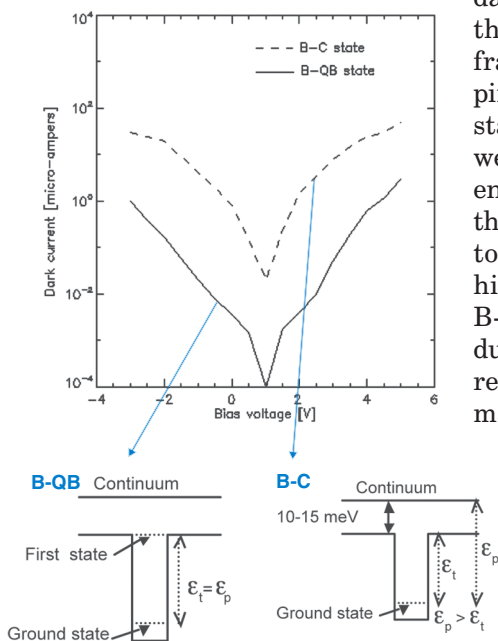


## Types of QWIPs (cont.)

Nevertheless, for temperatures over 45 K, the overall dark current in this type of detector is quite high since it is dictated by thermionic emissions of ground-state electrons directly into the energy continuum. This thermionic leakage of dark current depends exponentially on the bias voltage and it is governed by the Bose-Einstein probability distribution.



In **bound-to-quasi-bound (B-QB)** photodetectors, the design is engineered in such a way that the first excited energy state is at exactly the top of the well barriers of the QWs. Thermionic emissions dominate the production of dark currents within these long-wave infrared devices. Dropping the first excited state to the quantum well top causes the energy barrier of thermionic emissions to be 10–15 meV higher than in the B-C state, thus reducing the dark current by an order of magnitude.

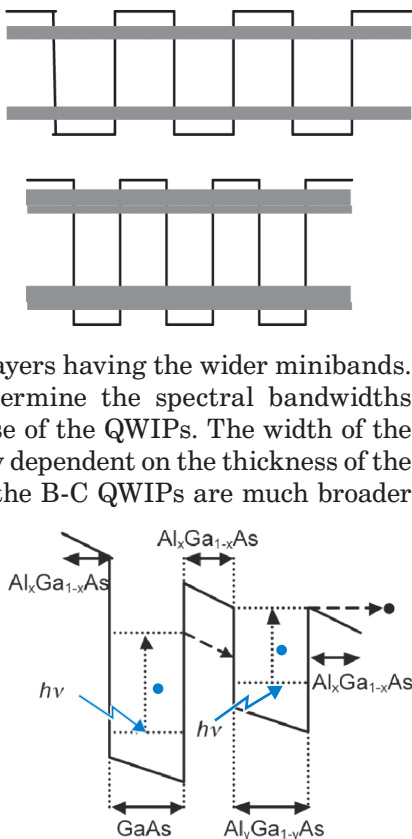


## Superlattices (SLs)

The wavefunctions decay exponentially within the barriers; however, if the barriers are thin enough (comparable to the de Broglie wavelength), perturbation will occur between the adjacent QWs.

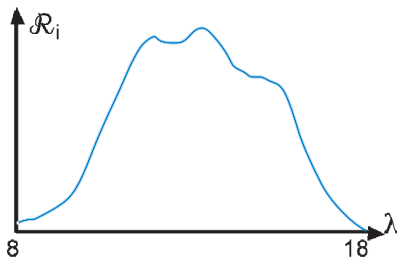
This perturbation causes the energy levels in the well to split by means of selection rules into two levels ( $\mathcal{E}$  and  $\mathcal{E} + \Delta\mathcal{E}$ ), where  $\Delta\mathcal{E}$  is the overlap energy resulting from the wavefunction interaction. The spread in energy levels (or minibands) varies with layer thickness, with thinner layers having the wider minibands. These energy bands determine the spectral bandwidths and the material response of the QWIPs. The width of the miniband is exponentially dependent on the thickness of the barriers. The spectra of the B-C QWIPs are much broader than the B-B or B-QB QWIPs, but their absorption is also significantly lower.

Large stark shifts can be obtained by introducing QWIPs with asymmetrical barriers. The electric field shifts of graded asymmetrical QWs are approximately an order of magnitude larger than those of symmetrical rectangular wells. They have much stronger bias-dependent behavior in escape probability, photoconductive gain, and in the responsivity spectrum. In addition, their spectral responsivity is narrower, and their cutoff wavelength is shorter compared to QWIPs operating in B-C transitions.



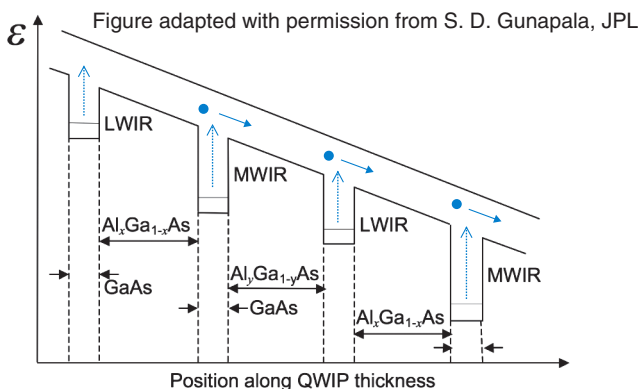
## Multispectral QWIPs

QWIPs can do typical narrow-band ( $\Delta\lambda/\lambda \approx 10\%$ ), two-band, three-band, four-band, and broadband detector arrays. These devices are built by stacking different quantum wells in a repeating sequence. Each sequence contains several quantum wells; the thicknesses and compositions of the barrier and well layers within each sequence can be tailored to optimize each well for a different pass band that partly overlaps one or more pass bands of the other wells in the sequence. The net effect of the multiple partial overlaps is a broader overall pass band or minibands for the QWIP device as a whole.



For example, the spectral responsivity of a device made with repeating three-well sequences and designed for peak responses at three different long wavelengths is shown.

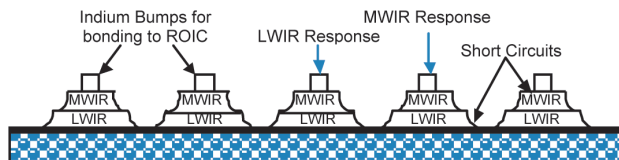
A QWIP designed to operate simultaneously at MWIR and LWIR spectral bands is shown below.



Two constraints arise with such a device: (1) the device must be supplied with two voltages, which cannot be acquired from the ROIC available currents; and (2) to switch on MWIR detection, a high bias must be supplied to the LWIR segment.

## Multispectral QWIPs (cont.)

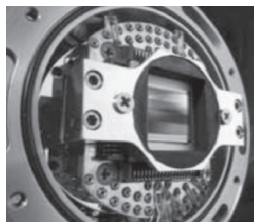
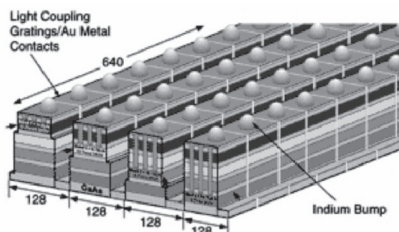
To overcome these difficulties, the device structure is built with  $(n + 1)$  terminal contacts per pixel to access the CMOS readout multiplexer. Then an interlace scheme is used to separate the two colors (i.e., odd-numbered rows for the MWIR QWIP and even-numbered rows for the LWIR QWIP). This interlaced short-circuiting of the MWIR and LWIR detectors eliminates complicated voltage tuning and the necessity for very high bias voltage to operate the LWIR QWIPs. The QWIP FPA is then hybridized to a CMOS multiplexer, which reads the alternating LWIR and MWIR rows, thus producing a two-color image of the same infrared scene target.



A four-band QWIP device structure with a  $640 \times 128$  pixel format of a four-band FPA is shown below. This device consists of a 15-period stack of  $4\text{--}5\text{ }\mu\text{m}$ , a 25-period stack of  $8.5\text{--}10\text{ }\mu\text{m}$ , a 25-period stack of  $10\text{ }\mu\text{m}$ , and a 30-period stack of  $13\text{--}15.5\text{ }\mu\text{m}$ . Each photosensitive MQW stack is separated by a heavily n-doped intermediate GaAs contact layer. The four-band QWIP structure is sandwiched between GaAs top and bottom n-doped contact layers and grown on a GaAs substrate. The light is incident from below.

Reprinted with permission from S. D. Gunapala, S. V. Bandara, et al., "640x512 pixel long-wavelength infrared narrowband, multiband, and broadband QWIP focal plane arrays," *IEEE Trans. Elect. Dev.* 50, 2353–2360 (2003).

Reprinted with permission from R. Sweda, "QWIPs: Multispectral mine clearance and medical," *III-Vs Review* 18(2), 44–47 (2005).



A QWIP's wavelength coverage can be bandgap engineered from  $3\text{--}20\text{ }\mu\text{m}$  spectral range. The longer wavelength limit can easily be extended using shallow quantum wells.

## Light Couplers

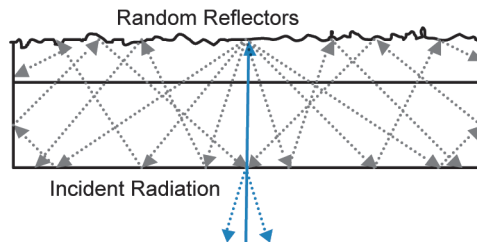
The structure of a QWIP FPA poses an optical-coupling problem because of a confluence of three considerations:

1. The direction through the thicknesses of the quantum wells is normal to the focal plane.
2. Quantum selection rules allow the detection of only that part of the incident light that is transverse electric along the direction through the thicknesses of the quantum wells (i.e., S-polarized to the focal plane).
3. The light to be detected is incident at approximately 90 deg to the focal plane, and thus only a small fraction of it is absorbed by the QWIP.

A QWIP device is fabricated so that the planes that bound the quantum-well layers are parallel to the broad outermost faces of the device; therefore, by virtue of this quantum mechanical selection rule, light incident normal to these faces is not detected. A light coupler is required to redirect incident light so that it traverses the QWIP layers in a direction that includes at least some vector component parallel to the planes that make the quantum wells. In other words, the function of a light coupler on a QWIP is to redirect normally incident light to oblique incidence.

### Random Reflectors

The purpose of random reflectors in QWIP light couplers is to divert the incoming light, as efficiently as possible, to propagate at large angles from the perpendicular.



Light that is not absorbed is reflected to a different structure on each bounce, thus improving the capability of the QWIP to trap light and absorb a larger number of photons.

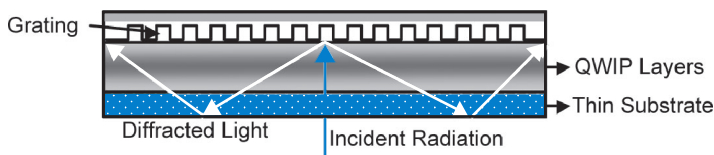
## Light Couplers (cont.)

Random reflectors are fabricated using lithographic processes, allowing feature sizes to be controlled to preserve pixel-to-pixel uniformity. The sizes of the random facets are proportionally scaled with the peak response wavelength of the QWIP; in which case, fewer sharp features are obtained for QWIPs operating in the mid-wave part of the spectrum, thus lowering the coupling efficiency of the device. Therefore, 2D periodic gratings are used for light coupling in the midwave infrared waveband.

### 2D Periodic Gratings

In 2D gratings, the periodicity repeats in two perpendicular directions on the detector plane, leading to the absorption of both polarizations of incident infrared radiation. Because a 2D grating exhibits the same periodicity along the  $x$ - $y$  directions in the focal plane array, it diffracts more light to higher angles, giving rise to increased coupling.

The difficulties in designing a grating for optical coupling are in choosing the proper dimensions of the grating to maximize the efficiency of the diffractive orders and in attaining suitable angles away from the perpendicular. Further improvement is attained when a thin substrate of AlGaAs is attached to the cavity and produces total internal reflection (TIR). The grating peak wavelength depends linearly on the grating period, but is independent of the depth of the groove.



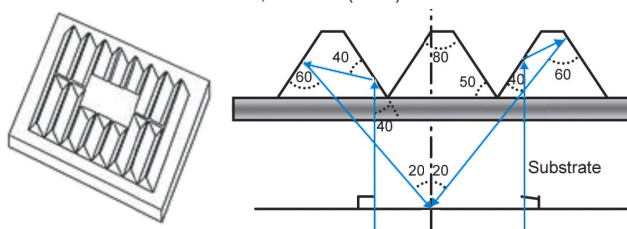
The standard approach of using gratings at the top of detectors for light diffraction has several shortcomings. For example, diffraction is less efficient for small detector pixels, which limits the array's spatial resolution. Also, each grating design is only effective for a specific wavelength, not for dual-band detection or broadband detection. The fine features in the grating design require stringent electron-beam lithography, which limits the production of these quantum devices and increases their cost.

## Light Couplers (cont.)

### Corrugated QWIPs (CQWIPs)

CQWIPs can be created within the detector design to allow total internal reflection to redirect light inside a detector. V-grooves are chemically etched through the active detector region down to the bottom contact layer to create a collection of angle facets within a single detector pixel. The normally incident light reflects at these angled sidewalls and travels at large oblique angles, allowing for efficient light absorption. Based on reflection rather than diffraction, the corrugated coupling scheme is both pixel-sized and wavelength-independent. These two important features improve the normalized detectivity of these corrugated devices by an order of magnitude.

Reprinted with permission from K. K. Choi et al.,  
 "Corrugated QWIP array fabrication and characterization,"  
*Proc. SPIE* **3287**, 118–127 (1998).

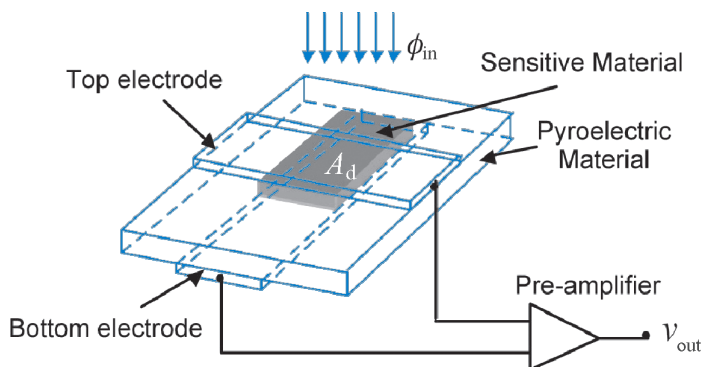


CQWIP structures expose the active layer; therefore, top metal contact cannot be used directly into the detector pixel. Two solutions are considered to address this problem: leave a reserved unetched center island for contact bonding, or isolate the active layers with dielectric polymers such as pyrozone and pyrodium.

CQWIPs are appropriate for both dual-band and broadband detection. Since the corrugations are created during the separation of the detector pixels, no extra processing steps are required for the fabrication of these corrugations. CQWIP devices increase the absorption and improve the sensitivity of the detector IRFPA. CQWIP production is rather simple and of low cost.

## Pyroelectric Detectors

**Pyroelectric detectors** are made of monoclinic ferromagnetic crystals that have a high degree of crystalline asymmetry, and therefore possess an electric dipole moment. Incident radiation heats the device, expanding the crystalline lattice and changing the dielectric constant, which alters the internal dipole moment, thus producing a change in the surface charge density and in the polarization state of the material. This affects the charge on the mounted metal electrodes and causes a current to flow in an external circuit. The electrodes must be placed on opposite faces of the crystal perpendicular to the polarization axis.



At a constant sample temperature, this internal polarization is neutralized by mobile charges on the surface of the sample; thus there is no voltage difference measured by the metal electrodes. A change in irradiance causes a temperature variation, thus producing a change in polarization, and consequently a measurable voltage difference occurs at the detector's output. As a result, pyroelectric detectors based on ferroelectric materials can only be used in an ac mode. In order to detect these small charges, low-noise high-impedance amplifiers are necessary.

The surface temperature is modulated in time by means of a radiation chopper. The chopper provides a continuous source of temperature variation that allows the material to generate charge while a scene is monitored.



## Pyroelectric Detectors (cont.)

This differential aspect provides a real-time calibration of offset non-uniformity, making the sensor essentially unaffected by severely dynamic thermal environments. High chopping frequencies (usually 25–60 Hz), are supported by pyroelectric devices in exchange for sensitivity. The polarization change in these ac-coupled devices is quite fast, but the thermal properties of the detector limit the high-frequency response.

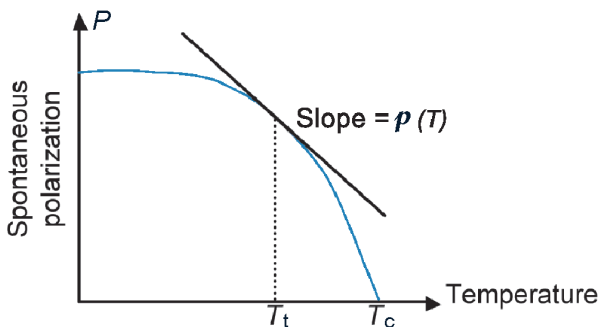
The pyroelectric coefficient  $\rho$  measures the rate of change of electric polarization with respect to temperature:

$$\rho = \frac{dP}{dT}, \quad \left[ \frac{\text{Coul}}{\text{cm}^2\text{K}} \right],$$

where  $P$  is the magnitude of internal polarization, and  $T$  is the temperature in Kelvin. An applied field induces an electrical displacement  $\mathbf{D}$  resulting in a more general coefficient:

$$\rho(\mathbf{E}) = \frac{d\mathbf{D}}{dT}.$$

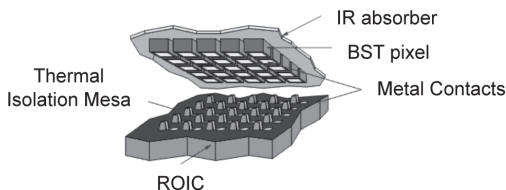
This produces a stable response in ferroelectric materials operated below and above their temperature phase transition (i.e., the Curie temperature  $T_c$ ). Once the material is heated above the Curie temperature, the ferroelectric material produces large pyroelectric coefficients.



In some cases, an absorption layer is applied in front of the responsive element to improve its absorption behavior.

## Pyroelectric Detectors—Mathematical Approach

At present, most of the pyroelectric FPA sensors are made from barium strontium titanate (BST), a crystal of ferroelectric ceramic material hybridized to a CMOS multiplexer circuit and mounted in contact with a thermoelectric cooler/stabilizer. The device's temperature set point is set close to the BST crystal's Curie temperature, or the narrow temperature band at which the crystal material exhibits the highest polarization relative to the detector's temperature change resulting from scene illumination.



Pyroelectric sensing materials are sensitive from  $0.2\text{ }\mu\text{m}$  to over  $100\text{ }\mu\text{m}$ , and their spectral range is limited only by the infrared window placed in front of the detector.

To investigate how a pyroelectric material produces an electrical current, it is good to start with Maxwell's equation known as Gauss's law in its differential form:

$$\nabla \cdot \mathbf{D} = \rho,$$

where  $\mathbf{D}$  is the electric displacement, and  $\rho$  is the electric charge. From elementary electrostatics,

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P},$$

where  $\mathbf{E}$  is the electric field,  $\mathbf{P}$  is the polarization, and  $\epsilon_0$  is the permeability of the material. Below the Curie temperature,  $\mathbf{P}$  is large compared to  $\epsilon_0 \mathbf{E}$ , thus  $\mathbf{D}$  and  $\mathbf{P}$  can often be used interchangeably, in which case

$$\nabla \cdot \mathbf{P} = \rho.$$

Applying Green's theorem and integrating over a cylindrically shaped volume that covers the area of one side of the detector that has an electrical lead, we arrive at the following equation:

### Pyroelectric Detectors—Mathematical Approach (cont.)

$$\int_{\text{Vol}} \rho dV = \int_{\text{Vol}} \nabla \cdot \mathbf{P} dV = \int_{\text{Sur}} \mathbf{P} dA_d = \mathbf{P} A_d.$$

When the ferroelectric material is aligned so that the intrinsic polarization is perpendicular to the surface through which the current flows, the detector current is

$$i = \frac{d\mathbf{P}A_d}{dt} = A_d \frac{d\mathbf{P}}{dT} \frac{dT}{dt} = A_d \rho \frac{dT}{dt}.$$

From this equation, it is clear that current flows through the device only when a temperature change  $dT/dt$  occurs, thus resulting in an ac detector.

The temperature difference  $\Delta T$  between the ferromagnetic material and the heat sink is related to the incident radiation by the heat balance equation:

$$H \frac{d(\Delta T)}{dt} + K\Delta T = \varepsilon \phi_e(t),$$

where  $K$  is the thermal conductance,  $H$  is the heat capacity,  $\varepsilon$  is the emissivity of the surface material, and  $\phi_e$  is the flux in energy-derived units.

Since the radiation is modulated at an angular frequency  $\omega$ , it can be expressed as an exponential function  $\phi_e = \phi_{e,0} e^{j\omega t}$ , in which case the heat balance equation provides the following solution:

$$|\Delta T| = \frac{\varepsilon \phi_{e,0} R_{th}}{\sqrt{1 + (\omega \tau_{th})^2}} \Rightarrow \tau_{th} \equiv \frac{H}{K} = R_{th} C_{th},$$

where  $\tau_{th}$  is the thermal time constant, and  $C_{th}$  and  $R_{th}$  are the thermal capacitance and thermal resistance respectively of the ferroelectric material.

Recognizing that  $\frac{d(\Delta T)}{dt} = \frac{dT}{dt}$ , we obtain

$$\left| \frac{dT}{dt} \right| = \frac{\omega \varepsilon \phi_{e,0} R_{th}}{\sqrt{1 + (\omega \tau_{th})^2}}.$$

The detector current flowing through the pyroelectric is

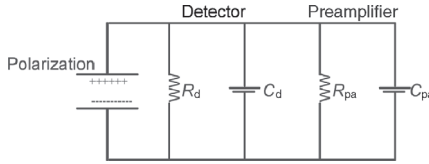
## Pyroelectric Detectors—Mathematical Approach (cont.)

$$i = \frac{\rho \omega \varepsilon \phi_{e,o} R_{th} A_d}{\sqrt{1 + (\omega \tau_{th})^2}}.$$

Thus the current responsivity is defined as

$$\mathcal{R}_i = \frac{i}{\phi_{e,o}} = \frac{\rho \omega \varepsilon R_{th} A_d}{\sqrt{1 + (\omega \tau_{th})^2}}.$$

The equivalent circuit of a pyroelectric element is shown:



The voltage across the pyroelectric device is the current times the parallel electrical impedance of the detector/preamplifier combination, yielding

$$v = \frac{iR}{\sqrt{1 + (\omega \tau_{RC})^2}},$$

where  $\tau_{RC}$  is the electrical time constant. The voltage responsivity is simply

$$\mathcal{R}_v = \frac{\rho \omega \varepsilon R_{th} R A_d}{\sqrt{1 + (\omega \tau_{th})^2} \sqrt{1 + (\omega \tau_{RC})^2}}.$$

At high frequencies, the voltage responsivity is inversely proportional to frequency, while at low frequencies it depends on the electric and thermal constants.

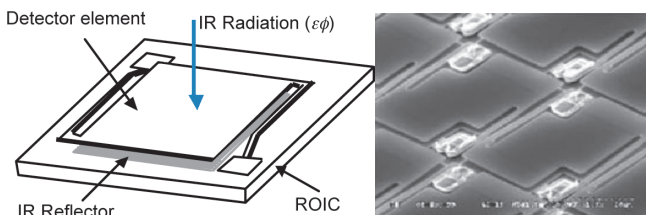
The dominant noise in pyroelectric detectors is most commonly Johnson noise, so the normalized detectivity may be calculated by

$$D^* = \frac{\mathcal{R}_v \sqrt{A_d \Delta f}}{v_j} = \frac{\rho \omega \varepsilon (R_{th} A_d) \sqrt{R A_d}}{\sqrt{4kT} \sqrt{1 + (\omega \tau_{th})^2} \sqrt{1 + (\omega \tau_{RC})^2}}.$$

## Microbolometers

Microbolometers are thermistor-based detectors that measure temperature change in a thermally isolated membrane, heated by infrared radiation absorbed on the surface. **Micro-electro-mechanical systems (MEMS)** technology is used to provide good thermal isolation from the silicon read-out integrated circuit (ROIC) substrate, and to obtain a high fill factor as well as good integrated circuit (IC) process compatibility.

Starting with a large silicon wafer, CMOS multiplexer (MUX) circuitry is fabricated by a commercial silicon foundry. The infrared detector foundry then deposits successive layers of silicon nitrate ( $\text{Si}_3\text{N}_4$ ) and silicon dioxide ( $\text{SiO}_2$ ) to fabricate a bridge structure. A layer of vanadium oxide ( $\text{VO}_x$ ), commonly known as vanadium pentoxide ( $\text{VO}_5$ ), amorphous silicon (a-Si), or yttrium-barium-copper-oxide ( $\text{YBa}_2\text{Cu}_3\text{O}_5$ ) is used to form a thin-film resistor over the surface of the bridge deck. Integral thin legs are released from the successive layers of  $\text{SiO}_2$ . These legs provide structural support to the detector material and electrical contact between the CMOS MUX and the film resistor on the bridge deck. Once all of the layers necessary to build the complete pixel structure have been deposited, a gas is used to dissolve away the sacrificial materials, and a freestanding suspended bridge is created. The wafer with individual detectors is tested, and the functioning parts are diced and packaged into vacuum (typically  $10^{-6}$  Torr) for thermal insulation. The thermal conductance of the isolating legs can be adjusted to match the desired time constant (i.e., 30/60 Hz frequencies).



The optical cavity is tuned for maximum absorption at a wavelength  $\lambda_m = 4nd$ , where  $d$  and  $n$  are respectively the thickness and index of refraction of the microbolometer material.

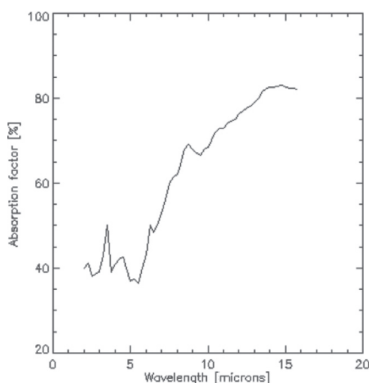
## Microbolometers (cont.)

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The infrared reflecting surface at the bottom of the optical cavity improves the thermal response of the device.

The two most commonly used IR radiation detecting materials in microbolometers are  $\text{VO}_x$  and a-Si. Other materials that have been investigated are a-Si carbide,  $\text{YBaCuO}$ , poly SiGe, and  $\text{GeSiO}$ . These materials exhibit large thermal resistance coefficients (TRCs) of  $2\text{--}5\% \text{K}^{-1}$ . Microbolometer technology requires the deposition of these high-TRC materials after CMOS fabrication. The post-CMOS surface micromachining processing is complicated and expensive, thus increasing the cost of these devices. Furthermore, one of the difficulties of microbridge structures is the limited optical fill factor ( $\sim 65\%$ ), because the detector area is reduced by the support legs.

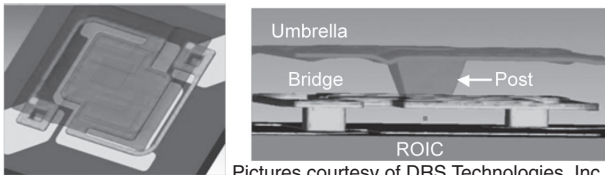
The pixel structure of the basic microbolometer FPAs allows a substantial percentage of infrared radiation to pass through and exit the TRC material without being absorbed. As a consequence, the decrement in performance of the thermal device is rather significant.



Basic microbolometer structures of conventional uncooled FPAs have been recently improved by DRS Sensors and Targeting Systems. The DRS approach consists of placing a high-fill-factor absorber ( $>90\%$ ) with a high resistivity on top of the microbridge. This patented top layer, called the “umbrella” structure, enhances the absorption in both MWIR and LWIR spectral regions.

### Microbolometers (cont.)

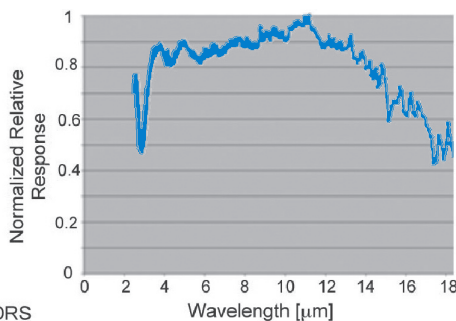
The absorption structure is formed of a layer of nickel-chromium (NiCr), or other highly resistive alloy or metal, sandwiched between two layers of  $\text{SiO}_2$  and supported by a central post, also formed of  $\text{SiO}_2$ . This post serves as a thermal connection between the absorbing layer composition and the transducer bridge.



Pictures courtesy of DRS Technologies, Inc.

The umbrella pixel structural design is strongly absorptive to radiation of a predetermined spectral band. It can be tailored to control the thermal conductance, heat capacitance, and to maximize the absorption of infrared radiation, thus minimizing the noise contributions and maximizing the responsivity. The combined spacing of the two existing gaps in this structure is  $\lambda/4n$ .

This innovative uncooled technology is suitable to absorb broadband radiation (i.e., MWIR and LWIR simultaneously). The maximum normalized spectral response of a  $17\text{-}\mu\text{m}$  pixel from  $3\text{--}18\text{ }\mu\text{m}$  was measured with a modified FTIR (below).



Courtesy of DRS

The most common array formats in the infrared market based on microbolometer technology are  $160 \times 120$  (low cost),  $320 \times 240$  (most common), and  $640 \times 480$  (high performance).

## Microbolometers—Mathematical Approach

The resistance in a microbolometer semiconductor material has an exponential dependence on temperature:

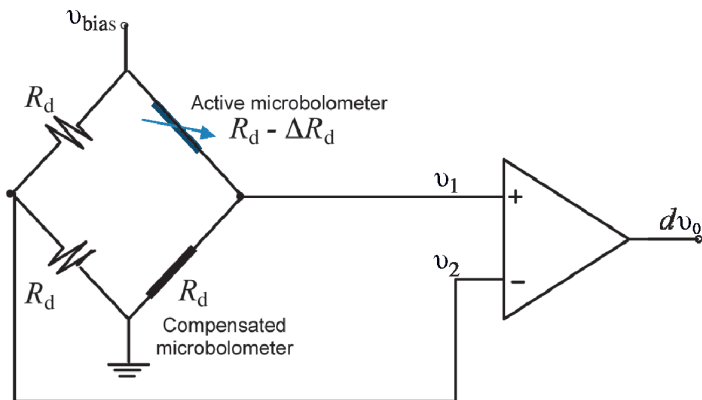
$$R_d = R_o e^{\left(\frac{1}{T} - \frac{1}{T_o}\right)},$$

where  $R_o$  is the ambient resistance at a nominal temperature  $T_o$ , and  $\mathcal{C}$  is a material characteristic constant. The resistance change that results from the optically induced temperature change is obtained by differentiation, yielding a TRC of

$$\alpha = \frac{1}{R_d} \frac{dR_d}{dT} = -\frac{\mathcal{C}}{T^2}.$$

When infrared radiation is absorbed into the microbolometer material, its temperature rises slightly, causing a small decrease in resistance. In order to produce an electrical current from this change in resistance, a bias voltage must be applied across the sensing device. This is accomplished by interfacing two identical microbolometer chips into a Wheatstone bridge.

When infrared radiation is incident on the active microbolometer, the detector resistance  $R_d$  decreases by  $\Delta R_d$ , thus unbalancing the bridge and producing an electrical current through the active device. The compensated microbolometer is shielded from infrared radiation and the operational amplifier amplifies the output signal.





### Microbolometers—Mathematical Approach (cont.)

Setting up the heat balance differential equation that relates the incident radiation and the heat loss,

$$H \frac{d\Delta T}{dt} + K \Delta T = \varepsilon \phi_e,$$

where  $H$  is the heat capacity [ $\text{J}\cdot\text{K}^{-1}$ ],  $K$  is the thermal conductance [ $\text{W}\cdot\text{K}^{-1}$ ],  $\varepsilon$  is the emissivity of the surface, and  $\phi_e$  is the IR flux in watts. Assuming a periodic radiant power incident on the microbolometer is sinusoidally varying (i.e.,  $\phi_e = \phi_{e,o} e^{j\omega}$ ); the heat balance differential equation provides the following solution:

$$|\Delta T| = \frac{\varepsilon \phi_{e,o}}{K \sqrt{1 + \omega^2 \tau_{th}^2}} \Rightarrow \tau_{th} \equiv \frac{H}{K} = R_{th} C_{th},$$

where  $\tau_{th}$  is the thermal time constant,  $R_{th}$  is the thermal resistance, and  $C_{th}$  is the thermal capacitance.

The transfer function of the electronic circuit is

$$dv_o = \frac{v_{bias}}{2} \frac{\Delta R_d}{2R_d - \Delta R_d} = \frac{v_{bias}}{2} \frac{\alpha \Delta T}{2 - \Delta T},$$

where  $dv_o$  is the change in output voltage due to the change in detector resistance  $\Delta R_d$  given by

$$\Delta R_d = \frac{dR_d}{dT} \Delta T = \alpha R_d \Delta T.$$

Substituting  $\Delta T$ , the radiation-induced change in output voltage  $dv_o$  is

$$dv_o = \frac{\alpha \varepsilon \phi_{e,o} v_{bias}}{4K \sqrt{1 + \omega^2 \tau_{th}^2} - 2\varepsilon \phi_{e,o}}.$$

The voltage responsivity is obtained by dividing both sides of the previous equation by  $\phi_{e,o}$ :

$$\mathcal{R}_v = \frac{\alpha \varepsilon v_{bias}}{4K \sqrt{1 + \omega^2 \tau_{th}^2} - 2\varepsilon \phi_{e,o}}.$$

While photon detectors are subject to BLIP conditions, microbolometer detectors are subject to temperature and background fluctuation noise limits.

## Microbolometers—Mathematical Approach (cont.)

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Whatever the heat-loss mechanisms (i.e., convection, conduction, and/or radiation) are, the ultimate sensitivity  $D^*$  is set by the temperature fluctuation noise

$$D_{\text{TF}}^* = \sqrt{\frac{\varepsilon A_d}{4kKT_d^2}}.$$

If the microbolometer is so well thermally isolated that the only heat mechanism is by radiation from the pixel to its surroundings, the thermal device is said to be background fluctuation noise limited. In this case the sensitivity of the device is given by

$$D_{\text{BF}}^* = \sqrt{\frac{\varepsilon}{8k\sigma(T_d^5 + T_{\text{bkg}}^5)}}.$$

The sensitivity is independent of the responsivity. The value of having a high responsivity is to minimize the contribution of the total system noise from the electronics in the ROIC from the focal plane array (FPA). Obviously, if the noise is dominated by the electronic noise, the microbolometer cannot be either fluctuation or background noise limited.

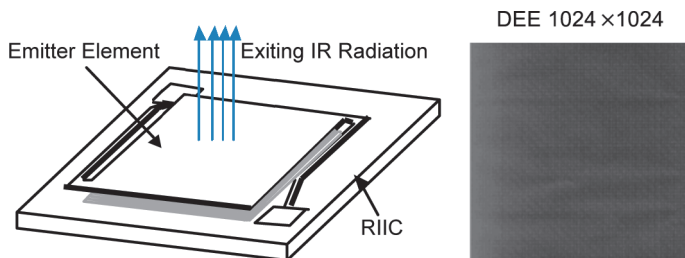
### Design elements for consideration in microbolometer performance:

- Readout noise from the CMOS devices.
- The thermal coefficient of resistance (TCR).
- Thermal isolation of the active device from the substrate.
- The vacuum level in the sealed detector package, eliminating the convective transfer of energy between pixels.
- Detector time constant or response time. Even though the detector must be thermally isolated, it must be able to transfer some energy to the substrate so that the device does not heat up due to constant radiation impinging on it. The thermal constant must be optimized to provide both high sensitivity and a small integration time.

## Infrared Dynamic Scene Simulators

What would happen if the process of a microbolometer were reversed? In this case, the resistive element would be connected to a read-in integrated circuit (RIIC) instead. It would result in an emitter array instead of the classical detector FPA, which is the principle of operation of a **dynamic scene simulator (DSS)**. The emitter element is called the digital emitter engine (DEE). These types of devices are used in laboratory dynamic infrared scenarios, thus reducing the cost of field testing. They also offer shorter development cycles for the infrared-based imaging systems by giving developers more immediate access to data.

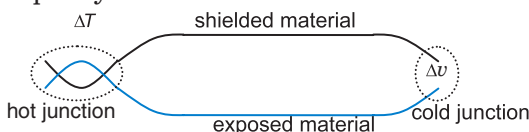
For example, the MIRAGE XL DSS from Santa Barbara Infrared, Inc. (SBIR), is an emitter array with a resolution of  $1024 \times 1024$  pixels. Its RIIC is designed to be mechanically mated to the suspended membrane, resulting in resistively heated micro-emitters. Each micro-emitter element is electronically connected to one of the  $1024 \times 1024$  RIIC unit cells, allowing individual control of the current driven through each emitter element during scene generation. As in the microbolometer, the electrical connection to the emitter from the RIIC also serves as the mechanical support for the emitter structure, and provides the thermal path for the heat flow during operation.



To generate the scene, the RIIC must be supplied with required bias voltage, timing, and formatter digital scene data (e.g., DVI digital video or NTSC/PAL analog video).

## Thermoelectric Detectors

A **thermoelectric detector** or **thermocouple** comprises two junctions between two dissimilar conductors having a large difference in their thermoelectric power (i.e., a large **Seebeck coefficient**  $\Theta$ ). The hot junction is an efficient absorber exposed to incident radiation, while the cold junction is purposely shielded.



The diagram above displays the Seebeck effect where two semiconductor materials are joined together at a hot point and a temperature,  $\Delta T$  is maintained between the hot and cold junctions; an open-circuit voltage  $\Delta v$  is developed between these materials at the cold point:

$$\Delta v = \Theta \Delta T \rightarrow \Theta \text{ } [\mu \text{V/K}]$$

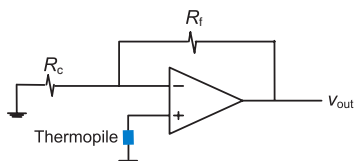
To obtain efficient and large electrical conductivity  $\sigma$ , both the thermal conductivity  $K$  and the Joulean heat loss must be minimized. This is achieved by maximizing the coefficient  $\sigma\Theta^2/K$  found in some heavily doped semiconductors.

The voltage output between the two dissimilar materials is increased by connecting numerous thermocouples in series; a device called a radiation **thermopile**. The responsivity of a thermopile is given by

$$\mathcal{R}_v = \frac{\varepsilon \Theta N}{K \sqrt{1 + \omega^2 \tau_{th}^2}},$$

where  $N$  is the number of thermocouples in electrical series.

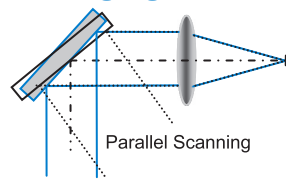
The thermopile device may then be interfaced to an operational amplifier circuit to increase the voltage to usable levels.



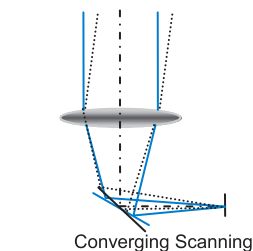
Thin-film techniques enable chip thermopiles to be fabricated as complex arrays with good reliability.

## Raster Scan Format: Single-Detector

Scanning mechanisms are often necessary in infrared systems to cover a 2D FOV with a reasonable number of detector elements, or when a substantial FOV is required. The many applications that need scanning usually depend on opto-mechanical elements to direct and focus the infrared radiation. There are two basic types of scanners: preobjective or **parallel beam scanners**, and post-objective or **converging beam scanners**.

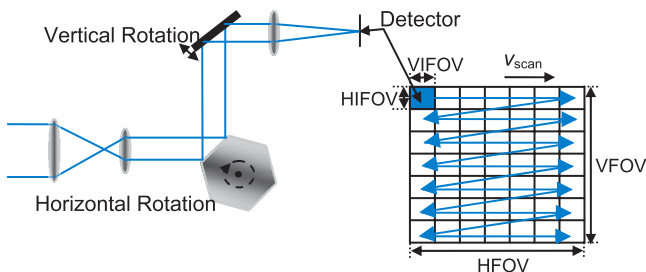


In parallel scanning, the scanning element is out in front of the final image-forming element, and must be the entrance pupil of the optical system. The converging scanner has the moving element between the final optical element and the image, and works on axis.



There are three basic scan formats: raster, parallel, and the staring focal plane array.

In the raster scan mechanisms, a single detector is scanned in two orthogonal directions in a 2D raster across the FOV. The moving footprint sensed by the detector is called the **instantaneous field of view (IFOV)**, and the specific time required for the footprint to pass across the detector is called the **dwell time** ( $\tau_{\text{dwell}}$ ).



One-hundred percent scan efficiency ( $\eta_{\text{scan}}$ ) is assumed. Scan inefficiencies include overlap between scan lines, over scanning of the IFOV beyond the region sensed, and finite retrace time to move the detector to the next line.

### Raster Scan Format: Single-Detector (cont.)

The number of horizontal lines to make up a 2D scene is given by

$$n_{\text{lines}} = \frac{\text{vertical field of view (VFOV)}}{\text{vertical instantaneous field of view (VIFOV)}}.$$

The time taken to scan one particular line is

$$\tau_{\text{line}} = \frac{\tau_{\text{frame}}}{n_{\text{lines}}} = \tau_{\text{frame}} \frac{\text{VIFOV}}{\text{VFOV}}.$$

The dwell time is the line time divided by the number of horizontal pixels contained in that line:

$$\tau_{\text{dwell}} = \frac{\tau_{\text{line}}}{\text{horizontal field of view (HFOV)} / \text{horizontal instantaneous field of view (HIFOV)}}.$$

The scan velocity and the dwell time can be written as

$$v_{\text{scan}} = \frac{\text{HFOV}}{\tau_{\text{line}}} \Rightarrow \tau_{\text{dwell}} = \frac{\text{HIFOV}}{v_{\text{scan}}}.$$

The dwell time can be also interpreted as the frame time divided by the total number of pixels within the 2D FOV:

$$\tau_{\text{dwell}} = \frac{\tau_{\text{frame}}}{(\text{VFOV}/\text{VIFOV}) \cdot (\text{HFOV}/\text{HIFOV})} = \frac{\tau_{\text{frame}}}{n_{\text{pixels}}},$$

where the frame time can be found by

$$\tau_{\text{frame}} = n_{\text{lines}} \cdot \tau_{\text{line}} = \frac{\text{VFOV}}{\text{VIFOV}} \cdot \frac{\text{HFOV}}{v_{\text{scan}}}.$$

The electronic bandwidth can be written in terms of the dwell time as

$$\Delta f = \frac{1}{2\tau_{\text{dwell}}}.$$

A scanning system that covers the entire FOV with a single detector considerably lowers the duration that the sensing element remains on a particular IFOV; hence resulting in a higher noise-equivalent bandwidth.

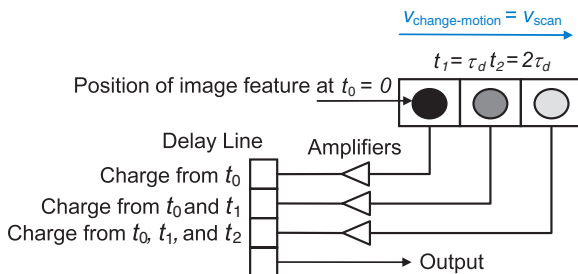
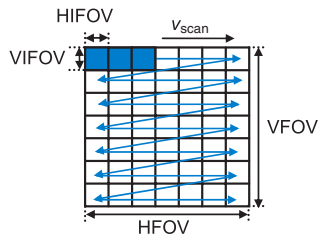
A longer dwell time is obtained using a multiple-detector system. In this case, the noise is reduced by the square root of the number of sensing elements, thus improving the SNR of the infrared system.

## Multiple-Detector Scan Formats: Serial Scene Dissection

Serial scan uses multiple sensors along the scan direction in such a way that each point in the image is scanned by all detectors.

The number of detectors used in a practical system varies between two and ten.

The main mechanism used to implement a serial scan is called **time delay and integration (TDI)**. TDI requires a synchronized delay line (typically a charge-coupled device), to move the signal charge along with the optical-scan motion. A particular IFOV is viewed  $n_d$  times;  $n_d$  being the number of detectors in series dissecting the overall system's FOV of the scanned system. The output charge from each detector is added together as the serial scan moves on. As a result, the amplitude of the added charge signal is incremented  $n_d$  times, and the uncorrelated noise, since it is added in quadrature, is intensified by a factor of  $\sqrt{n_d}$ . The overall SNR of the system is thereby improved by the square root of the number of sensor elements.



**Advantage:** The nonuniformity of the detectors is improved.

**Disadvantages:** High mirror speeds are necessary to cover the 2D FOV. TDI circuit increases the weight and power of the electronics subsystem.

**Assumption:** The  $\sqrt{n_d}$  increment in the SNR assumes that all the detectors are identical in noise and level responsivity. The practical result is around 10% short of the ideal.

## Multiple-Detector Scan Formats: Parallel Scene Dissection

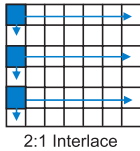
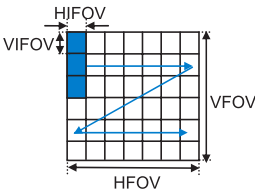
**Parallel scanning** uses multiple sensors in cross-scan directions. For a given fixed frame time, a slower scan is used since multiple vertical lines are covered at once.

If  $n_d < VFOV/VIFOV$ , a 2D raster is required, with the scan format such that any given detector will drop  $n_d \times VIFOV$ . If there are sufficient detectors to cover a full line-only horizontal scan motion is required.

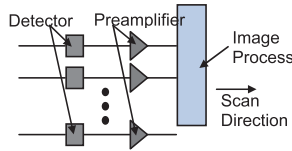
**Advantage:** Lower mirror speeds are required.

**Disadvantage:**  $D^*$  variations produce image nonuniformities.

In second-generation forward-looking infrared (FLIR) imagers, TDI/parallel scanning is used to perform 2:1 interlacing. A full line is stored and summed to the next line. Here, TDI is applied along the scan direction, and all the detectors are preamplified, processed, and displayed.



2:1 Interlace



For a system with a fixed frame time, an  $n_d$  sensor has a time line of

$$\tau_{\text{line}} = \frac{\tau_{\text{frame}}}{n_{\text{lines}} / n_d} = \tau_{\text{frame}} n_d \frac{VFOV}{VIFOV},$$

where longer dwell time is achieved by a factor of  $n_d$  yielding

$$\tau_{\text{dwell}} = \frac{\tau_{\text{line}}}{HFOV/HIFOV} = \frac{\tau_{\text{frame}} n_d}{(VFOV/VIFOV) \cdot (HFOV/HIFOV)} = \frac{\tau_{\text{frame}} n_d}{n_{\text{pixels}}}.$$

Bandwidth decreases inversely proportional to  $n_d$ , and the noise is proportional to the square root of the bandwidth, yielding

$$\Delta f = \frac{1}{2\tau_{\text{dwell}}} = \frac{n_{\text{pixels}}}{2\tau_{\text{frame}} n_d} \Rightarrow v_n \propto \sqrt{\Delta f} = \sqrt{\frac{n_{\text{pixels}}}{2\tau_{\text{frame}} n_d}}.$$

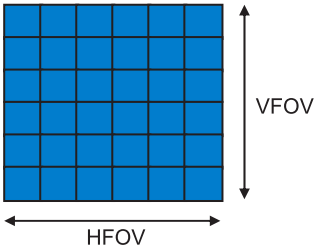
Therefore, the overall SNR increases by  $\text{SNR} \propto \sqrt{n_d}$ .



Staring Systems

**Staring systems** cover the 2D FOV in its entirety so the number of detector elements is equal to resolution elements in the image. As a result, the dwell time is equal to the frame time of the system increasing the SNR significantly:

$$\tau_{\text{dwell}} = \tau_{\text{frame}}$$



Each detector reduces the bandwidth because of the increment in the dwell time, so the SNR increases by a factor of  $\sqrt{n_d}$ . Nonuniformities and dead pixels are implicit in a staring array.

The SNR square root dependence can be used to compare the potential performance of the various system configurations. For example, a  $320 \times 256$  staring array produces a SNR that is higher by a factor of 25.3 in comparison to a parallel scanned system with a linear array of 128 detectors.

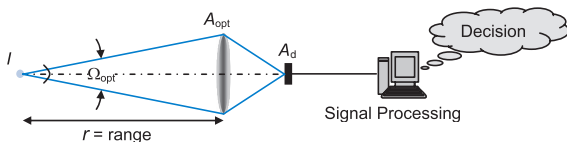
| Staring Systems                                   | Scanning Systems                                   |
|---------------------------------------------------|----------------------------------------------------|
| Good SNR                                          | Low SNR                                            |
| No moving parts                                   | Moving parts                                       |
| Uniformity problems                               | Good uniformity                                    |
| More complex electronically                       | More complex mechanically                          |
| Under-sampling problems                           | Prone to line-of-sight jitter                      |
| More prone to aliasing                            | Need more flux for a given SNR                     |
| Lower bandwidth for a given $\tau_{\text{frame}}$ | Higher bandwidth for a given $\tau_{\text{frame}}$ |
| $D^*$ for the array is lower                      | Good $D^*$ for individual detectors                |
| Expensive                                         | Cheaper                                            |

## Search Systems and Range Equation

**Search systems** are also called **detection, warning, or go-no-go systems**. Their intent is to detect and locate a target that has a prescribed minimum intensity within a prescribed search volume. They operate on an unresolved point-source basis (that does not fill the IFOV of the system); therefore, the spectral radiant intensity [W/ster·μm] is the variable of interest.

The principal concern is to assess the maximum SNR required for specified values of the probability of correct detection to minimize false-alarm rate. The result is a statistical decision about the existence or state of a target phenomenon within a search volume. Linear fidelity is unimportant for these systems because they do not produce an image.

The objective is to establish the maximum range at which an infrared search system can detect or track a point source target. The **range equation** states the distance at which a given point source can be detected and establishes the design tradeoffs available.



The amount of flux reaching the detector as a function of the radiant intensity is

$$\phi_d = I \cdot \Omega_{opt} \tau_{opt} \tau_{atm} = \frac{I \cdot A_{opt}}{r^2} \tau_{opt} \tau_{atm},$$

where  $\tau_{opt}$  is the optical transmittance and  $\tau_{atm}$  is the atmospheric transmittance between the point source and the search system. The signal voltage from the detector is given by

$$v_s = \mathcal{R}_v \cdot \phi_d = \mathcal{R}_v \frac{I \cdot A_{opt}}{r^2} \tau_{opt} \tau_{atm}.$$

The SNR is found by dividing each side of the equation by the rms value of the noise from the detector, yielding

$$SNR = \frac{v_s}{v_n} = \frac{\mathcal{R}_v}{v_n} \frac{I \cdot A_{opt}}{r^2} \tau_{opt} \tau_{atm}.$$

### Search Systems and Range Equation (cont.)

Using the definition of NEP, and recasting in terms of  $D^*$ :

$$\text{SNR} = \frac{1}{\text{NEP}} \frac{I \cdot A_{\text{opt}}}{r^2} \tau_{\text{opt}} \tau_{\text{atm}} = \frac{D^*}{\sqrt{A_d \Delta f}} \frac{I \cdot A_{\text{opt}}}{r^2} \tau_{\text{opt}} \tau_{\text{atm}},$$

where  $\Delta f$  is the noise-equivalent bandwidth. Recasting in terms of  $F/\#$  and the IFOV, and solving for the range,

$$r = \sqrt{\frac{ID^*}{\text{SNR} \sqrt{\Delta f}} \frac{\pi D_{\text{opt}}^2}{4f \sqrt{\Omega_d}} \tau_{\text{opt}} \tau_{\text{atm}}} = \sqrt{\frac{ID^*}{\text{SNR} \sqrt{\Delta f}} \frac{\pi D_{\text{opt}}}{4F/\# \sqrt{\Omega_d}} \tau_{\text{opt}} \tau_{\text{atm}}},$$

where  $D_{\text{opt}}$  is the diameter of the entrance pupil and  $f$  is the effective focal length of the optics. When the range equation is used to find the maximum detection or tracking range, the SNR is the minimum required for the system to work appropriately.

To analyze how the various factors affect the detection range, the range equation is regrouped in terms of optics, target and atmospheric transmittance, detector, and signal processing, respectively, yielding

$$r = \sqrt{\frac{\pi D_{\text{opt}} \tau_{\text{opt}}}{4F/\#}} \sqrt{I \tau_{\text{atm}}} \sqrt{D^*} \sqrt{\frac{1}{\text{SNR} \sqrt{\Omega_d \Delta f}}}.$$

In the first term, the diameter, the speed of the optics, and the transmittance characterize the optics. The first two terms are written separately to facilitate their independent solutions in the tradeoff process. The range is directly proportional to the square root of  $D_{\text{opt}}$ , where the bigger the optics, the more flux is collected. However, scaling up the entrance pupil changes the  $F/\#$  of the system, and requires a corresponding increase in both the focal length and the linear size of the detector to maintain the original FOV.

The second term contains the radiant intensity of the target and the transmittance along the line of sight. The amount of attenuation caused by the atmosphere, and the shot-noise contribution from the background can be optimized by choosing the best spectral band for the specific optical system. For example, if the emitting flux from the target is high, the spectral band that yields the best contrast is selected; however, if the flux is low, the spectral band that produces the optimum SNR is selected.

## Search Systems and Range Equation (cont.)

The third factor pertains to the characteristics of the detector. The range is proportional to the square root of the normalized detectivity of the detector. Therefore, an increment in the detection range can be achieved by enhancing the sensitivity of the system using serial TDI approaches, or by effectively shielding the detector from background radiation. Also notice that since the radiation is collected from a point source, increasing the area of the detector reduces the SNR of the system.

The final factor describes the range in terms of the signal processing parameters. It shows that decreasing either the FOV or the noise-equivalent bandwidth slowly increases the range because of the inverse fourth root dependence. The product  $\Delta f \Omega_d$  represents the angular scan rate in steradian per second. Increasing the integration time of the system averages away random noises resulting in longer detection ranges. The SNR in this type of system is interpreted as the minimum SNR required to reach a detection decision with an acceptable degree of certainty. For example, if the search system requires a higher SNR to improve the probability of correct detection, the system will have a shorter range.

The range equation for BLIP search systems is obtained by substituting  $D_{\text{BLIP}}^*$  for  $D^*$ , which in photon-derived units translates to

$$r_{\text{BLIP}} = \sqrt{\frac{\pi D_{\text{opt}} \tau_{\text{opt}}}{4}} \sqrt{\tau_{\text{atm}} I} \sqrt{\frac{\lambda_{\text{cutoff}}}{hc}} \sqrt[4]{\frac{2\eta}{\pi \int_0^{\lambda_{\text{cutoff}}} L_{\text{p,bkg}} d\lambda}} \sqrt{\frac{1}{\text{SNR} \sqrt{\Delta f} \sqrt{\Omega_d}}}.$$

Note that the  $F/\#$  term has dropped out of the equation. So a BLIP search system is influenced by the diameter of its optics but not by its speed.

There are several design concepts that can be used to decrease the background noise: the detector cold stop is designed so that it produces a 100% cold efficiency; a spectral filter is used to limit the spectral pass band to the most favorable region to the corresponding target flux; avoid the wavelengths at which the atmosphere absorbs strongly; minimize the emissivity of all the optical and opto-mechanical components, and, if necessary, cool the elements by the detector's angular subtense.

## Noise-Equivalent Irradiance

The **noise-equivalent irradiance (NEI)** is one of the main descriptors of infrared warning devices. It is the flux density at the entrance pupil of the optical system that produces an output signal equal to the system's noise (i.e., SNR = 1). It is used to characterize the response of an infrared system to a point source target.

The irradiance from a point source target is given by

$$E = \frac{\phi}{A_{\text{opt}}} = \frac{I \cdot \Omega_{\text{opt}}}{A_{\text{opt}}} = \frac{I}{r^2};$$

substituting the range expression and setting the SNR equal to 1,

$$\text{NEI} = E|_{\text{SNR}=1} = \frac{\sqrt{A_d \Delta f}}{A_{\text{opt}} D^* \tau_{\text{opt}} \tau_{\text{atm}}} = \frac{\text{NEP}}{A_{\text{opt}}} \frac{1}{\tau_{\text{opt}} \tau_{\text{atm}}}.$$

Recasting in terms of  $F/\#$  and the IFOV,

$$\text{NEI} = \frac{4F/\# \sqrt{\Omega_d \Delta f}}{\pi D_{\text{opt}} D^* \tau_{\text{opt}} \tau_{\text{atm}}}.$$

In BLIP conditions, the NEI is independent of the  $F/\#$ , yielding

$$\text{NEI}_{\text{BLIP}} = \frac{2hc}{\lambda_{\text{cutoff}} D_{\text{opt}} \tau_{\text{opt}} \tau_{\text{atm}}} \sqrt{\frac{2\Omega_d \Delta f \int_0^{\lambda_{\text{cutoff}}} L_{\text{p,bkg}} d\lambda}{\pi \eta}}.$$

NEI is especially useful when it is plotted as a function of wavelength. Such a plot defines the irradiance at each wavelength necessary to give a signal output equal to the system's rms noise. It can be interpreted either as an average value over the spectral measuring interval, or as the peak value.

Although NEI has a broader use in characterizing the performance of an entire system, it also may be used to evaluate the performance of a detector alone. In this case, it is defined as the radiant flux density necessary to produce an output signal equal to the detector noise. This term can also be used to compare the ability of differently sized sensor devices to detect a given irradiance.

$$\text{NEI} = \frac{E}{\text{SNR}} = \frac{\text{NEP}}{A_d}.$$

## Performance Specification: Thermal-Imaging Systems

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A **thermal imaging system (TIS)** collects, spectrally filters, and focuses the infrared radiation onto a multielement detector array. The detectors convert the optical signals into analog signals, which are then amplified, digitized, and processed for display on a monitor. The main function of a TIS is to produce a picture that maps temperature differences across an extended-source target; therefore, radiance is the variable of interest.

Two parameters are measured to completely specify a TIS and produce good thermal imagery: **thermal sensitivity** and **spatial resolution**. Spatial resolution is related to how small an object can be resolved by the thermal system, and thermal sensitivity concerns the minimum temperature difference discerned above noise level.

The **modulation transfer function (MTF)** characterizes both the spatial resolution and image quality of an imaging system in terms of spatial frequency response. The MTF is a major parameter used for system specifications and design analysis.

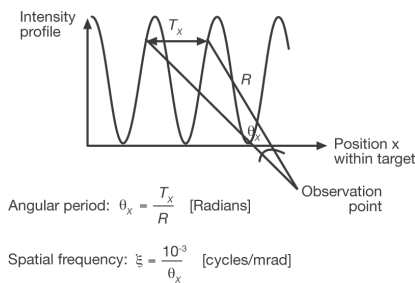
The **noise-equivalent temperature difference (NETD)** measures the thermal sensitivity of a TIS. While the NETD is a useful descriptor that characterizes the target-to-background temperature difference, it ignores the spatial resolution and image quality of the system.

The **minimum resolvable temperature difference (MRTD)** is a subjective measurement that depends on the infrared imaging system's spatial resolution and thermal sensitivity. At low spatial frequencies, the thermal sensitivity is more important; while at high spatial frequencies, the spatial resolution is the dominant factor. The MRTD combines both the thermal sensitivity and the spatial resolution into a single measurement. The MRTD is not an absolute value but is a perceivable temperature differential relative to a given background.

The **Johnson and TTP metric methodologies** are computer models that predict target-acquisition range performance of modern electro-optical imaging systems.

## Modulation Transfer Function (MTF) Definitions

**Spatial frequency** is defined as the reciprocal crest-to-crest distance (i.e., the spatial period) of a sinusoidal wavefront used as a basic function in the Fourier analysis of an object or image. Typically it is specified in cycles/mm in the image plane, and in angular spatial frequency (cycles/mil-radian) in object space. For an object located at infinity, these two representations are related through the focal length  $f$  of the image-forming optical system.



The **image quality** of an optical or electro-optical system is characterized either by the system's **impulse response** or its Fourier transform, the **transfer function**.

The impulse response  $h(x, y)$  is the 2D image form in response to an impulse or delta-function object. Because of the limitations imposed by diffraction and aberrations, the image quality produced depends on the wavelength distribution of the source, the  $F/\#$  at which the system operates, the field angle at which the point is located, and the choice of focus position.

A continuous object  $f(x, y)$  is decomposed using the shifting property of delta functions into a set of point sources, each with a strength proportional to the brightness of their original object at that location. The final image  $g(x, y)$  obtained is the superposition of the individual weighted impulse responses. This is equivalent to the convolution of the object with the impulse response:

$$g(x, y) = f(x, y) ** h(x, y),$$

where the double asterisk denotes a 2D convolution.

### MTF Definitions (cont.)

The validity of this requires **shift invariance** and **linearity**; a condition called **isoplanatism**. These assumptions are often violated in practice; however, to preserve the convenience of a transfer function analysis, the variable that causes nonisoplanatism is allowed to assume a set of discrete values. Each set has separate impulse responses and transfer functions. Although  $h(x, y)$  is a complete specification of image quality, additional insight is gained by use of the transfer function. A transfer-function analysis considers the imaging of sinusoidal objects, rather than point objects. It is more convenient than the impulse-response analysis because the combined effect of two or more subsystems can be calculated by a point-by-point multiplication of the transfer functions, rather than convolving the individual impulse responses.

Using the convolution theorem of Fourier transforms, the product of the corresponding spectra is given by

$$G(\xi, \eta) = F(\xi, \eta) \times H(\xi, \eta),$$

where  $F(\xi, \eta)$  is the object spectrum,  $G(\xi, \eta)$  is the image spectrum, and  $H(\xi, \eta)$  is the transfer function, which is the Fourier transform of the impulse response.  $\xi$  and  $\eta$  are spatial frequencies in the  $x$  and  $y$  directions, respectively.

The transfer function  $H(\xi, \eta)$  is normalized to have a unit value at zero spatial frequency. This normalization is appropriate for optical systems because the transfer function of an incoherent optical system is proportional to the 2D autocorrelation of the exit pupil, and the autocorrelation is the necessary maximum at the origin. In its normalized form, the transfer function  $H(\xi, \eta)$  is referred to as the **optical transfer function (OTF)**, which plays a key role in the theoretical evaluation and optimization of an optical system as a complex function that has both a magnitude and a phase portion:

$$\text{OTF}(\xi, \eta) = H(\xi, \eta) = |H(\xi, \eta)| \cdot e^{j\theta(\xi, \eta)}.$$

The absolute value or magnitude of the OTF is the MTF, while the phase portion of the OTF is referred to as the **phase transfer function (PTF)**. The system's MTF and PTF alter the image as it passes through the system.



### MTF Definitions (cont.)

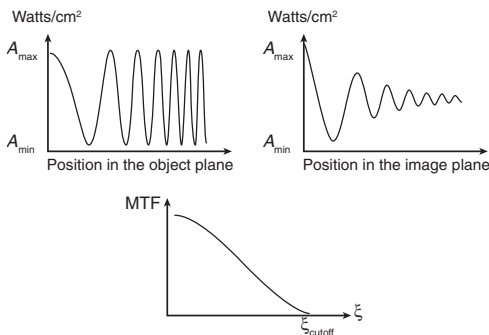
For linear phase-shift-invariant systems, the PTF is of no special interest since it indicates only a spatial shift with respect to an arbitrary selected origin. An image in which the MTF is drastically altered is still recognizable, where large nonlinearities in the PTF can destroy recognizability. PTF nonlinearity increases at high spatial frequencies. Since the MTF is small at high spatial frequencies, the linear phase-shift effect is diminished.

The MTF is then the magnitude response of the imaging system to sinusoids of different spatial frequencies. This response can also be defined as the attenuation factor in modulation depth:

$$M = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}},$$

where  $A_{\max}$  and  $A_{\min}$  refer to the maximum and minimum values of the waveform that describe the object or image in  $\text{W}/\text{cm}^2$  versus position. The modulation depth is actually a measure of visibility or contrast. The effect of the finite-size impulse response (i.e., not a delta function) of the optical system is to decrease the modulation depth of the image relative to that in the object distribution. This attenuation in modulation depth is a function of position in the image plane. The MTF is the ratio of image-to-object modulation depth as a function of spatial frequency:

$$\text{MTF}(\xi, \eta) = \frac{M_{\text{img}}(\xi, \eta)}{M_{\text{obj}}(\xi, \eta)}.$$



## Optics MTF: Calculations

The overall transfer function of an electro-optical system is calculated by multiplying individual transfer functions of its subsystems. The majority of thermal imaging systems operate with broad spectral bandpasses and detect noncoherent radiation. Therefore, classical diffraction theory is adequate for analyzing the optics of incoherent electro-optical systems. The OTF of diffraction-limited optics depends on the radiation wavelength and the shape of the entrance pupil. Specifically, the OTF is the autocorrelation of the entrance pupil function with entrance pupil coordinates  $x$  and  $y$  replaced by spatial frequency coordinates  $\xi$  and  $\eta$ , respectively. The change of variable for the coordinate  $x$  is

$$\xi = \frac{x}{\lambda d_i},$$

where  $x$  is the autocorrelation shift in the pupil,  $\lambda$  is the working wavelength, and  $d_i$  is the distance from the exit pupil to the image plane. The image-spaced cutoff frequency with a full width  $D$  is

$$\xi_{\text{cutoff}} = \frac{1}{\lambda F/\#},$$

which is when the autocorrelation reaches zero. The same analytical procedure is performed for the  $y$  coordinate.

A system whose image has no wave aberrations and is affected only by diffraction effects is called **diffraction limited**. The OTF for such a near-perfect system is purely real and nonnegative (i.e., MTF), and represents the best performance that the system can achieve, for a given  $F/\#$  and  $\lambda$ .

Consider the MTFs that correspond to diffraction-limited systems with square (width  $l$ ) and circular (diameter  $D$ ) exit pupils. When the exit pupil of the system is circular, the MTF is circularly symmetric, with  $\xi$  profile:

$$\text{MTF}(\xi) = \begin{cases} \frac{2}{\pi} \left\{ \cos^{-1}(\xi/\xi_{\text{cutoff}}) - \xi/\xi_{\text{cutoff}} \left[ 1 - (\xi/\xi_{\text{cutoff}})^2 \right] \right\}^{1/2} & \text{for } \xi \leq \xi_{\text{cutoff}} \\ 0 & \text{otherwise.} \end{cases}$$

## Optics MTF: Calculations (cont.)

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The square aperture has a linear MTF along the spatial frequency  $\xi$ , given by

$$\text{MTF}(\xi, \eta) = 1 - \frac{\xi}{\xi_{\text{cutoff}}}.$$

The MTF curve for a system with appreciable geometric aberrations is upwardly bounded by the diffraction-limited MTF curve. Aberrations broaden the impulse response  $h(x, y)$ , resulting in a narrower, lower MTF with a smaller integrated area. The area under the MTF curve relates to the **Strehl ratio (SR)**, which measures image quality degradation, and is the irradiance at the center of the impulse response, divided by that at the center of a diffraction-limited impulse response. Small aberrations reduce the intensity at the principal maximum of the diffraction pattern, or the diffraction focus, and that of the removed light is distributed to the outer parts of the pattern. Using the **central-ordinate theorem** for Fourier transforms, SR is written as the ratio of the area under the actual MTF curve to that under the diffraction-limited MTF curve, yielding

$$\text{SR} = \frac{\iint \text{MTF}_{\text{actual}}(\xi, \eta) d\xi d\eta}{\iint \text{MTF}_{\text{diff-limited}}(\xi, \eta) d\xi d\eta}.$$

The Strehl ratio falls between 0 and 1; however, its useful range is  $\sim 0.8$  to 1 for highly corrected optical systems. Geometrical-aberration OTF is calculated from ray-trace data by Fourier transforming the spot-density distribution without regard for diffraction effects. The OTF is obtained accurately if the aberration effects dominate the impulse-response size. The OTF of a uniform blur spot is written as

$$\text{OTF}(\xi) = \frac{J_1(\pi \xi d_{\text{blur}})}{\pi \xi d_{\text{blur}}},$$

where  $J_1(\cdot)$  is the first-order Bessel function, and  $d_{\text{blur}}$  is the diameter of the blur spot. The overall optics portion MTF of an infrared system is determined by multiplying the ray-trace data MTF with the diffraction-limited MTF of the proper  $F/\#$  and wavelength.

## Electronics MTF: Calculations

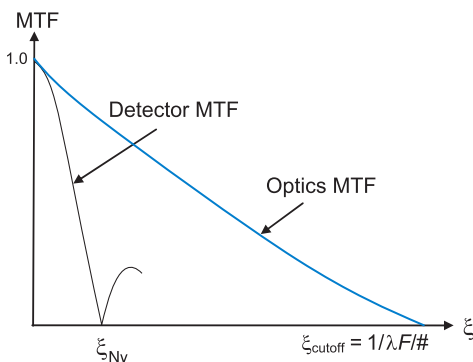
Two integral parts of modern infrared imaging systems are the electronic subsystems, which handle the signal and image processing functions, and the sensor(s) of the imaging system. Characterization of the electronic circuitry and components is well established by the temporal frequency in hertz. In order to cascade the electronic and optical subsystems, the temporal frequencies must be converted to spatial frequencies. This is achieved by dividing the temporal frequencies by the scan velocity of the imaging device. In contrast to the optical transfer function, the electronic MTF is not necessarily maximized at the origin, and can either amplify or attenuate the system MTF curve at certain spatial frequencies.

The detector MTF is expressed as

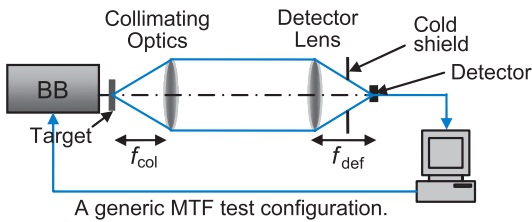
$$\text{MTF}_d(\xi, \eta) = \text{sinc}(d_h \xi) \text{sinc}(d_v \eta),$$

where  $d_h$  and  $d_v$  are the photosensitive detector sizes in the horizontal and vertical directions, respectively. Although the detector MTF is valid for all spatial frequencies, it is typically plotted up to its cutoff frequencies ( $\xi = 1/d_h$  and  $\eta = 1/d_v$ ). The spatial **Nyquist frequency** ( $\xi_{Ny}$ ) of the detector array must be taken into consideration to prevent aliasing effects.

It is the combination of the optical and electronic responses that produce the overall system MTF, even though the detector MTF usually becomes the limiting factor of the electro-optical system since, in general,  $\xi_{Ny} \ll \xi_{\text{cutoff}}$ .



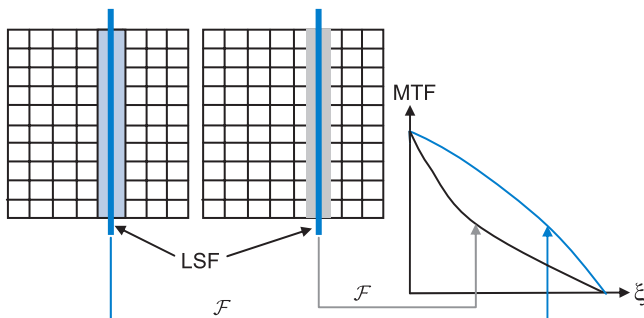
## MTF Measurement Setup and Sampling Effects



All optical and electro-optical components that make up the infrared-imaging system should be placed on a

vibration-isolated optical table. The aperture of the collimator should be large enough to overfill the aperture of the system under test. The optical axis of the infrared camera must be parallel to and centered on the optical axis of the collimator to ensure that its entrance pupil is perpendicular to the collimator optical axis. The display gain and brightness should be optimized prior to the start of the MTF measurements to assure that the display setting is not limiting the performance of the detector array.

**Sampling effects** alter the MTF and affect the fidelity of the image. The discrete location of the detectors in the staring array creates the sampling lattice. **Phasing effects** between the sampling lattice and the location of the target introduce problems at nearly all spatial frequencies. **Digitization** alters signal amplitude and distorts the pulse shape. Sampling causes sensor systems like **focal plane arrays (FPAs)** to have a particular kind of shift variance (i.e., spatial phase effects); in which case, they depend on the position of the target relative to the sampling grid to measure the MTF of the system.



## MTF Measurement Techniques: PSF and LSF

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Different measurement techniques can be used to assess the MTF of an infrared-imaging system. These include the measurement of different types of responses such as point-spread function, line-spread function, edge-spread function, sine-target response, square-target response, and noiselike target response. All targets, except the ones that are random, should be placed in a micro-positioning mount containing three degrees of freedom ( $x, y, \theta$ ) to account for phasing effects.

The imaging of a point source  $\delta(x,y)$  of an optical system has an energy distribution called the **point spread function (PSF)**. The 2D Fourier transform of the PSF yields the complete 2D OTF( $\xi, \eta$ ) of the system in a single measurement. The absolute value of the OTF gives the MTF of the system. The impulse response technique is practically implemented by placing a small pinhole at the focal point of the collimator. If the flux passing through the pinhole produces an SNR that is below a usable value, a slit target can be placed at the focal plane of the collimating optics; the output is called the **line spread function (LSF)**. The cross-section of the LSF is obtained by integrating the PSF parallel to the direction of the line source, because the line image is simply the summation of an infinite number of points along its length. The LSF only yields information about a single profile of the 2D OTF. Therefore, the absolute value of the Fourier transform of the LSF yields the 1D MTF( $\xi, \eta$ ) of the system.

To obtain other profiles of the MTF, the line target can be reoriented as desired. The slit angular subtense must be smaller than the IFOV with a value of 0.1 IFOV. The phasing effects are tested by scanning the line target relative to the sampling sensor grid until maximum and minimum signals are obtained at the sensor. The measurements are performed and recorded at different target positions, and averaging the output over all locations yields an average MTF. However, this average MTF is measured using a finite-slit aperture, in which case, this undesirable component is removed by dividing out the Fourier transform of the finite slit, yielding a more accurate MTF.

## MTF Measurement Techniques: ESF and CTF

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The MTF is also obtained from an **edge spread function (ESF)**, the response of the system under test to an illuminated knife-edge target. There are two advantages in using this target over the line target: it is simpler to build than a narrow slit, and there is no MTF correction. The edge is differentiated to obtain the line-spread function and is then Fourier transformed. However, the derivative operation accentuates the system noise presented in the data, which can corrupt the resulting MTF. The edge must be straight with no raggedness. To increase the SNR for both the line and edge-spread techniques, the 1D Fourier transform is averaged over all the rows of the image. In addition, reducing the system gain reduces noise, and the target signal can be increased if possible.

The MTF is also obtained by measuring the system's response to a series of sine-wave targets, where the image modulation depth is measured as a function of spatial frequency. Sinusoidal targets can be fabricated on photographic films or transparencies for the visible spectrum; however, they are not easy to fabricate for the testing of infrared systems due to materials limitations. A less expensive, more convenient target is the bar target, a pattern of alternate bright and dark bars of equal width. The square-wave response is called the **contrast transfer function (CTF)**, and is a function of the **fundamental spatial frequency**  $\xi_t$  of the specific bar target under test. The CTF is measured on the peak-to-valley variation of the image irradiance, and is defined as

$$\text{CTF}(\xi_t) = \frac{M_{\text{square-response}}(\xi_t)}{M_{\text{input-square-wave}}(\xi_t)}.$$

The CTF is higher than the MTF at all spatial frequencies because of the contribution of the odd harmonics of the infinite-square wave test pattern to the modulation depth in the image. The CTF is expressed as an infinite series of MTFs. A square wave can be expressed as a Fourier-cosine series. The output amplitude of the square wave at frequency  $\xi_t$  is an infinite sum of the input cosine amplitudes modified by the system's MTF:

## MTF Measurement Techniques: ESF and CTF (Cont.)

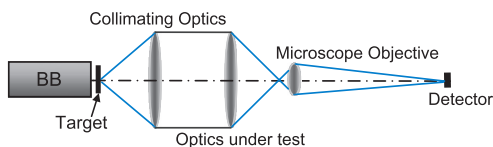
$$\text{CTF}(\xi_t) = \frac{4}{\pi} \left[ \text{MTF}(\xi_t) - \frac{1}{3} \text{MTF}(3\xi_t) + \frac{1}{5} \text{MTF}(5\xi_t) - \frac{1}{7} \text{MTF}(7\xi_t) + \dots \right],$$

conversely, the MTF can be expressed as an infinite sum of CTFs as

$$\text{MTF}(\xi_t) = \frac{\pi}{4} \left[ \text{CTF}(\xi_t) + \frac{1}{3} \text{CTF}(3\xi_t) - \frac{1}{5} \text{CTF}(5\xi_t) + \frac{1}{7} \text{CTF}(7\xi_t) + \dots \right].$$

Optical systems are characterized with three- and four-bar targets and not by infinite square-wave cycles. Therefore, the CTF might be slightly higher than the CTF curve for an infinite square wave. For bar targets with a spatial frequency above one-third the cutoff frequency, where the MTF approaches zero, the MTF is equal to  $\pi/4$  times the measured CTF. Electronic nonlinearity, digitization effects, and sampled-scene phase effects can make these MTF and CTF measurements difficult.

The MTF of optics is measured, without including the detector MTF, by placing a microscope objective in front of the detector FPA. The microscope objective is used as a relay lens to reimage the system's response formed by the optics under test onto the FPA with the appropriate magnification. Here, the detector is no longer the limiting component of the imaging system, since its MTF response becomes appreciably higher than the optical MTF curve. The microscope objective must be of high quality to reduce degradation of the measured response function, and have high enough NA to capture the entire image-forming cone angle.



Imaging systems containing a detector FPA are nonisoplanatic, and their responses depend on the location of the deterministic targets relative to the sampling grid, introducing problems at nearly all spatial frequencies. The use of random-target techniques for measuring the MTF of a digital imaging system tends to average out the phase effects.



## MTF Measurement Techniques: Noiselike Targets

Using noiselike test targets of known spatial frequency content allows measurement of the shift-invariant MTF because the target information is positioned randomly with respect to the sampling sites of the digital imaging system. The MTF of the system is calculated because the input power density  $\text{PSD}_{\text{input}}(\xi)$  of the input random pattern is known, and an accurate estimate of the output power density  $\text{PSD}_{\text{output}}(\xi)$  is made from the FPA response. The MTF is then calculated from the following relationship:

$$\text{MTF}(\xi) = \sqrt{\frac{\text{PSD}_{\text{output}}(\xi)}{\text{PSD}_{\text{input}}(\xi)}}$$

This approach is commonly used to characterize time domain electrical networks, and its application to the MTF testing of digital imaging systems provides an average of the shift variation, which eases alignment tolerances and facilitates MTF measurements at spatial frequencies beyond Nyquist.

Two different techniques are used for the generation of random targets: laser speckle and transparency-based noise targets. The former is used to characterize the MTF of FPAs alone, while the latter is used to characterize the MTF of a complete imaging system (i.e., imaging optics together with the FPA).

A laser speckle pattern of known PSD is generated by the illustrated optical train.



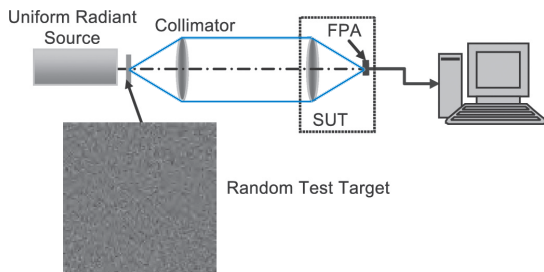
The integrating sphere produces a uniform irradiance with a spatially random phase. The aperture following the integrating sphere (typically, a double slit), determines the  $\text{PSD}_{\text{input}}(\xi)$  of the speckle pattern at the FPA, which is proportional to the aperture transmission function.

## MTF Measurement Techniques: Noiselike Targets (cont.)

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The spatial frequency of the resulting narrowband speckle pattern can be tuned by changing the aperture-to-focal-plane distance  $z$ . The MTF is calculated from the relative strength of the sideband center of the  $\text{PSD}_{\text{output}}(\xi)$ .

To characterize a complete imaging system, a 2D uncorrelated random pattern with uniform band-limited white-noise distribution is created using a random generator computer algorithm. This random gray-level pattern is printed onto a transparency and placed in front of a uniform radiant extended source, producing a 2D radiance pattern with the desired input power spectrum  $\text{PSD}_{\text{input}}$ .



The output spectral density is estimated by imaging the target through the optical system onto the FPA. The output data is then captured by a frame grabber and processed to yield the output power spectrum  $\text{PSD}_{\text{output}}(\xi)$  as the absolute value squared of the Fourier transform of the output image data. The MTF is then calculated using the equation from the previous page.

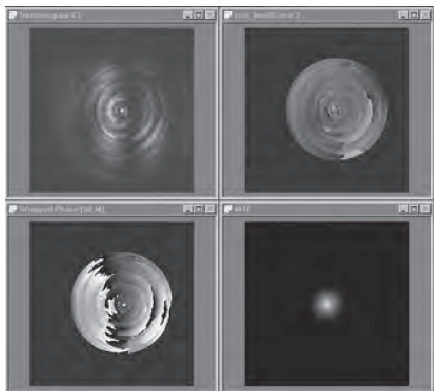
In the infrared region, the transparency must be replaced by a random thermoscene made of a chrome deposition on an infrared material substrate. Microlithographic processes enable production of square apertures of various sizes on a 2D matrix to achieve the desirable random pattern. To avoid diffraction-induced nonlinearities of transmittance, the minimum aperture size must be five times the wavelength.

## MTF Measurement Techniques: Interferometry

**Common-path interferometers** may be employed for measuring the transfer functions of optical systems. An interferogram of the wavefront exiting the system is reduced to find the phase map. The distribution of amplitude and phase across the exit pupil contains the necessary information for calculating the OTF by pupil autocorrelation.

The performance of a lens at specific conjugates can be measured by placing the optical element in one of the arms of an interferometer. The process begins by computing a single wrapped phase map from the resultant wavefront information or **optical path difference (OPD)** exiting the pupil of the system under test.

The wrapped phase map is represented in multiples of  $2\pi$  with phase values ranging from  $-\pi$  to  $\pi$ . Removal of the  $2\pi$  modulus is accomplished by using an unwrapping algorithm, thus producing an unwrapped phase map also known as the surface map. The PSF is obtained by multiplying the



conjugate Fourier transform of the surface map data (i.e., element by element multiplication of the amplitude complex function). The inverse Fourier transform of the PSF yields the complex OTF with the modulus that corresponds to the MTF of the optical system.

In summary, the MTF is a powerful tool used to characterize an imaging system's ability to reproduce signals as a function of spatial frequency. It is a fundamental parameter that determines (1) where the limitations of performance in optical and electro-optical systems occur, and (2) which crucial components must be enhanced to yield a better overall image quality. It guides system design and predicts system performance.

## Noise-Equivalent Temperature Difference (NETD)

**Noise-equivalent temperature difference (NETD)** is the target-to-background temperature difference that produces a peak signal-to-rms-noise ratio of unity. Its analytical formula is given by:

$$\text{NETD} = \frac{4}{\pi} \left[ \frac{(F/\#)^2 \sqrt{\Delta f}}{D^* \sqrt{A_d} \partial L / \partial T} \right],$$

where  $\Delta f$  is the electronic bandwidth,  $D^*$  and  $A_d$  are respectively the normalized detectivity and the effective area of the detector, and the partial derivative of the radiance with respect to temperature is the radiance contrast. This equation applies strictly to detector-limited situations.

A smaller NETD indicates better thermal sensitivity. For the best NETD,  $D^*$  is peaked near the wavelength of maximum exitance contrast of the source. A smaller  $F/\#$  collects more flux, yielding a more accurate estimated NETD. A smaller electronic bandwidth yields a larger dwell time, obtaining a smaller noise voltage and lowering the NETD. A larger detector area gives a larger IFOV, collecting more flux and resulting in a better NETD. The drawback of NETD as a system-level performance descriptor is that while the thermal sensitivity improves for larger detectors, the image resolution deteriorates. Thus, while the NETD is a sufficient operational test, it cannot be applied as a designed criterion.

When the system operates under BLIP conditions, the equation for NETD becomes

$$\text{NETD}_{\text{BLIP}} = \frac{2\sqrt{2}}{\sqrt{\pi}} \frac{hc}{\lambda_{\text{cut}}} \frac{F/\#}{\partial L / \partial T} \sqrt{\frac{\Delta f \cdot \int_0^{\lambda_{\text{cut}}} L_{\text{p,bkg}} d\lambda}{\eta A_d}}$$

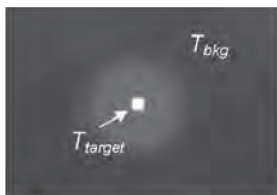
where  $\lambda$  is the wavelength,  $h$  is Planck's constant,  $c$  is the speed of light in vacuum,  $L_{\text{bkg}}$  is the background radiance, and  $\eta$  is the quantum efficiency of the detector. Notice that the NETD is inversely proportional to the square root of the quantum efficiency and proportional to the square root of the in-band background flux. Under BLIP conditions, it has a linear dependence on  $F/\#$  rather than a square dependence as in the detector-limited condition case.

## NETD Measurement Technique

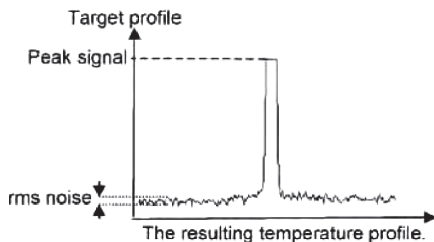
The NETD measurement is usually carried out using a square target. The size of the square must be several times the detector angular substance (i.e., several IFOVs) of the extended source to ensure that the spatial response of the system does not affect the measurement. This target is usually placed in front of an extended area blackbody source, so that the temperature difference between the square target and the background is several times the expected NETD to ensure a response that is clearly above the system noise. The peak signal and rms noise data are obtained by capturing, averaging, and taking the standard deviation of several images. The NETD is then calculated from the experimental data as follows:

$$\text{NETD} = \frac{\Delta T}{\text{SNR}},$$

where  $\Delta T = T_{\text{target}} - T_{\text{bkg}}$ , and the SNR is the signal-to-noise ratio of the thermal system.



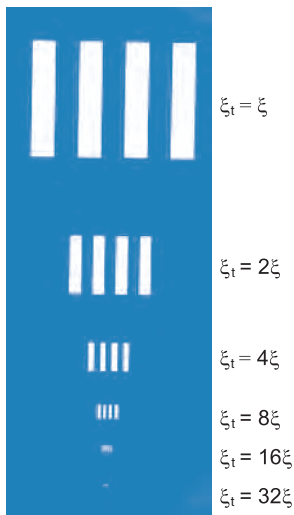
The NETD test imaged pattern



Care must be taken to ensure that the system is operating linearly and that no noise sources are included. Because of the dependence of noise on bandwidth, and to obtain the proper dwell time and bandwidth, the NETD must be measured with the system running at its full operational scan rate.

## Minimum Resolvable Temperature Difference (MRTD)

The **minimum resolvable temperature difference (MRTD)** simultaneously characterizes both the spatial resolution and the thermal sensitivity. It is a subjective measurement for which the SNR-limited thermal sensitivity is outlined as a function of spatial frequency. Conceptually, the MRTD is the image SNR required for an observer to resolve four-bar targets at several **fundamental spatial frequencies**  $\xi_t$ , so that the bars are just discernable by a trained observer with unlimited viewing time. The noise-limited rationale is essential in this case, because an infrared imaging system displays its utmost sensitivity when the highest noise is visible to the observer (i.e., gain is increased to compensate for adverse atmospheric and/or scene conditions).



The bars and spaces of the target are blackbody radiators of different but uniform temperatures, and the bar height is seven times the bar width.

These tests depend on decisions made by the observer. The results vary with training, motivation, and visual capacity, as well as the environmental setting. Because of the considerable inter- and intra-observer variability, several observers are required. The underlying distribution of observer responses must be known, so that the individual responses can be appropriately averaged together.

MRTD is a better system-performance descriptor than the MTF alone because the MTF measures the attenuation in modulation depth, without regard for a noise level. MRTD is also a more complete measurement than the NETD because it accounts for both spatial resolution and noise level. Therefore, the MRTD is a useful overall analytical and design tool that is indicative of system performance.

## MRTD: Calculation

MRTD measures the ability to resolve detail in imagery, and is directly proportional to the NETD and inversely proportional to the MTF:

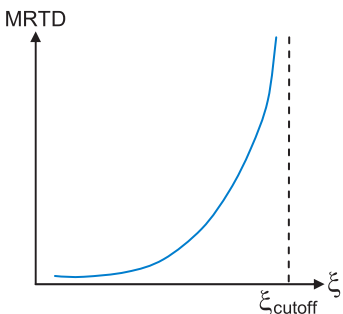
$$\text{MRTD} \propto \frac{\text{NETD} \xi_t \sqrt{\text{HIFOV} \cdot \text{VIFOV}}}{\text{MTF}(\xi_t) \sqrt{\tau_{\text{eye}} \cdot \tau_{\text{frame}}}},$$

where  $\xi_t$  is the spatial frequency of the target being observed,  $\tau_{\text{eye}}$  is the integration time of the human eye,  $\tau_{\text{frame}}$  is the frame time, MTF is the transfer function of the system at that particular target frequency, and HIFOV and VIFOV are the horizontal and vertical IFOVs of the system, respectively.

The derivation of an exact analytical expression for MRTD is complex because of the number of variables in the calculation; therefore, computer-aided performance models such as the NVTherm model are used. Substituting the NETD equation into the MRTD equation yields

$$\text{MRTD} \propto \frac{\xi_t \sqrt{\text{HIFOV} \cdot \text{VIFOV}}}{\text{MTF}(\xi_t) \sqrt{\tau_{\text{eye}} \cdot \tau_{\text{frame}}}} \times \frac{(F/\#)^2 \sqrt{\Delta f}}{D^* \sqrt{A_d} \partial L / \partial T}.$$

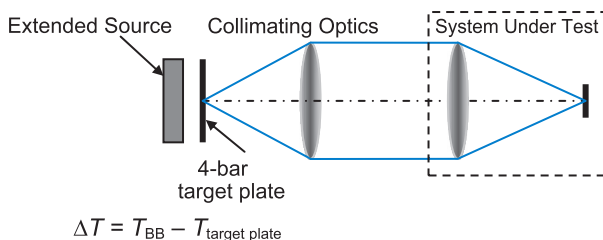
MRTD depends on the same variables as NETD (i.e.,  $F/\#$ ,  $\Delta f$ ,  $D^*$ , and radiance contrast). However, the thermal performance of the system cannot be increased by increasing the area of the detector or IFOV; the MTF decreases at higher frequencies. Therefore, the amount of  $\Delta T$  required for a 4-bar target to be discernable increases as the size of the bars decrease. The MRTD increases when MTF decreases, but the



MRTD increases more rapidly due to the extra factor  $\xi_f$  in the numerator. The effect of the observer is included in the factor  $\tau_{\text{eye}} \tau_{\text{frame}}$ . Increasing the frame rate gives more observations within the temporal integration time of the human eye, and then the eye-brain system tends to average out some of the noise, leading to a lower MRTD.

## MRTD Measurement Technique

A generic MRTD test configuration is shown:



The four-bar target is located in front of the blackbody source at the focal plane of the collimator to collimate the radiation from each point of the surface of the target. To achieve high spatial frequency, the MRTD setup is mounted on a vibration-isolated optical table. Since the MRTD is a detection criterion for noisy imagery, the gain of the infrared imaging system must be set high enough that the image is noisy. Infrared imaging systems are subject to sampling effects. The MRTD does not have a unique value for each spatial frequency, but has a range of values depending on the location of the target with respect to the detector array. Therefore, the targets must be adjusted to achieve the best visibility. An observer must count the number of bars to ensure that all four are present and discernable. The targets should range from low spatial frequencies to just past the system cutoff, and span the entire spatial frequency response.

Problems associated with the MRTD measurements include the related distance between the display screen and the observer, background brightness, and strain. The contrast sensitivity increases with background radiance; however, during the MRTD tests, the observer can adjust the system's gain and level, and monitor the brightness and contrast to optimize the image for detection criterion. Inconsistencies between the results obtained by different observers can occur, and over a long period of time, the human eye-brain sensitivity decreases, causing unreliability. The use of the MRTD is also somewhat limited because all the field scenes are spectrally selective (i.e., emissivity is a function of wavelength), while most MRTD tests are performed with extended-area blackbodies.



### MRTD Measurement: Automatic Test

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It is of practical interest to measure the MRTD without the need of a human observer. Automatic tests or objective tests are desirable because of an insufficient number of trained personnel and because the subjective test is time consuming. The MRTD equation can be written as

$$\text{MRTD} = \mathcal{K}(\xi_t) \frac{\text{NETD}}{\text{MTF}(\xi_t)},$$

where the constant of proportionality and any spatial-frequency-dependent terms, including the effect of the observer, are taken up into the function  $\mathcal{K}(\xi_t)$ . To characterize the average effects of the observer for a given display and viewing geometry, an MRTD curve is measured for a representative sample of the system under test. Along with the MRTD data, the NETD and MTF are measured and recorded for the system. From these data, the function  $\mathcal{K}(\xi_t)$  can be determined, and subsequent tests of similar systems can be performed without the observer.

A comprehensive automatic laboratory test station, which provides the means to measure the performance of an infrared imaging system, and a field-tester apparatus measuring the FLIR parameters of an Apache helicopter shown below.



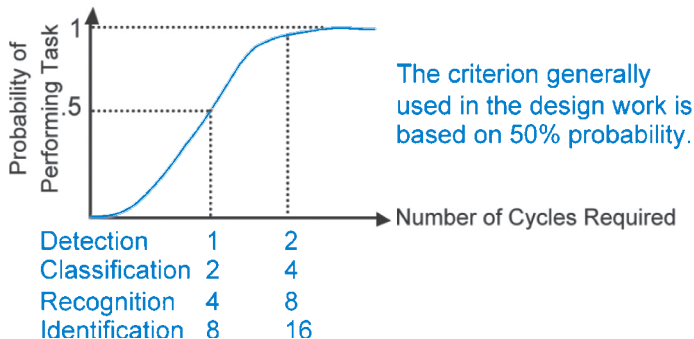
Pictures courtesy of CI Systems, Inc.

## Johnson Metric Methodology

The **Johnson metric** is a metric in a computer model that accounts for both the thermal sensitivity and the spatial resolution of a thermal imaging system. It provides a way to describe real targets in terms of simpler square-wave visual patterns.

Johnson placed sets of square-wave patterns with a selection of bar widths alongside eight military targets and a man. These square arrangements have the same apparent temperature difference  $\Delta T$  as the military targets and are viewed under the same conditions. When it was determined that a particular target could be detected, classified, recognized, or identified, the corresponding just-resolved bar target of the set was noted as well. Johnson concluded that it was possible to characterize the performance of an average observer for a particular decision task in terms of the number of just-resolved cycles that would fit across the minimum dimensions of the target.

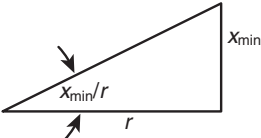
By normalizing the number of cycles in the bar pattern by the minimum dimension of the target, it was found that the spatial resolution required for a particular decision task was approximately constant.



### Johnson Metric Methodology (cont.)

A more complex decision task requires a higher level of detail, and hence a finer image resolution is necessary. These resolutions are based on a 50% probability of correctly determining the level of discrimination.

Once the required numbers of cycles for a particular task are determined, the required angular spatial frequency in cycles/rad is calculated by

$$\xi = \frac{n_{\text{cycles}}}{x_{\text{min}} / r}$$


where  $n_{\text{cycles}}$  is the number of cycles,  $x_{\text{min}}$  is the minimum target dimension,  $r$  is the range; and therefore,  $x_{\text{min}}/r$  is the angular subtense of the target.

To discriminate a target with 50% probability, two IFOVs are required per cycle of the highest spatial frequency (i.e., corresponding to one cycle on target). Therefore, we can write the IFOV in terms of the Johnson parameters:

$$\frac{1}{2 \times \text{IFOV}} = \frac{n_{\text{cycles}}}{x_{\text{min}} / r} = \frac{f}{2\sqrt{A_d}},$$

$$\Rightarrow \frac{f}{\sqrt{A_d}} = \frac{2r (N50)}{x_{\text{min}}},$$

where  $f$  is the effective focal length of the optical system and  $A_d$  is the area of the detector. Johnson determined that the N50 values needed to predict the discrimination of the target with 50% probability are 1, 4, and 8 cycles for detection, recognition, and identification, respectively. However, these values were never explicitly defined, and other values can be experimentally determined.

The Johnson methodology multiplies the resolving power of an imager (cycles/mrad) by the target size [mrad] to get the number of cycles (N50) on target. These values were established without removing the effect of chance.

## Johnson Criteria Flaws

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In the Johnson methodology, sensor spatial frequency at a particular temperature characterizes the sensor for target acquisition purposes. The performance of the imager can be determined exclusively by the Johnson frequency ( $\xi_J$ ) visible at the average target-to-background contrast. When the Johnson method works, it is because an increase in  $\xi_J$  represents an improvement in the contrast rendition at all spatial frequencies. However, with focal plane arrays (FPAs),  $\xi_J$  is more an indicator of sample rate than of image quality. In addition, the Johnson metric fails to accurately predict the effect of noise on task performance. The observer appears to require more sensor resolution when the resolution is noise limited as opposed to spatial-frequency-response limited. Also, because the Johnson metric is based on the system response at a single frequency, it cannot predict the effect of varying the image frequency spectrum through digital processing.

Due to Johnson methodology insensitivities to sampled imager characteristics in the vicinity of the limiting frequency, the Johnson metric is fundamentally flawed. These flaws make predictions for many modern imaging systems inaccurate.

The Night Vision and Electronic Sensors Directorate of the U.S. Army (NVESD) has developed a new target acquisition metric to better predict the performance of modern electro-optical imagers. The targeting task performance (TTP) metric provides better accuracy than the Johnson criteria. The TTP methodology extends the capabilities of the Johnson model to sampled imagers and digital image enhancement. For example, the TTP metric can be applied in circumstances where the sensor noise is not white (i.e., spectrally weighted). This means that, unlike the Johnson criteria, the TTP metric accurately predicts the performance of digital image enhancements like high frequency boost. The contrast limitation of vision is incorporated in the TTP metric model as well, thus allowing the modeling of image intensifiers, CCDs, CMOS, and thermal focal plane arrays.

## Targeting Task Performance (TTP) Metric Methodology

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The computer model presented in this section was developed by the Night Vision & Electronic Sensors Directorate (NVESD).

A scene consists of many luminance levels. The eye achieves an integrated view of objects connecting lines and surfaces. These lines and surfaces do not share particular brightness through their extent. Perspective is gained from converging lines, which might vary in both sharpness and luminance with increasing range. Slight changes in hue or texture can provide hints as to the distance and orientation of an object and possibly indicate the nature of the surface attributes. The discrimination of a target requires the ability to distinguish small differences in shades of gray, and not just the ability to identify black symbols on a white background (i.e., acute vision). A quantification of visual performance requires that the resolution be measured for all shades of gray in the image.

An imager provides advantages over human sight. The target can be magnified, and the imager allows the viewer to perceive radiation beyond the visible spectrum. However, aberrations and noise associated with the imager limit the viewer's performance. Also, human vision performance depends upon display luminance and contrast.

The TTP model takes into consideration hardware characteristics and human vision. These interactions establish how well the scene is perceived through the imager and how well the observer is able to discriminate targets. It predicts the effects of blur, noise, contrast, and display characteristics on target acquisition and range performance.

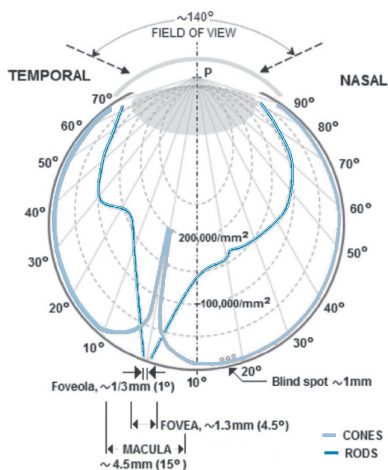
Two major software models were developed by NVESD using the TTP metric model:

- SSCamIP models reflected-light imagers such as CCD and CMOS arrays and image intensifiers.
- NVThermIP models thermal focal plane arrays.

## Human Vision—Distribution of Retinal Photoreceptors

The limitations of human vision are important when predicting the target performance of an electro-optical sensor. The three modes of eye function under different illuminance levels, photopic (bright light), scotopic (low light conditions) and mesopic (intermediary), result from the specific response of its two types of photoreceptor cells, cones and rods. Their activity is specific to retinal illuminance level, which is determined by the brightness level of the object observed, as well as brightness of the background and surroundings.

Rods cannot distinguish colors, but are responsible for scotopic monochrome (black-and-white) vision; they work well in dim light but saturate at illumination levels above 10 fL. Rods are distributed throughout the retina but there are none at the fovea and none at the blind spot. Rod density is greater in the peripheral retina than in the central retina. Cones are responsible for color vision and cease to be important below 0.01 fL. They are mostly concentrated in and near the fovea, and only a few are present at the sides of the retina. With adequate illumination, objects are seen most sharply when their images fall on this spot, as when one looks at an object directly. When rods and cones are stimulated by light, the nerves send off impulses to the brain.

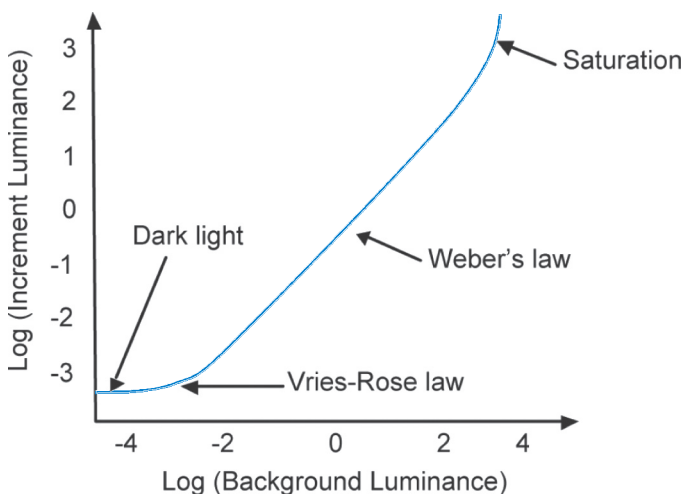


## Human Vision (cont.)

Eye behavior changes significantly with luminance and with angular eccentricity from the fovea.

The sensitivity of the eye is measured by determining the absolute intensity threshold, (i.e., the increment luminance  $\Delta L$ ) of a small disk of light against a uniform background.

The eye response to signal luminous intensity is logarithmic for the most part according to **Weber's law**, which states that the ratio of the increment signal to the background signal at threshold is constant. The response changes for very high-level stimuli, due to saturation, and for very low-level stimuli, due to the increased role of neural noise (dark light). At very low brightness levels, rod response follows the square root law (de Vries-Rose Law), which is limited by quanta fluctuation.



Sometimes display luminance is kept low because the operator chooses to maintain dark adaptation. For example, during night flight, military pilots flying without goggles set instrumentation displays to between 0.1 to 0.3 fL. This permits reasonable viewing of the instruments while maintaining dark adaptation in order to see outside the aircraft.

## Contrast Threshold Function (CTF)

While MRTD describes resolution and noise limitations of an imaging system, there are also limitations associated with human vision. Our ability to discern low-contrast patterns varies with the size of the pattern, or its spatial frequency. The **contrast threshold function (CTF)** is a measure of the minimum contrast needed to discern the presence of a sinusoidal pattern at a given spatial frequency.

Each individual has a different  $CTF_{eye}$  curve. Relatively speaking, higher amounts of minimum contrast are needed at extremely low and extremely high spatial frequencies.

$CTF_{eye}$  is similar to the MRTD function used to characterize infrared systems. The observer views sinusoidal patterns on an electronic display spanning different frequencies with constant illumination while the contrast is lowered, until the observer is unable to discern the sine-wave patterns.



In this image, the spatial frequency increases from left to right, and contrast varies from 100% at the top to less than 1% at the bottom. Since human perception of contrast is frequency dependent, the alternating bars in the middle appear taller.

The contrast of an object or its representation in an image is arrived at by comparing the maximum and minimum visual luminance ( $L_v$ ) values

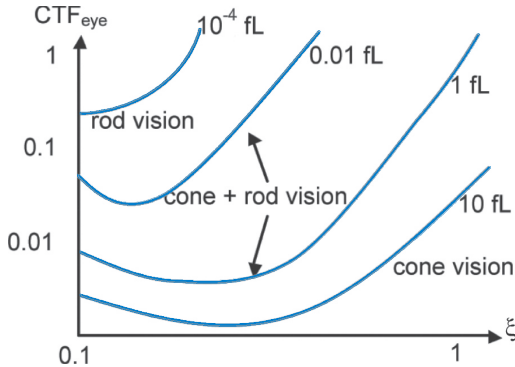
$$\text{Contrast} = \frac{L_{v,\max} - L_{v,\min}}{L_{v,\max} + L_{v,\min}}.$$

The denominator represents twice the average value of the luminance. This contrast ratio has a tremendous impact on the overall image quality. A higher contrast significantly increases the amount of image content conveyed to the human observer.



### CTF (cont.)

The  $CTF_{eye}$  curve provides the threshold contrast value at all spatial frequencies.



The limiting resolution is the spatial frequency  $\xi$  where the  $CTF_{eye}$  curve crosses unity contrast. It provides the smallest detail that will be visible at that light level, and this detail is only visible at the highest possible contrast. The eye loses the ability to see small contrast changes at high frequency as cone vision is lost. The  $CTF_{eye}$  curve raises as light level decreases, resulting in lower limiting resolution and loss of the ability to see small contrast differences at any spatial frequency. For luminance levels above 0.01 fL, people have better threshold vision in middle spatial frequencies than at low and high spatial frequencies.

The contrast threshold function in the presence of external noise ( $CTF_n$ ) is found empirically by

$$CTF_n^2 = \kappa (N^2 + N_{eye}^2),$$

where  $N$  and  $N_{eye}$  are sensor and eye noises, and  $\kappa$  is a calibration factor fitted to experimental data.

$CTF_n$  is the contrast threshold function of the sensor-eye system, assuming that the sensor has an MTF of one for all of the spatial frequencies that can be seen by the eye. As with naked-eye vision,  $CTF_n$  depends on radiation level and spatial frequency.

### CTF (cont.)

Setting  $N = 0$ , the  $\text{CTF}_{\text{eye}}$  is obtained at each specific frequency and light level:

$$\text{CTF}_{\text{eye}}^2 = \kappa^2 N_{\text{eye}}^2 \Rightarrow \kappa^2 = \frac{\text{CTF}_{\text{eye}}^2}{N_{\text{eye}}^2},$$

$$\text{CTF}_{\text{n}}^2 = \kappa^2 N^2 + \text{CTF}_{\text{eye}}^2 = \text{CTF}_{\text{eye}}^2 \left( 1 + \frac{N^2}{N_{\text{eye}}^2} \right).$$

Since  $N$  and  $N_{\text{eye}}$  are both referenced to the same display luminance, the above equation can be written as

$$\text{CTF}_{\text{n}}^2 = \text{CTF}_{\text{eye}}^2 \left( 1 + \frac{\sigma_{\text{n}}^2}{\sigma_{\text{eye}}^2} \right),$$

where  $\sigma_{\text{n}}$  is the rms noise filtered on by the display, and  $\sigma_{\text{eye}}$  is the rms visual noise expressed at the display. Both  $\sigma_{\text{n}}^2$  and  $\sigma_{\text{eye}}^2$  have units of  $\text{K}^2 \cdot \text{sec}$ .

$\sigma_{\text{eye}}$  is directly proportional to the average display scene contrast temperature  $\text{SCN}_{\text{tmp}}$  (i.e., Weber's law):

$$\sigma_{\text{eye}}^2 = \frac{\text{SCN}_{\text{tmp}}^2}{\alpha^2}.$$

thus,  $\text{CTF}_{\text{n}}$  can then be written as

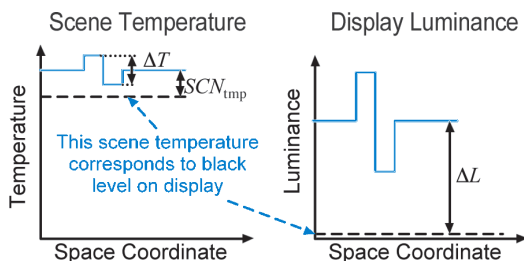
$$\text{CTF}_{\text{n}} = \text{CTF}_{\text{eye}} \left( 1 + \frac{\alpha^2 \sigma_{\text{n}}^2}{\text{SCN}_{\text{tmp}}^2} \right)^{1/2},$$

where  $\alpha = 169.6$  root-Hz is an empirically derived calibration factor relating eye noise to average scene temperature as seen on the display. This value is a constant regardless of the system or environment modeled.

The **Scene Contrast Temperature ( $\text{SCN}_{\text{tmp}}$ )** [K] is the term used in thermal imagers to define the radiometric temperature in the scene needed to raise the display luminance from zero to average displayed luminance. Zero luminance on the display corresponds to minimum visible scene radiant energy, not to zero radiant energy.

### CTF (cont.)

The value of  $SCN_{tmp}$  is determined by the settings of the sensor's gain and offset. Guidance regarding the appropriate value of this model parameter is given in the NVThermIP user's guide.



When using visible sensors, the term luminance or scene contrast illumination is used instead. In this case, the units of  $\sigma_n^2$  and  $\sigma_{eye}^2$  are footlamberts<sup>2</sup>·sec.

The  $CTF_{sys}$  of an imaging system is the naked-eye  $CTF_{eye}$  degraded by a wide variety of factors associated with the imaging system. It is important to realize that the signal is blurred by every component in the imaging processing chain (optics, detector, electronics, display, and eye), while the noise is only blurred by the components subsequent to photodetection. This is because most of the system noise is introduced in the photodetector.  $CTF_{sys}$  is obtained by dividing  $CTF_n$  by the system's MTF to account for the variety of MTF losses associated with the sensor system:

$$CTF_{sys} = \frac{CTF_{eye}}{MTF_{sys}(\xi)} \left( 1 + \frac{\alpha^2 \sigma_n^2}{SCN_{tmp}^2} \right)^{1/2} = \frac{CTF_{eye}}{MTF_{sys}(\xi)} (1 + C^2)^{1/2}.$$

$CTF_{sys}$  is a unitless figure-of-merit and specifies a system's performance under two main conditions:

- $C^2 > 1$ : the noise is visible to the observer, such as in uncooled sensors and second-generation FLIRs.
- $C^2 < 1$ : the noise is not discernible to the observer, such as in cryogenic photon detectors operating with fast optics. In this case, the system performance is limited only by the  $CTF_{eye}$  and the image blur.

**CTF (cont.)**

The  $CTF_{sys}$  equation is 1D. In the TTP model, imaging systems are analyzed in both the horizontal and vertical coordinates separately. Even though this approximation is seldom satisfied, the error associated with it is rather small, thus making this assumption valid in most cases.

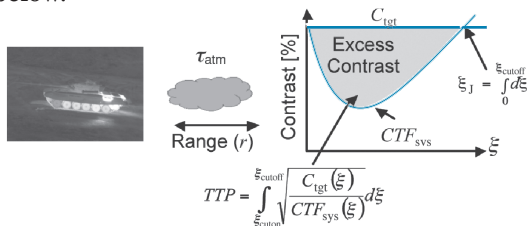
The effect of display glare affects targeting performance. For example, sunlight reflecting off of the display represents a reduction in modulation depth at all spatial frequencies, thus decreasing the system's MTF.



Electronic and digital processing can boost intermediate and high spatial frequencies, improving the displayed representation of scene detail and enhancing target acquisition performance. The sensor gain and offset are adjusted to get optimal acquisition performance, and these parameters determine the scene contrast temperature.

## Target Acquisition Performance

The procedure for predicting field range performance is shown below.



At the sensor, the target-to-background contrast is defined as

$$C_{\text{tgt}} = \tau_{\text{atm}} \frac{\sqrt{(\mu_{\text{tgt}} - \mu_{\text{bkg}})^2 + \sigma_{\text{tgt}}^2}}{2(\text{SCN}_{\text{tmp}})} = \tau_{\text{atm}} \frac{\Delta T_{\text{RSS}}}{2(\text{SCN}_{\text{tmp}})},$$

where  $\mu_{\text{tgt}}$  and  $\mu_{\text{bkg}}$  are the corresponding mean temperatures of the target and background respectively,  $\sigma_{\text{tgt}}$  is the standard deviation of the target temperature, and  $\tau_{\text{atm}}$  is the atmospheric transmission. The numerator of  $C_{\text{tgt}}$  is an estimate of  $L_{\text{v,max}} - L_{\text{v,min}}$  and  $2(\text{SCN}_{\text{tmp}})$  estimates  $L_{\text{v,max}} + L_{\text{v,min}}$  in the contrast equation.

$C_{\text{tgt}}$  establishes the signal available at each single spatial frequency, and that signal is compared to the observer's threshold at that particular frequency.  $C_{\text{tgt}}$  is measured close to the target, so that sensor blur does not affect the accuracy of measuring the target variance. The model treats the "target" signal as having a uniform distribution of spatial frequencies (i.e., white noise). Remember that the model is predicting the ability to discriminate between targets, which means that all spatial frequencies are equally important in telling targets apart.

The Johnson methodology uses the limiting frequency  $\xi_J$  visible at the average target contrast to estimate acquisition performance. In the Johnson method,

$$\xi_J = \int_0^{\xi_{\text{cutoff}}} d\xi,$$

where  $\xi_J$  is the frequency where  $C_{\text{tgt}} = CTF_{\text{sys}}$ .

The TTP methodology generalizes the Johnson methodology. The TTP metric is calculated separately for both the

## Target Acquisition Performance (cont.)

horizontal and vertical dimensions by

$$\text{TTP}_H = \int_{\xi_{\text{cuton}}}^{\xi_{\text{cutoff}}} \sqrt{\frac{C_{\text{tgt}}}{\text{CTF}_{\text{sys}}(\xi)}} d\xi \quad \text{and} \quad \text{TTP}_V = \int_{\eta_{\text{cuton}}}^{\eta_{\text{cutoff}}} \sqrt{\frac{C_{\text{tgt}}}{\text{CTF}_{\text{sys}}(\eta)}} d\eta,$$

where the limits of the integrals are determined by the intersection points of  $C_{\text{tgt}}$  with the  $\text{CTF}_{\text{sys}}$  curve. Here,  $\text{CTF}_{\text{sys}}(\xi)$  and  $\text{CTF}_{\text{sys}}(\eta)$  are the system contrast threshold functions in the horizontal and vertical directions. The TTP metric gives weight to the amount that the threshold is exceeded at each spatial frequency, making the TTP metric sensitive to image qualities not quantified by the Johnson methodology. The geometric mean of  $\text{TTP}_H$  and  $\text{TTP}_V$  values is used to find the overall TTP value:

$$\text{TTP} = \sqrt{\text{TTP}_H \cdot \text{TTP}_V} \quad [\text{cycles/mrad}].$$

The number of cycles on target is then calculated by

$$V_{\text{resolved}} = \text{TTP} \cdot \frac{\sqrt{A_{\text{tgt}}}}{r}.$$

When predicting the probability of accomplishing a task for a staring system at a certain range, sampling artifacts reduce the resolved cycles. A correction factor for the out-of-band spurious response (i.e., signal beyond the half-sampled frequency) is applied:

$$V_{\text{sampling}} = V_{\text{resolved}} \sqrt{1 - 0.58 \text{SSR}_{H,\text{out}}} \sqrt{1 - 0.58 \text{SSR}_{V,\text{out}}},$$

where the factor 0.58 comes from an empirical fit of perception data, and the amount of degradation or spurious response ratio is given by

$$\text{SSR}_{H,\text{out}} = \frac{\int_{\frac{\nu}{2}}^{\infty} \sqrt{\sum_{n \neq 0} \text{MTF}_{H,\text{pre}}^2(\xi - n\nu) \text{MTF}_{H,\text{post}}^2(\xi)} \cdot d\xi}{\int_0^{\infty} \text{MTF}_{H,\text{sys}}^2(\xi) \cdot d\xi}$$

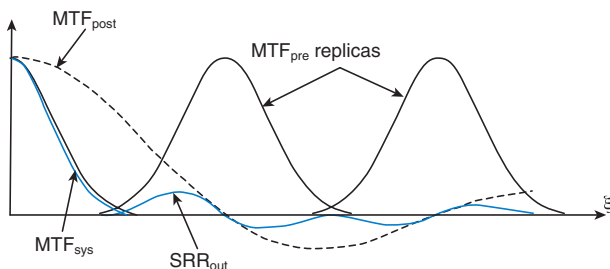
$$\text{SSR}_{V,\text{out}} = \frac{\int_{\frac{\mu}{2}}^{\infty} \sqrt{\sum_{n \neq 0} \text{MTF}_{V,\text{pre}}^2(\eta - n\mu) \text{MTF}_{V,\text{post}}^2(\eta)} \cdot d\eta}{\int_0^{\infty} \text{MTF}_{V,\text{sys}}^2(\eta) \cdot d\eta}$$

## Target Acquisition Performance (cont.)

where

- $n$  is the summation index ( $\dots, -2, -1, 1, 2, \dots$ ),
- $\xi$  is the horizontal spatial frequency,
- $\upsilon$  is the horizontal sample frequency,
- $\text{MTF}_{\text{H,pre}}$  is the pre-sampled MTF (optics detector and line-of-sight jitter),
- $\text{MTF}_{\text{H,post}}$  is the horizontal post-sampled MTF (display, digital processing, and the eye-brain filter),
- $\text{MTF}_{\text{H,sys}}$  is the horizontal system's MTF,
- $\eta$  is the vertical spatial frequency,
- $\mu$  is the vertical sample frequency,
- $\text{MTF}_{\text{V,pre}}$  is the vertical pre-sampled MTF,
- $\text{MTF}_{\text{V,post}}$  is the vertical post-sampled MTF,
- $\text{MTF}_{\text{V,sys}}$  is the vertical system's MTF.
- Most imagers have different sampling spacings in the horizontal and vertical dimensions; and for that reason,  $\text{SSR}_{\text{out}}$  is calculated in both coordinates independently.
- $\text{MTF}_{\text{pre}}$  is replicated at multiples of the sample frequency (i.e., the  $n \neq 0$  terms). The spurious response is the  $\text{MTF}_{\text{pre}}$  replicas filtered by  $\text{MTF}_{\text{post}}$ .

A notional plot of the sampled imager response function is shown as follows:



In real imagers,  $\text{MTF}_{\text{post}}$  limits the upper frequency content visible to the observer to approximately 2.5 times the sample frequency. The replicas centered on spatial frequencies above twice the sample frequency are filtered out. In the  $\text{MTF}_{\text{pre}}$  aliased regions, the signals from different replicas are added in quadrature before integration.

## Probability of Task Performance

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The probability that the observer can discriminate between targets depends on how alike the targets look. The discrimination task is quantified by specifying the number of V50 cycles in the TTP model, where V50 is the number of cycles needed for achieving a 50% probability of correct task accomplishment. The V50 values in the TTP model have the same units and are analogous to the N50 values used in the Johnson criteria, and they are determined in the field by experiment and experience. These N50 and V50 values are used to quantify task difficulty.

Johnson determined that the N50 values needed to predict the discrimination of a target with 50% probability are 1, 4, & 8 cycles for detection, recognition, and identification respectively. He used the critical dimension (minimum height- $x_{\min}$ ) of the target. However, in 1990 NVESD found that the square root of the area of the target ( $A_{\text{tgt}}^{1/2}$ ) was a better descriptor, and these standard sets of values evolved to 0.75 for detection, 3 for recognition, and 6 for identification. The reason for these modifications is that NVESD computer models predicted range performance of staring arrays (i.e., 2D systems) rather than the historical 1D 1<sup>st</sup> generation FLIR systems.

An empirically derived target transfer probability function (TTPF) is an equation that converts the ratio of N to N50 or the ratio of V to V50 into probabilities of task performance.

TTPF<sub>J</sub> associated with the Johnson criteria is given by

$$\text{TTPF}_J = P = \frac{(N/N50)^{E_J}}{1 + (N/N50)^{E_J}}, \text{ where } E_J = 1.75 + 0.35(N/N50),$$

while TTPF<sub>TTP</sub> associated with the TTP criteria takes the form

$$\text{TTPF}_{\text{TTP}} = P = \frac{(V/V50)^{E_{\text{TTP}}}}{1 + (V/V50)^{E_{\text{TTP}}}} \text{ where } E_{\text{TTP}} = 1.51 + 0.24(V/V50).$$



### Probability of Task Performance (cont.)

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The ultimate goal of the TTP model is to predict how well an observer can discriminate targets. Measurements related to recognizing or identifying field targets are based on forced-choice experiments, typically three choices for recognition and twelve choices for identification. There are usually 10–20 well-trained observers, 3–4 ranges, and a good variety of targets and different aspect angles. The images of the targets are degraded by blurring, adding noise, or reducing contrast. The observers are then asked to discriminate the targets based on degraded images, and an ensemble probability as a function of measured range is obtained.

With modern staring arrays, N50 and V50 values are routinely compared, and they do not follow a particular pattern. Their ratio of cycles on target can vary as low as 0.75 and as high as 8, depending on the sampling frequency, MTF, rms noise, contrast, etc. Therefore, the best way to acquire the V50 values is directly from data field experimentation. However, when the N50 values are the only readily available data, an approximate ratio of 1:2.7 is used to relate the Johnson cycles to the TTP cycles on target. This number increases as the probability due to chance is removed. The N50 uncorrected data is rectified by the TTP model by removing the probability of chance before using experimental data to calibrate the model. The data corrected for chance is described by

$$P_{\text{corrected}} = \frac{P_{\text{measured}} - P_{\text{chance}}}{P_{\text{observer}} - P_{\text{chance}}},$$

where  $P_{\text{chance}}$  is the probability of correctly recognizing or identifying the target just by chance and  $P_{\text{measured}}$  is the field-measured probability as a function of range. Most of the military subjects used in these types of experiments are trained to proficiency ( $P_{\text{observer}}$ ) of between 90% and 95% probability. Experienced enlisted military subjects perform at 100% proficiency. Whether this base error rate should be included in the model is a matter of judgment. If the above formulae are followed, the TTP model accurately predicts observer performance.

### N50 to V50 Conversion (Example)

The method for finding the equivalent V50 from a 12-forced-choice identification experiment yielding an  $N50_{ID} = 7.8$  is illustrated assuming a perfect observer (i.e.,  $P_{observer} = 1$ ).

The probability of chance is  $P_{chance} = \frac{1}{12} = 0.083$ .

This probability of chance is removed using

$$P_{corrected} = \frac{P_{measured} - P_{chance}}{1 - P_{chance}} = \frac{0.5 - 0.083}{1 - 0.083} = 0.455.$$

Using the empirical equation of the TTPF associated with the Johnson criteria,

$$0.445 = \frac{(N/7.8)^{E_J}}{1 + (N/7.8)^{E_J}} \quad \text{where} \quad E_J = 1.75 + 0.35(N/7.8)$$

$$\Rightarrow N = 7.15.$$

The scale factor is  $\frac{7.8}{7.15} = 1.1$ , in which case,

$$V50_{ID} = 2.7 \times 1.1 \times N50_{ID} = 23.2.$$

In the case of target detection, there is virtually no chance that an observer discovers a target by chance. Therefore

$$V50_{det} = 2.7 \times N50_{det} = 2.7 \times 0.75 = 2.025.$$

The conversion factors based on the number of choices in the experiment are listed in table below. As the number of choices increases, the impact using uncorrected data to established N50 decreases to one.

| Number of Choices | N50 multiplier |
|-------------------|----------------|
| 3                 | 1.79           |
| 4                 | 1.43           |
| 5                 | 1.30           |
| 6                 | 1.23           |
| 8                 | 1.16           |
| 10                | 1.12           |
| 12                | 1.10           |
| 20                | 1.05           |

## Acquisition Level Definitions

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**Detection:** The determination that an object or location in the FOV may be of military interest such that the military observer takes an action to look closer, alters a search in progress, changes magnification, selects a different sensor, or cues a different sensor.

**Classification:** The object is discriminated by class, such as wheeled or tracked vehicles, rotary- or fixed-wing aircraft, and human being or other animal.

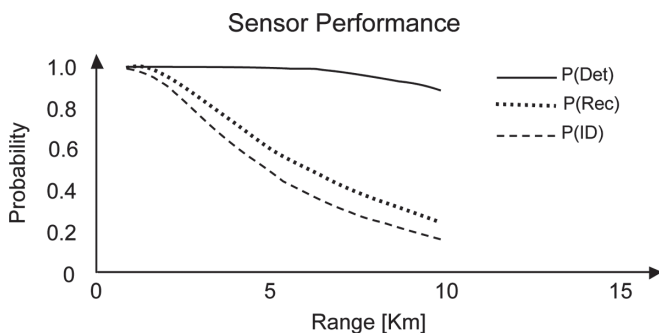
**Recognition:** 1. For vehicles and weapons platforms, the object can be distinguished by category within a class (e.g., tracked or wheeled air defenses or personnel carrier; artillery or tanks or utility vehicles; a dozer or an excavator). 2. For humans, the perception of individual elements (a combination, or a lack of, equipment, hand-held objects), and/or posture that can be distinguished to the extent that the human is determined to be of special military interest (e.g., wearing head gear or carrying hand-held objects).

**Identification:** 1. For military vehicles and weapon systems, the object is discerned by model (e.g., MIA 1 or T80). 2. For commercial vehicles, the object is discerned by model types (e.g., four-door utility vehicle or two-door pickup truck, a four-door sedan or a two-door coupe). 3. For humans, the perception of individual elements, such as clothing, equipment, hand-held objects, posture and/or gender that can be distinguished to the extent that the human is determined to be armed or potentially combatant. Examples include rifle or shovel, uniformed infantry or non-uniformed civilian, armored head gear or construction helmet.

**Feature Identification:** 1. Commercial vehicles can be distinguished by make and model (e.g., Ford Mustang GT or Toyota Camry). 2. Individual elements of clothing, equipment, hand-held objects, and/or gender that can be discriminated by name or country/region of origin (e.g., M16 or Kalachnikov, revolver or cell phone, facial recognition/ID).

## TTP Summary

The basics of a new target acquisition metric for predicting target performance was explained. This new computer model called “the targeting task performance (TTP) metric,” developed by NVESD fixes the shortcomings of the Johnson criteria. The TTP metric can be applied directly to sampled imagers, intensifiers, and imagers that exhibit colored noise due to frequency boost. It predicts the CTF of the system and uses it to calculate the target acquisition range performance. An example of a resultant range performance plot is illustrated below.



The NVTherm\_IP model also calculates the horizontal and vertical pre- and post-sample MTFs, as well as the system’s and turbulence MTFs. It also predicts the 2D MRTDs of the system.

There are still many remaining challenges associated with modeling and predicting target acquisition performance. A primary challenge is that although the field performance and model predictions match well, a robust human/sensor visual acuity test for under-sampled imagers is still unavailable. Furthermore, sophisticated staring array sensors, multiband thermal imagers, and advanced signal processing being driven by new applications, such as urban fighting and force protection/anti-terrorism (FP/AT), are not predicted by any of the current computer models yet.

## Equation Summary

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### Thin lens equations:

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q} \quad \text{Gaussian} \qquad x_{\text{obj}} x_{\text{img}} = f^2 \quad \text{Newtonian}$$

### Thick lens equation:

$$\frac{1}{f_{\text{eff}}} = (n-1) \left[ \frac{1}{R_1} - \frac{1}{R_2} + \frac{(n-1)t}{n} \frac{1}{R_1 R_2} \right]$$

### Lateral or transverse magnification:

$$\mathcal{M} = -\frac{q}{p} = \frac{h_{\text{img}}}{h_{\text{obj}}}$$

### Area or longitudinal magnification:

$$\mathcal{M}^2 = \frac{A_{\text{img}}}{A_{\text{obj}}} = \left( -\frac{q}{p} \right)^2$$

### F-number and numerical aperture:

$$F / \# \equiv \frac{f_{\text{eff}}}{D_{\text{enp}}} \qquad NA \equiv \eta \sin \alpha$$

$$F / \# = \frac{1}{2 \cdot \tan(\sin^{-1} NA)} \qquad NA = \sin \left( \tan^{-1} \frac{1}{2F / \#} \right)$$

$$F / \# \cong \frac{1}{2NA} \quad \text{paraxial approximation}$$

### Field of view:

$$\text{FOV}_{\text{half-angle}} = \theta_{1/2} = \left| \tan^{-1} \frac{h_{\text{obj}}}{p} \right| = \left| \tan^{-1} \frac{h_{\text{img}}}{q} \right|$$

$$\text{FOV}_{\text{full-angle}} = \theta = \frac{d}{f} \quad \text{paraxial approximation}$$

### Diffraction-limited expressions:

$$d_{\text{diff}} = 2.44 \lambda F / \# \quad \text{blur spot} \qquad \beta = 2.44 \frac{\lambda}{D} \quad \text{angular blur}$$

### Refractive index:

$$n = \frac{c}{v}$$

## Equation Summary (cont.)

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### Law of reflection:

$$\theta_i = \theta_r$$

### Snell's law:

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

### Fresnel equations (normal incidence):

$$\rho = \left( \frac{n_2 - n_1}{n_1 + n_2} \right)^2 \quad \tau = \frac{4n_1 n_2}{(n_1 + n_2)^2}$$

### Internal transmittance:

$$\tau_{\text{internal}} = \frac{\phi(z)}{\phi_{\text{inc}}} = e^{-\alpha z}$$

### External transmittance:

$$\tau_{\text{external}} = \tau^2 e^{-\alpha z} = \tau^2 \tau_{\text{internal}}$$

### Reciprocal relative dispersion or Abbe number:

$$V = \frac{n_{\text{mean}} - 1}{\Delta n} = \frac{n_{\text{mean}} - 1}{n_{\text{final}} - n_{\text{initial}}}$$

### Relative partial dispersion:

$$P = \frac{n_{\text{mean}} - n_{\text{initial}}}{n_{\text{final}} - n_{\text{initial}}}$$

### Solid angle equations:

$$\Omega = 4\pi \sin^2 \frac{\theta_{\text{max}}}{2}$$

$$\Omega = \frac{a}{r^2} \quad a \ll r^2 \quad \text{paraxial approximation}$$

### Fundamental equation of radiation transfer:

$$\partial^2 \phi = L \partial A_s \cos \theta_s \partial \Omega_d$$

$$\phi \cong L A_s \cos \theta_s \Omega_d \quad \text{finite quantities}$$

### Intensity:

$$I = \frac{\partial \phi}{\partial \Omega_d} = L A_s \cos \theta_s$$

## Equation Summary (cont.)

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### Exitance and Radiance:

$$M = \frac{\partial \phi}{\partial A_s} \equiv \frac{\phi}{A_s} \text{ finite quantity} \quad M = \pi L \text{ Lambertian radiator}$$

### Irradiance:

$$E_{\text{extended source}} = \frac{\partial \phi}{\partial A_d} = \pi L \sin^2 \theta_{\max} = \frac{\pi L}{4(F/\#)^2 + 1}$$

$$E_{\text{point source}} = \frac{\phi}{A_d} \times 0.84 = \frac{0.84 I \Omega_{\text{optics}}}{\frac{\pi}{4} d_{\text{diff}}^2} = \frac{0.84 I \Omega_{\text{optics}}}{\frac{\pi}{4} [2.44 \lambda (F/\#)]^2}$$

### $A\Omega$ product or optical invariant:

$$A_s \Omega_d = A_d \Omega_s$$

### Planck radiation law:

$$M_{e,\lambda} = \frac{2\pi h c^2}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda kT}\right) - 1} \quad [\text{watt}/\text{cm}^2 \cdot \mu\text{m}]$$

$$M_{p,\lambda} = \frac{2\pi c}{\lambda^4} \frac{1}{\exp\left(\frac{hc}{\lambda kT}\right) - 1} \quad [\text{photon}/\text{sec} \cdot \text{cm}^2 \cdot \mu\text{m}]$$

### Rayleigh-Jeans radiation law:

$$\frac{hc}{\lambda kT} \ll 1$$

$$M_{e,\lambda} \cong \frac{2\pi c kT}{\lambda^4} \quad M_{p,\lambda} \cong \frac{2\pi kT}{\lambda^3 h}$$

### Wien's radiation law:

$$\frac{hc}{\lambda kT} \gg 1$$

$$M_{e,\lambda} \cong \frac{2\pi h c^2}{\lambda^5} \exp\left(-\frac{hc}{\lambda kT}\right) \quad M_{p,\lambda} \cong \frac{2\pi c}{\lambda^4} \exp\left(-\frac{hc}{\lambda kT}\right)$$

### Stefan-Boltzmann law:

$$M_e = \sigma_e T^4 \quad \sigma_e = 5.7 \times 10^{-12} \text{ watt}/\text{cm}^2 \text{K}^4$$

$$M_p = \sigma_p T^3 \quad \sigma_p = 1.52 \times 10^{-11} \text{ photon}/\text{sec} \cdot \text{cm}^2 \text{K}^3$$

## Equation Summary (cont.)

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### Wien's displacement law:

$$\lambda_{\max, e} T = 2898 \quad [\mu\text{m} \cdot \text{K}] \quad \lambda_{\max, p} T = 3662 \quad [\mu\text{m} \cdot \text{K}]$$

### Peak exitance contrast:

$$\lambda_{\text{peak-contrast}} T = 2410 \quad [\mu\text{m} \cdot \text{K}]$$

### Emissivity:

$$\varepsilon(\lambda, T) = \frac{M_{\lambda, \text{source}}(\lambda, T)}{M_{\lambda, \text{BB}}(\lambda, T)}$$

### Kirchhoff's law:

$$\text{Absorbance} = \alpha(\lambda, T) \equiv \varepsilon(\lambda, T) = \text{Emittance}$$

### Power Spectral Density:

$$\text{PSD} = N(f) = \mathcal{F}\{c_n(\tau)\} = \int_{-\infty}^{\infty} c_n(\tau) e^{-j2\pi f\tau} d\tau$$

### Noise-Equivalent Bandwidth:

$$\text{NE}\Delta f \equiv \frac{1}{G^2(f_0)} \int_0^{\infty} |G(f)|^2 df$$

$$\text{NE}\Delta f = \frac{1}{2\tau} \quad \text{square function}$$

$$\text{NE}\Delta f = \frac{1}{4\tau} \quad \text{exponential function}$$

### Shot noise:

$$i_{n, \text{shot}} = \sqrt{2q\bar{i}\Delta f}$$

### Generation/recombination noise:

$$i_{n, G/R} = 2qG\sqrt{\eta E_p A_d \Delta f + g_{th} \Delta f} \equiv 2qG\sqrt{\eta E_p A_d \Delta f}$$

$$i_{n, G/R} = \sqrt{2} i_{n, \text{shot}} G$$



## Equation Summary (cont.)

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### Johnson noise:

$$v_{n,j} = \sqrt{4kTR} \quad i_{n,j} = \sqrt{\frac{4kT}{R}}$$

$$i_{n,j} = \sqrt{i_d^2 + i_L^2} = \sqrt{4kT\Delta f \left[ \frac{T_d}{R_d} + \frac{T_L}{R_L} \right]} \quad \text{for cryogenic detector conditions}$$

### 1/f noise:

$$i_{n,f} = \sqrt{\mathcal{K} \frac{i^\alpha \Delta f}{f^\beta}}$$

### Temperature noise:

$$\overline{\Delta T^2} = \frac{4kKT^2}{K^2 + (2\pi f)^2 C^2}$$

### Responsivity (frequency and spectral):

$$\mathcal{R} = \frac{\text{output signal}}{\text{input flux}}$$

$$\left| \mathcal{R}_{v,e}(f) \right| = \frac{\mathcal{R}_0}{\sqrt{1 + (2\pi f \tau)^2}} \quad \mathcal{R}_{v,e}(\lambda) = \mathcal{R}_{v,e}(\lambda_{\text{cutoff}}) \frac{\lambda}{\lambda_{\text{cutoff}}}$$

### Noise-equivalent power (NEP):

$$NEP = \frac{v_n}{\mathcal{R}_v} = \frac{v_n}{v_{\text{sig}} / \phi_{\text{sig}}} = \frac{\phi_{\text{sig}}}{SNR} \quad [\text{watt}]$$

### Specific or normalized detectivity ( $D^*$ ):

$$D^* = \frac{\sqrt{A_d \Delta f}}{NEP} = \frac{\sqrt{A_d \Delta f}}{\phi_d} SNR = \frac{\sqrt{A_d \Delta f}}{v_n} \mathcal{R}_v \quad [\text{cm}\sqrt{\text{Hz}}/\text{watt}]$$

### D-double star ( $D^{**}$ ):

$$D^{**} = \sin\theta \cdot D^*$$

## Equation Summary (cont.)

### Photodiodes under BLIP conditions:

$$\text{SNR}_{\text{PV}} = \frac{\phi_{\text{e,sig}} \eta q \frac{\lambda}{hc}}{\sqrt{2q^2 \phi_{\text{e,bkg}} \eta \frac{\lambda}{hc} \Delta f}}$$

$$\text{NEP}_{\text{PV,BLIP}}(\lambda) = \sqrt{\frac{hc}{\lambda} \frac{2\phi_{\text{e,bkg}} \Delta f}{\eta}}$$

$$D_{\text{PV,BLIP}}^*(\lambda_{\text{cutoff}}) = \frac{\lambda_{\text{cutoff}}}{hc} \sqrt{\frac{\eta}{2 \int_0^{\lambda_{\text{cutoff}}} E_{\text{p,bkg}}(\lambda) d\lambda}}$$

$$= F / \# \frac{\lambda_{\text{cutoff}}}{hc} \sqrt{\frac{2\eta}{\pi \int_0^{\lambda_{\text{cutoff}}} L_{\text{p,bkg}}(\lambda) d\lambda}},$$

### Photodiodes under JOLI conditions:

$$\text{SNR}_{\text{PV,JOLI}} = \frac{q\eta\phi_{\text{e,sig}} \frac{\lambda}{hc}}{4k\Delta f \left( \frac{T_{\text{d}}}{R_{\text{d}}} + \frac{T_{\text{L}}}{R_{\text{L}}} \right)} \cong \frac{q\eta\phi_{\text{e,sig}} \frac{\lambda}{hc}}{4k\Delta f \frac{T_{\text{d}}}{R_{\text{d}}}} R_{\text{L}} \gg R_{\text{d}}$$

$$D_{\text{PV,JOLI}}^* = \frac{\lambda q \eta}{2hc} \sqrt{\frac{R_{\text{d}} A_{\text{d}}}{kT_{\text{d}}}}$$

### Photoconductive detectors under BLIP conditions:

$$\text{NEP}_{\text{PC,BLIP}} = \frac{2hc}{\lambda G} \sqrt{\frac{E_{\text{p,bkg}} A_{\text{d}} \Delta f}{\eta}} \quad D_{\text{PC,BLIP}}^* = \frac{\lambda}{2hc} G \sqrt{\frac{\eta}{E_{\text{p,bkg}}}}$$

### Photoconductive detectors under JOLI conditions:

$$\text{NEP}_{\text{PC,JOLI}} = \frac{i_{\text{j}}}{\mathcal{R}_{\text{i,PC}}} = \frac{\sqrt{4k\Delta f T / R_{\text{eq}}}}{\frac{\lambda q \eta}{hc} G}$$

### Indium antimonite detectors:

$$\mathcal{E}_{\text{gap}}(x) = 0.233 - (6.498 \times 10^{-5})T - (5.0826 \times 10^{-7})T^2$$

$$i = i_{\text{o}} (e^{qv/kT} - 1) - i_{\text{ph}} + \frac{v}{R_{\text{s}}}$$

## Equation Summary (cont.)

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### Mercury cadmium telluride detectors:

$$\begin{aligned}\mathcal{E}_{\text{gap}}(x) = & -0.302 + 1.93x \\ & + (5.35 \times 10^{-4})T(1-2x) - 0.801x^2 + 0.832x^3\end{aligned}$$

$$\lambda_c = \frac{1}{-0.2435 + 1.5565x + (4.3145 \times 10^{-4})T(1-2x) - 0.646x^2 + 0.671x^3}$$

$$\frac{\partial \lambda_c}{\lambda_c^2} = (1.5565 - 8.629 \times 10^{-4}T - 1.292x + 2.013x^2)\partial x$$

### Quantum well infrared photodetectors (QWIPs):

The de Broglie wavelength:

$$\Lambda = \frac{h}{2\pi p}$$

Schrödinger equation:

$$-\frac{h^2}{8\pi^2 m_e^*} \frac{d^2 \psi}{dx^2} = \mathcal{E} \psi$$

Eigenfunctions:

$$\psi_i = \sqrt{\frac{2}{l_w}} \sin\left(\frac{\pi x}{l_w} i\right)$$

Discrete energy levels:

$$\mathcal{E}_i = \frac{h^2}{8m_e^* l_w^2} i^2$$

Intersubband transition:

$$\Delta \mathcal{E} = \mathcal{E}_2 - \mathcal{E}_1 = \frac{3h^2}{8m_e^* l_w^2}$$

## Equation Summary (cont.)

### Pyroelectric detectors:

$$\mathcal{R}_i = \frac{i}{\phi_{e,o}} = \frac{\rho \omega \varepsilon R_{th} A_d}{\sqrt{1 + (\omega \tau_{th})^2}}$$

$$\mathcal{R}_v = \frac{\rho \omega \varepsilon R_{th} R A_d}{\sqrt{1 + (\omega \tau_{th})^2} \sqrt{1 + (\omega \tau_{RC})^2}}$$

$$D^* = \frac{\mathcal{R}_v \sqrt{A_d \Delta f}}{v_j} = \frac{\rho \omega \varepsilon (R_{th} A_d) \sqrt{R A_d}}{\sqrt{4kT} \sqrt{1 + (\omega \tau_{th})^2} \sqrt{1 + (\omega \tau_{RC})^2}}$$

### Microbolometers:

$$\mathcal{R}_v = \frac{\alpha \varepsilon V_{bias}}{4K \sqrt{1 + \omega^2 \tau_{th}^2} - 2\varepsilon \phi_{e,o}}$$

$$D_{TF}^* = \sqrt{\frac{\varepsilon A_d}{4kKT_d^2}} \quad \text{temperature fluctuation}$$

$$D_{BF}^* = \sqrt{\frac{\varepsilon}{8k\sigma(T_d^5 + T_{bkg}^5)}} \quad \text{background fluctuation}$$

### Thermoelectric detectors:

$$\mathcal{R}_v = \frac{\varepsilon \Theta N}{K \sqrt{1 + \omega^2 \tau_{th}^2}}$$

### SNR in scanning and staring systems:

$$\text{SNR} \propto \sqrt{\text{number of sensor elements}} = \sqrt{n_d}$$

### Range equation:

$$r = \sqrt{\frac{\pi D_{opt} \tau_{opt}}{4(F/\#)}} \sqrt{I \tau_{atm}} \sqrt{D^*} \sqrt{\frac{1}{\text{SNR} \sqrt{\Omega_d \Delta f}}}$$

$$r_{BLIP} = \sqrt{\frac{\pi D_{opt} \tau_{opt}}{4}} \sqrt{\tau_{atm} I} \sqrt{\frac{\lambda_{cut}}{hc}} \sqrt[4]{\frac{2\eta}{\pi \int_0^{\lambda_{cut}} L_{p,bkg}}} \sqrt{\frac{1}{\text{SNR} \sqrt{\Delta f} \sqrt{\Omega_d}}}$$

## Equation Summary (cont.)

---

### Noise-equivalent irradiance (NEI):

$$\text{NEI} = \frac{4F/\#\sqrt{\Omega_d \Delta f}}{\pi D_{\text{opt}} D^* \tau_{\text{opt}} \tau_{\text{atm}}}$$

$$\text{NEI}_{\text{BLIP}} = \frac{2hc}{\lambda_{\text{cutoff}} D_{\text{opt}} \tau_{\text{opt}} \tau_{\text{atm}}} \sqrt{\frac{2\Omega_d \Delta f \int_0^{\lambda_{\text{cutoff}}} L_{\text{p,bkg}} d\lambda}{\pi \eta}}$$

### Modulation depth:

$$M = \frac{A_{\text{max}} - A_{\text{min}}}{A_{\text{max}} + A_{\text{min}}}$$

### MTF definition:

$$\text{MTF}(\xi, \eta) = \frac{M_{\text{img}}(\xi, \eta)}{M_{\text{obj}}(\xi, \eta)}$$

### Strehl ratio (SR):

$$\text{SR} = \frac{\iint \text{MTF}_{\text{actual}}(\xi, \eta) d\xi d\eta}{\iint \text{MTF}_{\text{diff-limited}}(\xi, \eta) d\xi d\eta}$$

### Detector MTF—calculation:

$$\text{MTF}_d(\xi, \eta) = \text{sinc}(d_h \xi) \text{sinc}(d_v \eta)$$

### Optics MTF—calculations (square & circular exit pupils):

$$\text{MTF}(\xi, \eta) = 1 - \frac{\xi}{\xi_{\text{cutoff}}}$$

$$\text{MTF}(\xi) = \begin{cases} \frac{2}{\pi} \left\{ \cos^{-1}(\xi/\xi_{\text{cutoff}}) - \xi/\xi_{\text{cutoff}} \left[ 1 - (\xi/\xi_{\text{cutoff}})^2 \right]^{1/2} \right\} & \text{for } \xi \leq \xi_{\text{cutoff}} \\ 0 & \text{otherwise} \end{cases}$$

## Equation Summary (cont.)

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### MTF measurement techniques:

Point spread function response:

$$\text{MTF}(\xi, \eta) = |\mathcal{F}\{\text{PSF}\}|$$

Line spread function response:

$$\text{MTF}(\xi, \eta) = |\mathcal{F}\{\text{LSF}\}|$$

Edge spread function response:

$$\text{MTF}(\xi, \eta) = \left| \mathcal{F} \left\{ \frac{d(\text{ESF})}{dx} \right\} \right| = |\mathcal{F}\{\text{LSF}\}|$$

Bar target response:

$$\text{CTF}(\xi_f) = \frac{M_{\text{square-response}}(\xi_f)}{M_{\text{input-square-wave}}(\xi_f)}$$

$$\text{CTF}(\xi_f) = \frac{4}{\pi} \left[ \text{MTF}(\xi_f) - \frac{1}{3} \text{MTF}(3\xi_f) + \frac{1}{5} \text{MTF}(5\xi_f) - \frac{1}{7} \text{MTF}(7\xi_f) + \dots \right]$$

$$\text{MTF}(\xi_f) = \frac{\pi}{4} \left[ \text{CTF}(\xi_f) + \frac{1}{3} \text{CTF}(3\xi_f) - \frac{1}{5} \text{CTF}(5\xi_f) + \frac{1}{7} \text{CTF}(7\xi_f) + \dots \right]$$

Random noise target response:

$$\text{MTF}(\xi) = \sqrt{\frac{\text{PSD}_{\text{output}}(\xi)}{\text{PSD}_{\text{input}}(\xi)}}$$

## Equation Summary (cont.)

### Noise-equivalent temperature difference (NETD):

$$\text{NETD} = \frac{4}{\pi} \left[ \frac{(F/\#)^2 \sqrt{\Delta f}}{D^* \sqrt{A_d} \partial L / \partial T} \right]$$

$$\text{NETD}_{\text{BLIP}} = \frac{2\sqrt{2}}{\sqrt{\pi}} \frac{hc}{\lambda_{\text{cut}}} \frac{F/\#}{\partial L / \partial T} \sqrt{\frac{\Delta f \cdot \int_0^{\lambda_{\text{cut}}} L_{\text{p,bkg}} d\lambda}{\eta A_d}}$$

$$\text{NETD} = \frac{\Delta T}{\text{SNR}}$$

### Minimum resolvable temperature difference (MRTD):

$$\text{MRTD} \propto \frac{\xi_t \sqrt{\text{HIOFV} \cdot \text{VIFOV}}}{\text{MTF}(\xi_t) \sqrt{\tau_{\text{eye}} \cdot \tau_{\text{frame}}}} \times \frac{(F/\#)^2 \sqrt{\Delta f}}{D^* \sqrt{A_d} \partial L / \partial T}$$

$$\text{MRTD} = \mathcal{K}(\xi_t) \frac{\text{NETD}}{\text{MTF}(\xi_t)}$$

### Johnson criteria:

Number of Johnson cycles on target:

$$N_{\text{resolved}} = \frac{f \cdot x_{\min}}{2r \sqrt{A_d}}$$

Limiting frequency visible at the average target contrast:

$$\xi_J = \int_0^{\xi_{\text{cutoff}}} d\xi$$

Johnson target transfer probability function (TTPF<sub>J</sub>):

$$\text{TTPF}_J = P = \frac{(N/N50)^{E_J}}{1 + (N/N50)^{E_J}} \quad \text{where } E_J = 1.75 + 0.35(N/N50)$$

### Contrast:

$$\text{Contrast} = \frac{L_{\text{v,max}} - L_{\text{v,min}}}{L_{\text{v,max}} + L_{\text{v,min}}}$$

## Equation Summary (cont.)

### Targeting task performance (TTP) metric model:

Contrast threshold function:

$$\text{CTF}_{\text{sys}} = \frac{\text{CTF}_{\text{eye}}}{\text{MTF}_{\text{sys}}(\xi)} \left( 1 + \frac{\alpha^2 \sigma_n^2}{\text{SCN}_{\text{tmp}}^2} \right)^{1/2}$$

Target-to-background contrast:

$$C_{\text{tgt}} = \tau_{\text{atm}} \frac{\sqrt{(\mu_{\text{tgt}} - \mu_{\text{bkg}})^2 + \sigma_{\text{tgt}}^2}}{2(\text{SCN}_{\text{tmp}})} = \tau_{\text{atm}} \frac{\Delta T_{\text{RSS}}}{2(\text{SCN}_{\text{tmp}})}$$

TTP target acquisition performance:

$$\text{TTP}_{\text{H}} = \int_{\xi_{\text{cuton}}}^{\xi_{\text{cutoff}}} \sqrt{\frac{C_{\text{tgt}}}{\text{CTF}_{\text{sys}}(\xi)}} d\xi \quad \text{and} \quad \text{TTP}_{\text{V}} = \int_{\eta_{\text{cuton}}}^{\eta_{\text{cutoff}}} \sqrt{\frac{C_{\text{tgt}}}{\text{CTF}_{\text{sys}}(\eta)}} d\eta$$

$$\text{TTP} = \sqrt{\text{TTP}_{\text{H}} \cdot \text{TTP}_{\text{V}}} \quad [\text{cycles/mrad}]$$

Number of cycles on target:

$$V_{\text{resolved}} = \text{TTP} \cdot \frac{\sqrt{A_{\text{tgt}}}}{r}$$

Number of cycles on target corrected for sampling:

$$V_{\text{sampled}} = V_{\text{resolved}} \sqrt{1 - 0.58 \text{SSR}_{\text{H,out}}} \sqrt{1 - 0.58 \text{SSR}_{\text{V,out}}}$$

Amount of degradation or spurious response ratio:

$$\text{SSR}_{\text{H,out}} = \frac{\int_{\frac{\nu}{2}}^{\infty} \sqrt{\sum_{n \neq 0} \text{MTF}_{\text{H,pre}}^2(\xi - n\nu) \text{MTF}_{\text{H,post}}^2(\xi)} \cdot d\xi}{\int_0^{\infty} \text{MTF}_{\text{H,sys}}^2(\xi) \cdot d\xi}$$



### Equation Summary (cont.)

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$$\text{SSR}_{V,\text{out}} = \frac{\int_{\mu/2}^{\infty} \sqrt{\sum_{n \neq 0} \text{MTF}_{V,\text{pre}}^2(\eta - n\mu) \text{MTF}_{V,\text{post}}^2(\eta)} \cdot d\eta}{\int_0^{\infty} \text{MTF}_{V,\text{sys}}^2(\eta) \cdot d\eta}$$

TTP target transfer probability function ( $\text{TTPF}_{\text{TTP}}$ ):

$$\text{TTPF}_{\text{TTP}} = P = \frac{(V/V50)^{E_{\text{TTP}}}}{1 + (V/V50)^{E_{\text{TTP}}}}$$

where

$$E_{\text{TTP}} = 1.51 + 0.24(V/V50)$$



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## Index

---

- $1/f$  noise, 45, 47, 54, 62, 65, 68, 71, 153  
 $\Delta f$ , xvi, 152
- Abbe number, xvi, 21, 150  
 absorption, 15, 21, 23, 75–76, 80, 82, 86, 87, 89, 93–95  
 absorption coefficient, xvi, 15, 69  
 afocal systems, 10  
 alloy composition, 68–69  
 aluminum, 16, 40  
 amorphous silicon, xi, 93–94  
 angular magnification, xiv, 10  
 aperture stop (AS), 6, 10–11, 30  
 area magnification, 4, 29, 149  
 astronomical telescope, 10  
 autocorrelation, 46, 112, 114, 123  
 $A\Omega$  product, 28, 29, 151  
 axial ray, 6
- back focal length, xi, 5, 9  
 back focal points, 3, 5  
 background-limited infrared photodetector (BLIP), 51, 62–63, 65, 154  
 barium strontium titanate (BST), 90  
 blackbody (BB), 32–34, 35, 37–38, 41–42, 58, 60, 70, 125, 128  
 blackbody responsivity, xv, 58  
 blur spots, 12–13, 115, 149  
 bolometer, 44  
 bound-to-bound (B-B), 80  
 bound-to-continuum (B-C), 80
- bound-to-quasi-bound (B-QB), 81  
 brass, 40  
 brick, 40  
 brightness temperature, xv, 41–42
- carbon, 40  
 cardinal points, 5  
 cavity radiation, 32  
 central-ordinate theorem, 46, 115  
 chemical vapor deposition (CVD), 77  
 chief ray, 3, 5–6, 14  
 classification, 147  
 cold shield, 11  
 cold stop, 11, 108  
 cold-stop efficiency, 11  
 color temperature, xv, 42  
 common path interferometers, 123  
 concrete, 40  
 cones, 32, 134  
 contrast, 2, 37, 42, 113, 124, 128, 132–133, 136–137, 141  
 contrast threshold function, 136–140, 160  
 contrast transfer function, 119–120  
 converging beam scanner, 101  
 cooled detectors, 66  
 copper, 40  
 corrugated QWIP (CQWIP), 87  
 cryogenic temperatures, 11, 53, 62, 70  
 Curie temperature, xv, 89–90  
 current responsivity, xv, 55, 92
- $D^{**}$ , 63, 153  
 $D_{PV,BLIP}^{*}$  63

## Index

---

- de Broglie wavelength, 78, 82, 155
- detection, 55, 75–77, 85–87, 106–108, 144, 146–147
- detection systems, 106
- detectivity, 44, 60, 87, 92, 153
- detector MTF calculation, 116, 120, 157
- detector output voltage, 58
- Dewar, 11, 70
- diameter, xi–xii, 7, 11–13, 107–108, 114–115
- diffraction, xvi, 12–13, 86–87, 111, 114–115, 122
- diffraction limited, xi, 12–13, 31, 114–115
- diffraction-limited expressions, 149
- digitization, 117, 120
- dispersion, 20–22, 150
- double-layer heterostructure junction (DLHJ), 70–71
- durable coatings, 17
- dwell time, xvii, 101–102, 104–105, 124–125
- dynamic scene simulator (DSS), 99
- edge-spread function (ESF) response, 118–120, 158
- effective focal lengths, xii, 5, 9, 107, 131
- electromagnetic radiation, 1
- electromagnetic spectrum, 1
- electron-injected avalanche photodiode (EAPD), 74
- emission, 2, 23, 38
- emissivity, xvi, 38–42, 108, 128, 152
- enhanced coatings, 17
- enlarging lenses, 9
- entrance pupil, xi, xviii, 6–7, 10, 13, 26, 29, 101, 107, 114, 117
- erecting telescope, 10
- exit pupil, xi, xviii, 6, 10–11, 112, 114, 123, 157
- exitance, xiv, 14, 25–27, 33–38, 41–43, 70, 124, 151
- extended-area source, 30–31
- external transmittance, xvii, 16, 150
- F-number, xii, 6–8, 11–13, 31, 107–109, 124, 149
- feature identification, 147
- field lens, 11
- field of view (FOV), 2, 8, 101–105, 149
- field stop, 8, 11
- first and second principal points, 5
- flux, xvii, 8, 11–12, 25–27
- flux collection efficiency, 2
- flux transfer, 8, 25, 28–29, 31
- focal plane array (FPA), 66–67, 70, 73, 84–86, 98, 101, 117, 120–121, 132
- footprint, xi, 8, 30, 101
- frequency range, 1, 46
- Fresnel equation (normal incidence), 15, 150
- front focal length, xii, 5
- front focal points, 5
- fundamental equation of radiation transfer, 27–28, 31, 150
- fundamental spatial frequency, 119



## Index

---

- Galilean telescope, 10
- Gaussian lens equation, 3–4
- generation-recombination (G/R) noise, 47, 52, 65
- glass, 5, 19–20, 40
- go/no go systems, 106
- Golay cells, 44
- gold, 16, 40
- “good absorbers are good emitters,” 39
- hard coatings, 17
- high-density vertically integrated photodiode (HDVIP), 70, 73–74
- human skin, 40
- human vision, 133–136
- identification, 147
- image irradiance, xii, 30–31, 119
- image quality, 2, 8, 12–13, 110–112, 123, 132, 136
- immersion lens, 11
- impulse response, xiii, 47, 49, 56, 111–113, 115, 118
- index of refraction, 11, 21, 23
- indium antimonite photodiodes, 67, 154
- infrared focal plane array (IRFPA), 66, 72–73
- infrared imaging systems, 2, 116, 128
- instantaneous field of view (IFOV), 101–104, 127, 131
- intensity, 26–27, 30–31, 106, 150
- internal transmittance, xvii, 16, 150
- iron, 40
- irradiance, 25–26, 30–31, 43, 56, 59, 88, 109, 121, 151
- isoplanatism, 112
- Johnson criteria, 110, 131–132, 144, 146, 148, 159
- Johnson-limited noise performance (JOLI), 63
- Johnson metric, 130, 132
- Johnson noise, 47, 53, 62, 63, 65, 92, 153
- Keplerian telescope, 10
- Kirchhoff’s law, 39, 152
- lacquer, 40
- Lambertian disc, 30
- Lambertian radiator, 27, 151
- lateral magnification, 4, 149
- law of reflection, 15, 150
- line spread function (LSF), 118–119
- linearity, 55, 112
- liquid phase epitaxy (LPE), 74
- long-wave infrared (LWIR), 1, 58, 81
- longitudinal magnification, 4, 29, 149
- lubricant, 40
- magnification, xiv, 4, 6, 8, 120, 133, 147
- marginal ray, 6–7, 30
- material dispersion, 15, 21–22
- mean, 21, 45–46, 50
- mercury cadmium telluride (HgCdTe) photodetectors, 68, 70, 72, 75–76, 155
- mesopic, 134
- metals and other oxides, 40
- micro-electro-mechanical systems (MEMS), 93

## Index

---

- microbolometers, 54, 93–98, 156
- midwave infrared (MWIR), 1, 17, 23, 69, 72, 83–84, 94
- minimum resolvable temperature difference (MRTD), 110, 126, 159
- mirrors, 16
- modulation depth, 113, 119, 126, 140, 157
- modulation transfer function (MTF), 74, 110–123, 126–127, 139–140, 157–158
- molar fraction, 68–69
- molecular beam epitaxy (MBE), 71–72, 77
- MTF measurement techniques, 118–123, 158
- multiple quantum well (MQW), 77, 79, 84
  
- N50 values, 131, 144–146
- narcissus effect, 14
- Newtonian lens equation, 3
- nickel, 40
- nodal points, 5
- noise-equivalent bandwidth, xiv, 47–49, 60, 102, 107, 108, 152
- noise-equivalent irradiance (NEI), 109, 157
- noise-equivalent power (NEP), 44, 59–60, 63, 107, 153
- noise-equivalent temperature difference (NETD), 110, 124, 159
- non-uniformity correction (NUC), 67
- nonmetallic materials, 40
- normalized detectivity, xi, 60, 155
  
- numerical aperture (NA), 6–7, 11, 120, 149
- Nyquist frequency, 116
  
- objective lens, 9
- oil, 39, 40
- open circuit, 54, 61, 64
- optical aberrations, 8, 12–13
- optical axis, 3–5, 11, 117
- optical invariant, 28, 29, 151
- optical path difference (OPD), 123
- optical transfer function (OTF), 112, 114–115, 118, 123
- optics MTF calculations, 114–115, 157
- out-of-band spurious response, 142–143, 160
  
- paint, 40
- paper, 40
- parallel beam scanner, 101
- parallel scanning, 101, 104–105
- paraxial approximation, 3, 7, 12, 29
- peak exitance contrast, 37, 42
- phase transfer function (PTF), 112–113
- phasing effects, 117–118
- photoconductive (PC) detector, 64–65, 68, 155
- photoconductive detector under BLIP conditions, 154
- photoconductive detector under JOLI conditions, 154
- photoconductor, 44, 52, 54, 64–65
- photodiode, 44, 61–62, 67–68, 70–76

## Index

---

- photon detector, 11, 44, 57–58, 60, 75, 97, 139
- photons, 1, 25, 36, 44
- photopic, 134
- photovoltaic (PV) detectors, 54, 61
- Planck's equation, 2, 33, 35
- Planck's radiation equation, 36
- Planck's radiation law, 33–35
- plaster, 40
- point source, 12, 26, 30–31, 107–111
- point spread function (PSF), 118, 123, 158
- power spectral density (PSD), 46–48, 53, 121–122, 152
- primary and secondary principal planes, 5
- probability of task performance, 144–145
- protected, 17
- pyroelectric, 44, 91–92
- pyroelectric coefficient, 89
- pyroelectric detector, 88–92, 156
- quantum well, 77–81, 83–85
- quantum well infrared photodetector (QWIP), 77–87, 155
- radiance, 14, 25, 27–28, 30, 124, 151
- radiation temperature, xv, 41
- radiation transfer, 27–28, 31, 150
- radiometry, 25–27
- random reflectors, 85–86
- range equation, 106–108, 156
- Rayleigh-Jeans radiation law, 36
- read-in integrated circuit (RIIC), 99
- read-out integrated circuit (ROIC), 66, 70–74, 83, 93, 98
- reciprocal relative dispersion, 21, 150
- recognition, 147
- reflection loss, 15–16
- refractive index, xiv, 5, 7, 15, 19, 76, 149
- relative partial dispersion, 21, 150
- resolution, 2
- responsive time constant, 56
- responsivity (frequency, spectral, blackbody, and *K*-factor), xv, 44, 55–60, 69, 153
- reverse bias, 61, 74
- rods, 134
- rupture modulus, 15
- sampling effects, 117, 128
- sand, 40
- scan noise, 14
- scanning and staring systems, 105, 156
- scattered flux density, 23
- scene contrast temperature, xv, 138–140
- Schrödinger equation, 79, 155
- scotopic, 134
- search systems, 11, 106–108
- Seebeck coefficient, xvi, 16, 100
- self-radiation, 2, 14
- shading, 14
- shift invariance, 112
- short circuit, 61, 84
- shot noise, 47, 50–52, 62–65, 152

## Index

---

- signal-to-noise ratio (SNR),
  - 25, 51, 53, 59–60, 62,
  - 104–109, 156
- silver, 16, 40
- Snell's law, 3, 15, 150
- soil, 40
- solid angle, xvii, 24–26
- solid angle equations, 150
- spatial frequency, 111–115
- spatial resolution, 110
- specific detectivity, xi, 60, 155
- spectral radiometric
  - quantities, 32
- spectral responsivity, xv, 44, 57–58
- stainless steel, 40
- standard deviation, xvii, 45, 125, 141
- staring systems, 105, 156
- steel, 40
- Stefan-Boltzmann constant, xvii, 35
- Stefan-Boltzmann law, 35–36, 41, 151
- steradians, 24
- stray radiation, 11
- Strehl ratio (SR), 115
- superconductors, 44
  
- target acquisition performance, 140–143, 148, 160
- target transfer probability
  - function (TTPF), 144, 159, 161
- targeting task performance
  - metric (TTP), 110, 132–133, 140–142, 144–145, 148, 159–161
- telecentric stop, 6
- telecentric system, 6
  
- temperature, 2, 14, 23, 32–39, 41–42
- temperature noise, 54, 153
- terrestrial telescope, 10
- thermal conductivity, 15, 18, 100
- thermal detector, 44–45, 57
- thermal equations in
  - photon-derived units, 36
- thermal expansion, 15, 70
- thermal imaging system
  - (TIS), 110, 130
- thermal noise, 11, 53
- thermal resistance coefficient
  - (TRC), 94, 96
- thermal sensitivity, 110, 124, 126
- thermistor, 93
- thermocouple, 100
- thermoelectric detector, 100, 156
- thermopile, 44, 100
- thick lens equation, 149
- thin lens, 3
- thin lens equation, 149
- throughput, 28, 66
- time delay and integration
  - (TDI), 103–104
- tin, 40
- total internal reflection
  - (TIR), 86
- transfer function, 49, 55, 97, 111–112, 114
- transmission range, 15
- transport length, 80
- transverse magnification, 4, 149
- triple-layer heterostructure
  - junction (TLHJ), 70, 72
- two-dimensional (2D)
  - periodic gratings, 86

## Index

---

- type of radiation, 1
- umbrella structure, 94
- V50 values, 144–146
- vanadium oxide ( $\text{VO}_x$ ), 93
- variance or mean-square, xvii, 45–46
- voltage responsivity, xv, 55, 92, 97
- warning systems, 106
- water, 40
- water-erosion resistance, 15
- wavelength range, 1
- Weber's law, 135, 138
- white noise, 46–48, 52
- Wien's displacement law, 35–37, 152
- Wien's radiation law, 36, 151
- wood, 40
- yttrium barium copper oxide, 9



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